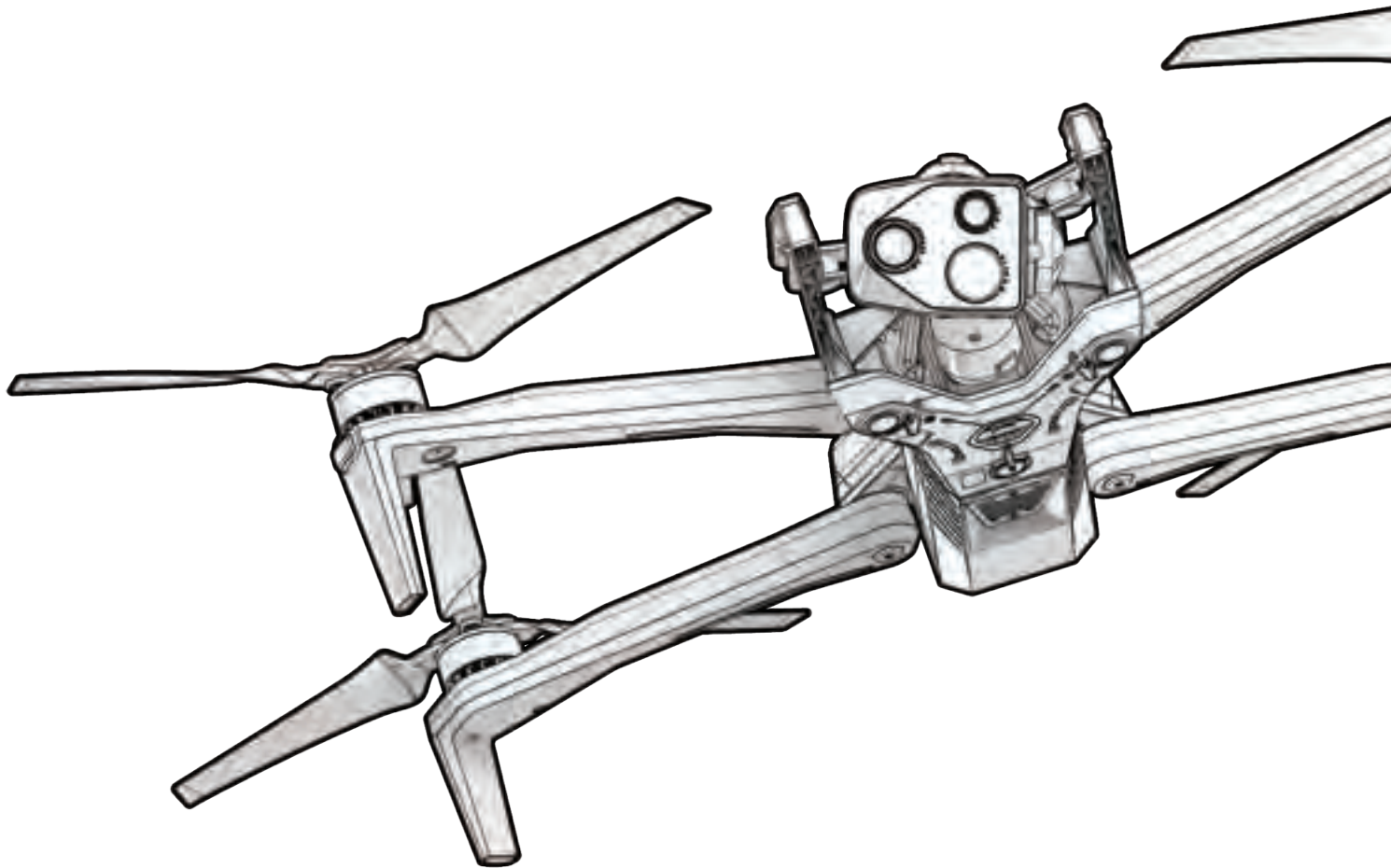


Skydio X10

Operator Manual





WARNING: Please read all documentation provided with your Skydio X10, including but not limited to the X10 Safety Guidelines in the Safety and Operating Guide: www.skydio.com/safety. Failure to follow any instructions or recommendations in our documentation may void the Skydio Limited Warranty.

Notice Regarding Printed Copies

This manual is subject to updates. For the most current procedures, safety information, and operational guidance, refer to the online version of the manual on our [Support site](#). **Printed or locally stored copies may become outdated and should not be relied upon as the sole source of truth.**

It is the responsibility of the operator to ensure they are referencing the most recent version available online.

Additional Resources

For the latest information about Skydio and our products, visit: www.skydio.com

Scan the QR codes to view more information about flying with Skydio X10.



Getting Started with
Skydio X10



Flying with
Skydio X10



Skydio X10
Maintenance



Skydio X10 Safety and
Operating Guide



Skydio Support



Skydio Legal

For legal, warranty and intellectual property information, visit:
www.skydio.com/legal

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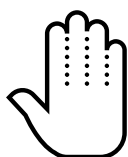
Specifications

Legal

Safety Guidelines



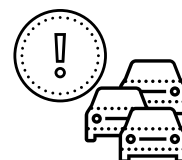
WARNING: To avoid injury or damage to your drone, read the Skydio X10 Safety Guidelines in the Safety and Operating Guide.



Keep your fingers away from moving propellers at all times.



Use caution around reflective surfaces (e.g., still water or mirrors) and small obstacles (e.g., thin branches, utility lines, or chain link fencing)



Skydio X10 does not avoid moving objects (e.g., vehicles).



Skydio X10 obstacle avoidance can be impaired when in low light and poor visibility when flying without NightSense. Fly with extreme caution under these conditions.



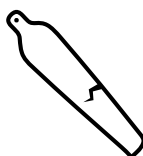
Before flying over water, ensure your drone has a strong GPS signal. Launch and land over a dry surface.



Skydio X10 is IP55 rated and able to fly in light to moderate precipitation with obstacle avoidance disabled. Skydio X10 Controller is IP54 rated.



Clean all of the cameras before each flight so Skydio X10 can see clearly.



Check your propeller blades for damage before each flight.



Follow all civil aviation authority regulations, as well as all local, state, and federal laws.

Warnings

- Do not operate directly over people and vehicles without following all required regulations and garnering any required Certificates of Waiver or Authorization (COA).
- Fly with extreme caution and care around moving obstacles including but not limited to other aerial vehicles, cars, and/or animals.
- Skydio obstacle avoidance may be degraded around transparent or reflective surfaces, windows, mirrors, or still water greater than 23 in (58 cm) wide. Fly with caution.
- The pilot in command (PIC) is solely responsible for: a) managing altitude, range, and battery level and b) following all civil aviation authority regulations, as well as all local, state, and federal laws.
- Adhere to all in-app alerts, warnings, and recommendations such as landing in clear and safe areas.
- Propeller blades are sharp—handle with extreme caution and care especially when the propeller blades are spinning as serious injury and/or damage may occur.
- Obstacle avoidance is disabled during launching and landing. Exercise extreme caution and care to avoid injury and/or damage.
- When using the flashlight on the VT300-L sensor package, do not stare directly into the light at any range for any extended period of time.
- Ensure your landing area is flat, stable, and clear of obstacles.
- Skydio should not be used or handled by a person under the age of 16 years.
- Never fly near or interfere with crewed aircraft operations.
- Never fly under the influence of drugs or alcohol.
- Do not use artificial lighting to brighten the launch or landing zone during night flights. Enhancing illumination in a localized area, such as using vehicle headlights, can cause the drone to misinterpret ambient lighting conditions. This may result in a sudden transition into Attitude Mode after launching, increasing the risk of drift, loss of control, or cause serious bodily harm or injury. For safe night operations, follow best practices for Low Light flight and ensure the drone is prepared for consistent lighting conditions throughout the mission.
- You must maintain a distance of at least 10 ft (3 m) between Skydio X10 and emitters (including cell towers) to reduce the risk of electromagnetic interference (EMI). EMI may lead to camera failures, potentially impacting situational awareness of the remote pilot during operation. Disruptions to both controlled and autonomous flight may lead to a total loss of the vehicle and pose a risk of serious bodily injury.

Preflight

- Skydio X10 navigates visually using cameras so it is essential to keep all of the cameras clean. Use the included microfiber cleaning cloth (or a similar type of microfiber cloth) to ensure camera lenses are free of dust and dirt before every flight.
- Ensure all propellers are firmly attached and free of nicks, cracks, or other visible damage. Never fly with damaged propellers.
- Keep your fingers away from spinning propellers at all times.
- Ensure all 4 arms are fully deployed prior to initiating flight. Failure to do so may result in unstable flight and/or a loss of control.
- Skydio X10 uses magnets to retain the battery which may attract metallic debris that could interfere with the connection of the battery to the drone.
 - Prior to installing the battery, inspect the battery connection pins and the battery bay to ensure that they are undamaged and free of debris.
 - Verify the battery is fully seated in the drone prior to launching.
- Do not fly with any batteries with enclosures that are cracked, swollen, gouged, dented, or otherwise substantially physically deformed.
- Safely handle and dispose of any batteries in accordance with all local laws and regulations.
- Batteries should not be stored in extreme environmental conditions.
- Ensure the Skydio X10 Controller has adequate battery life remaining to complete your intended flight.
- Ensure you have set your Return and Lost Connection behaviors before flying.
- Remove the sensor package lock before flying.
- Inspect the chassis and entire drone for damage and debris prior to flight.

Environment

- Skydio X10 is IP55 rated providing protection from limited dust ingress and light to moderate precipitation conditions; it is recommended to not fly in heavy dust conditions or heavy precipitation.
- The Skydio X10 Controller is IP54 rated providing protection from limited dust ingress and light precipitation conditions; it is recommended to not be used in heavy dust conditions or moderate to heavy precipitation.
- Flight in icy conditions is not supported and may result in the loss of your drone.
- Flight in lightning is not supported and may result in the loss of your drone.
- Ensure the flight environment has good initial visibility and will have good visibility throughout the duration of the flight.
- Do not hand launch or hand land during windy days, when Low Light flight is enabled, when flying at night, or extreme environmental conditions as serious injury and/or damage may occur.
- Fly cautiously over bodies of water as low relative-altitude flight may degrade or impair autonomous flight performance. Before flying over bodies of water, ensure your drone has a strong GPS signal. Fly at least 10 ft (3 m) above the surface of the water.
 - Failure to acquire strong GPS prior to flight over water may result in erratic flight and/or emergency landing and total loss of the drone.
- Launch and land over dry surfaces. Use extreme caution and care when launching or landing from moving vessels.
- Skydio X10 requires good visibility to retain its obstacle avoidance capabilities. Obstacle avoidance can also be impaired when in low light (without NightSense) and poor visibility. Fly with extreme caution and care under these conditions.
- Skydio does not recommend flying the X10 under the following conditions which can result in serious injury and/or damage including total loss of the drone:
 - Wind speeds (including gusts) at or above 28 mph (45 km/h)
 - Temperatures less than -4°F (-20°C) or more than 113°F (45°C)
- The Skydio X10 battery features self-warming technology. When flying in temperatures below 32°F (0°C), prewarm batteries before launching. Battery endurance may be degraded when operating near temperature limits below -4°F (-20°C) and above 113°F (45°C).

Flying Safely

- Your Skydio drone only avoids obstacles that are not in motion.
 - Cars, boats, people, animals, drones, crewed aircraft, or other moving objects may not be avoided.
- In the event that your X10 collides with an object, it will attempt to stabilize and continue flying.
- Keep your fingers away from the propellers anytime they are spinning such as during launch, flight, and landing.
- Skydio X10 can't see certain visually challenging obstacles. Do not fly around thin branches, telephone or power lines, ropes, netting, wires, chain link fencing, or other objects less than 0.5 inch (1.3 centimeters) in diameter. This type of crash is not covered under the Skydio Limited Warranty.
- Do not intentionally try to crash Skydio X10.
- The chassis of Skydio X10 may become hot to the touch in high-temperature environments or direct sunlight, even when powered off. The metal frame may also become hot if powered on while on the ground for long periods of time. Handle with extreme caution and care.
- Do not fly over bodies of water if Skydio X10 indicates a GPS quality warning.
- Exercise extreme caution and care when the sun is low on the horizon as it can temporarily blind the Skydio X10 cameras depending on the angle of flight. Your drone may be cautious or jerky when flying directly toward the sun.
- Skydio X10 may provide an indication, such as displaying an alert to land, if it encounters an issue or determines the environment is not safe for flying. Fly to the nearest safe area and land immediately.
- Flying at high altitudes may significantly increase the time required to return and safely land the Skydio X10. The pilot is solely responsible for managing altitude, range and battery level at all times.
- Be sure to read/watch all flight tutorials and safety-related materials and pay close attention to any in-app messages.
- Keep your hands on the controller joysticks to maintain control throughout flight.

- In preparation for landing, stop active autonomous Flight Skills and fly to a clear and stable area. Avoid areas with people, animals, and moving objects. Try to avoid areas with lots of fine pebbles, sand, rocks, or similar materials.
 - The lights on X10 will turn yellow as the drone descends below 10 ft (3 m) indicating that obstacle avoidance is disabled.
 - **WARNING:** Do not attempt to hand catch Skydio X10 before obstacle avoidance is disabled automatically during landing. Attempting to hand catch Skydio X10 while obstacle avoidance is still active will cause it to attempt to avoid your hand and may result in Skydio X10 impacting yourself or another nearby object, resulting in serious injury and/or damage.
 - While Skydio X10 is landing you may nudge the drone forward, backward, left, or right using the Skydio X10 Controller joysticks.
 - Always monitor Skydio X10 during landing and be prepared to use the “nudge” feature or cancel the landing if Skydio X10 is landing in an undesirable location. Use extreme caution and care when landing on elevated platforms, such as the roof of a car or truck, as the Skydio X10 may move laterally to avoid the platform before descending to the 10 ft (3 m) threshold.

Regulations

- You are solely responsible for your Skydio X10 at all times.
- Always follow [FAA](#) and country-specific civil aviation authority regulations, as well as local, state, and federal laws and regulations when operating your Skydio X10.
- Download the [FAA Drone Safety Guide](#) (if located in the United States).
- Check resources including but not limited to [knowbeforeyoufly.org](https://www.knowbeforeyoufly.org) for more information.
- Do not fly in an environment where the use of drones is restricted or not authorized.
- **Maintain visual line of sight at all times**, unless you have received express permission to fly beyond visual line of sight from a civil aviation authority such as the FAA.
- Skydio drones sold in the United States are Remote ID-compliant.



Getting Started

Welcome to your Skydio X10! Review the basic hardware and accessories that come with your drone before your first flight.

This section covers

Skydio X10 Starter Kit

Skydio X10 Hardware

Skydio X10 Controller Hardware

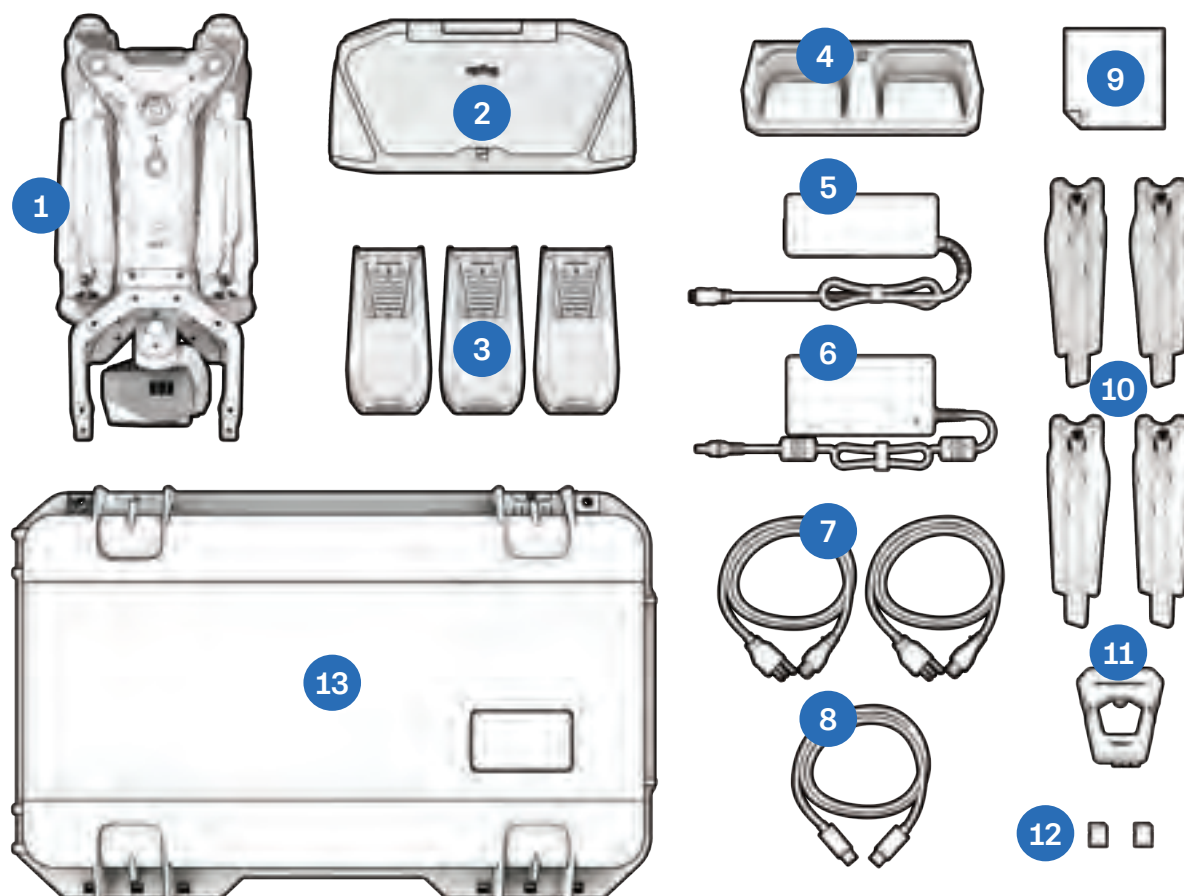
Skydio Autonomy Features

Skydio Connect

Charging

Skydio Cloud Setup

Skydio X10 Starter Kit

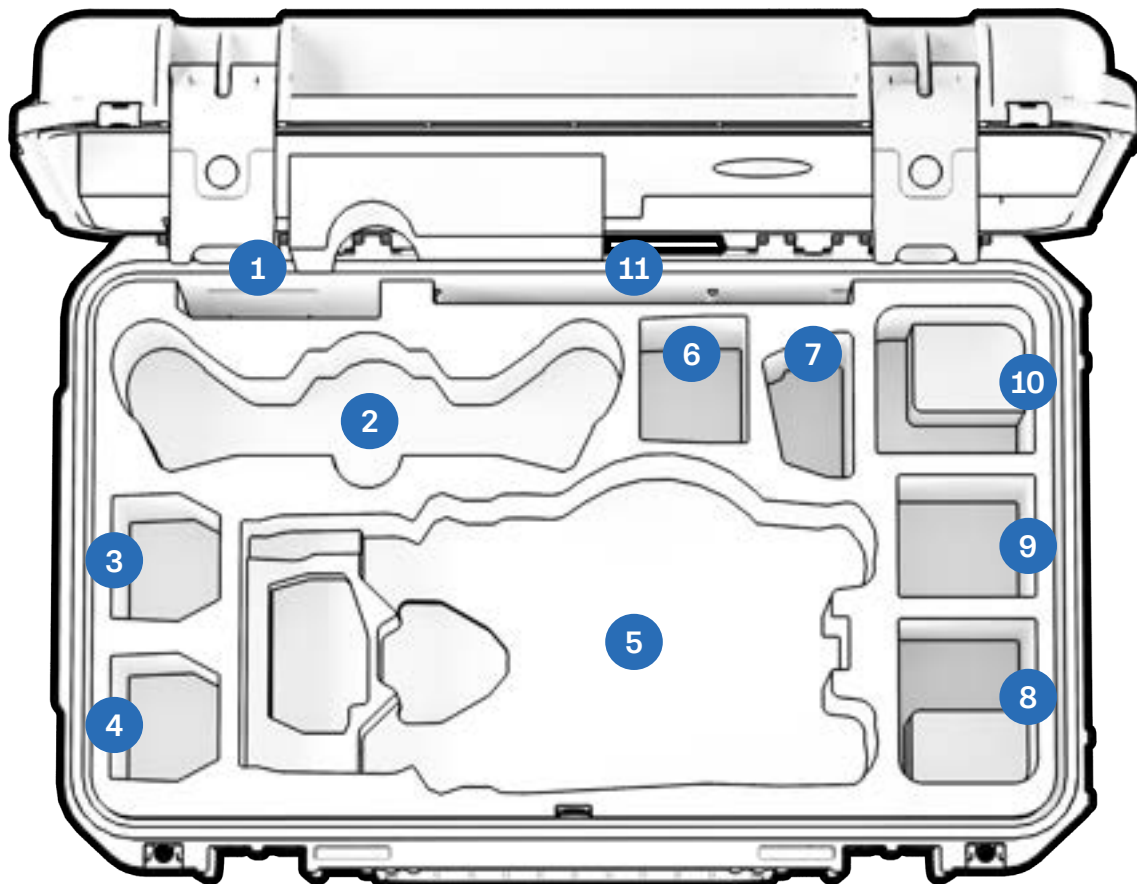


1. Skydio X10 and sensor package
2. Skydio X10 Controller with chosen Skydio Connect option
3. Batteries (3)
4. Skydio X10 Dual Charger
5. 100 W power supply (USB-C)
6. 230 W fast power supply (barrel jack)
7. Power cables (2)
8. USB-C to USB-C pairing cable
9. Microfiber cleaning cloth
10. Spare propeller sets (4)
11. Sensor package lock
12. 256 GB microSD cards, pre-installed (2)
13. Starter Case (hard shell)



Scan for more information about the kits available for purchase.

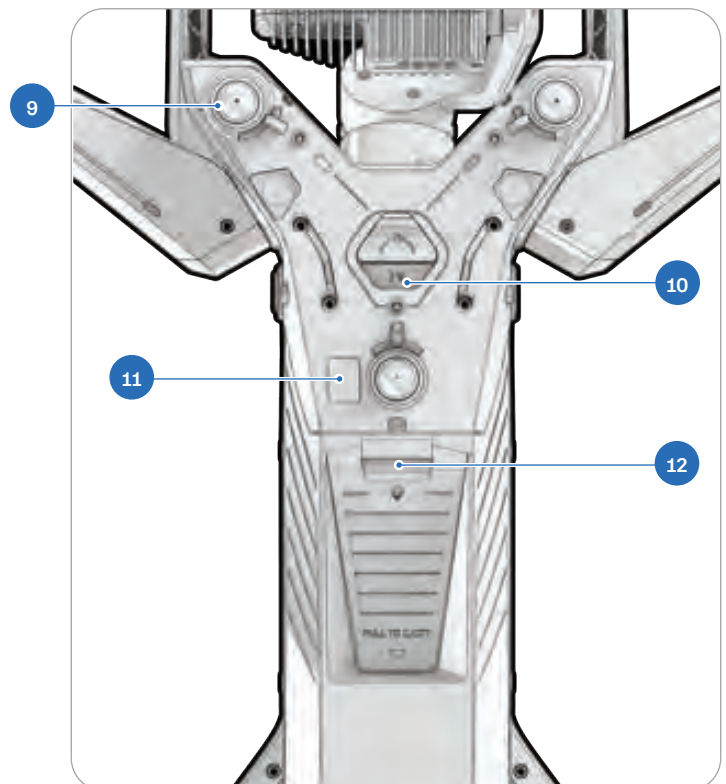
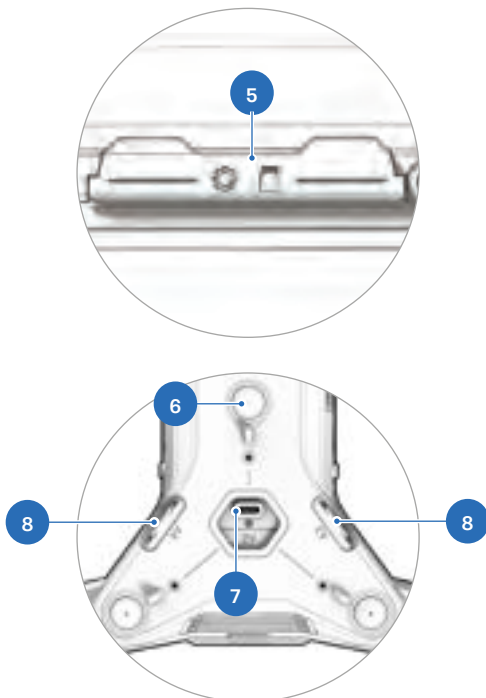
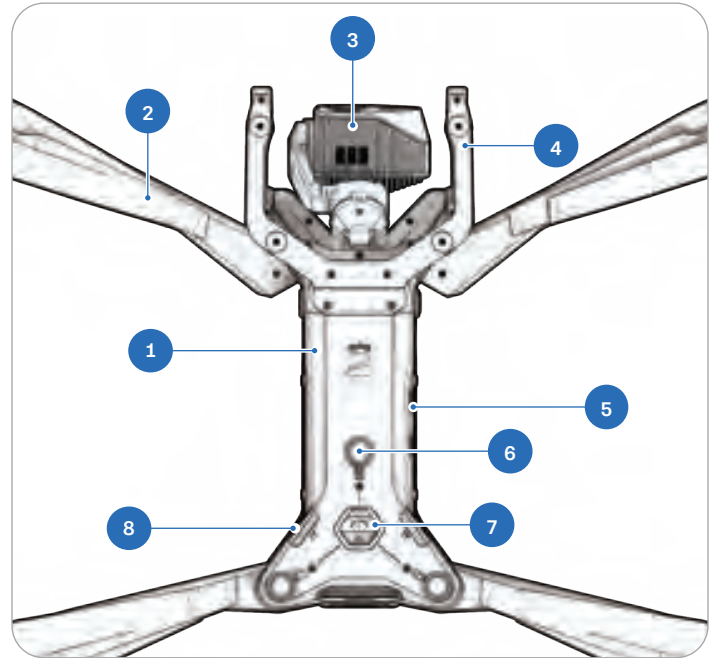
Skydio X10 Starter Case Layout



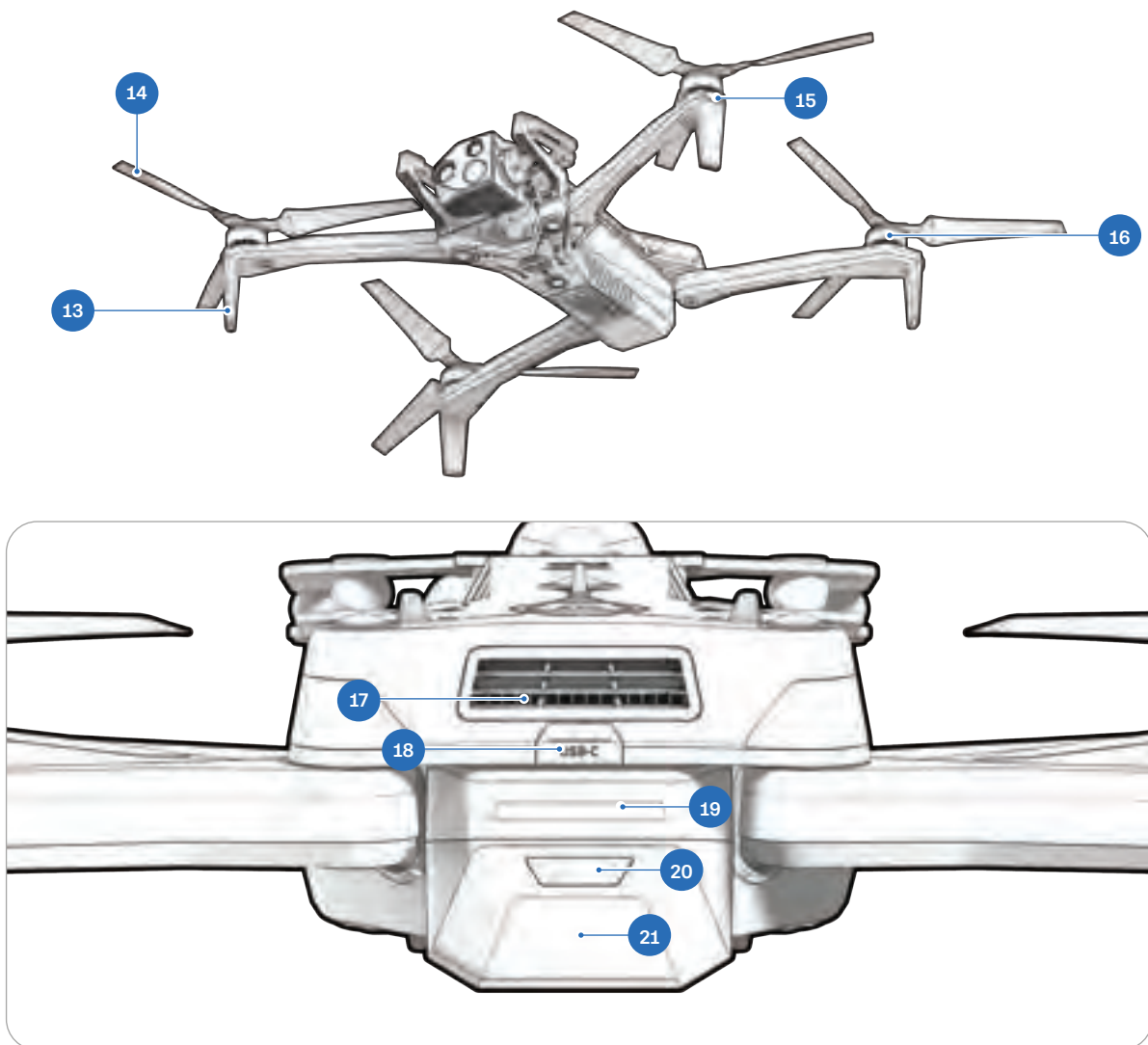
- | | |
|---|---|
| 1. Propellers | 7. X10 Dual Charger |
| 2. X10 Controller | 8. Flex space: 100W Power Supply or Attachment |
| 3. X10 Battery | 9. Flex space: Attachment or 100W Power Supply only (battery not recommended) |
| 4. X10 Battery | 10. Flex Space: 230W Power Supply or Attachment |
| 5. Skydio X10 Drone | 11. Quick Start Guide and other documents |
| 6. Flex space: X10 Battery or 100W Power Supply | |

Skydio X10 Hardware

1. Chassis
2. Arm (4)
3. Sensor package
4. Sensor package frame
5. Log and Media card slots (2)
6. Top navigation cameras (3)
7. Top attachment bay (A2)
8. Side attachment bay (A3, A4)
9. Bottom navigation cameras (3)
10. Bottom attachment bay (A1)
11. Time of flight sensor
12. Parachute strap location (optional attachment)



Getting Started



13. Landing feet/antennas

14. Propeller blades

15. RGB/strobe lights

16. Propeller motors

17. Cooling fan/outlet

18. USB-C charge port

19. Battery lights

20. Power button

21. Battery



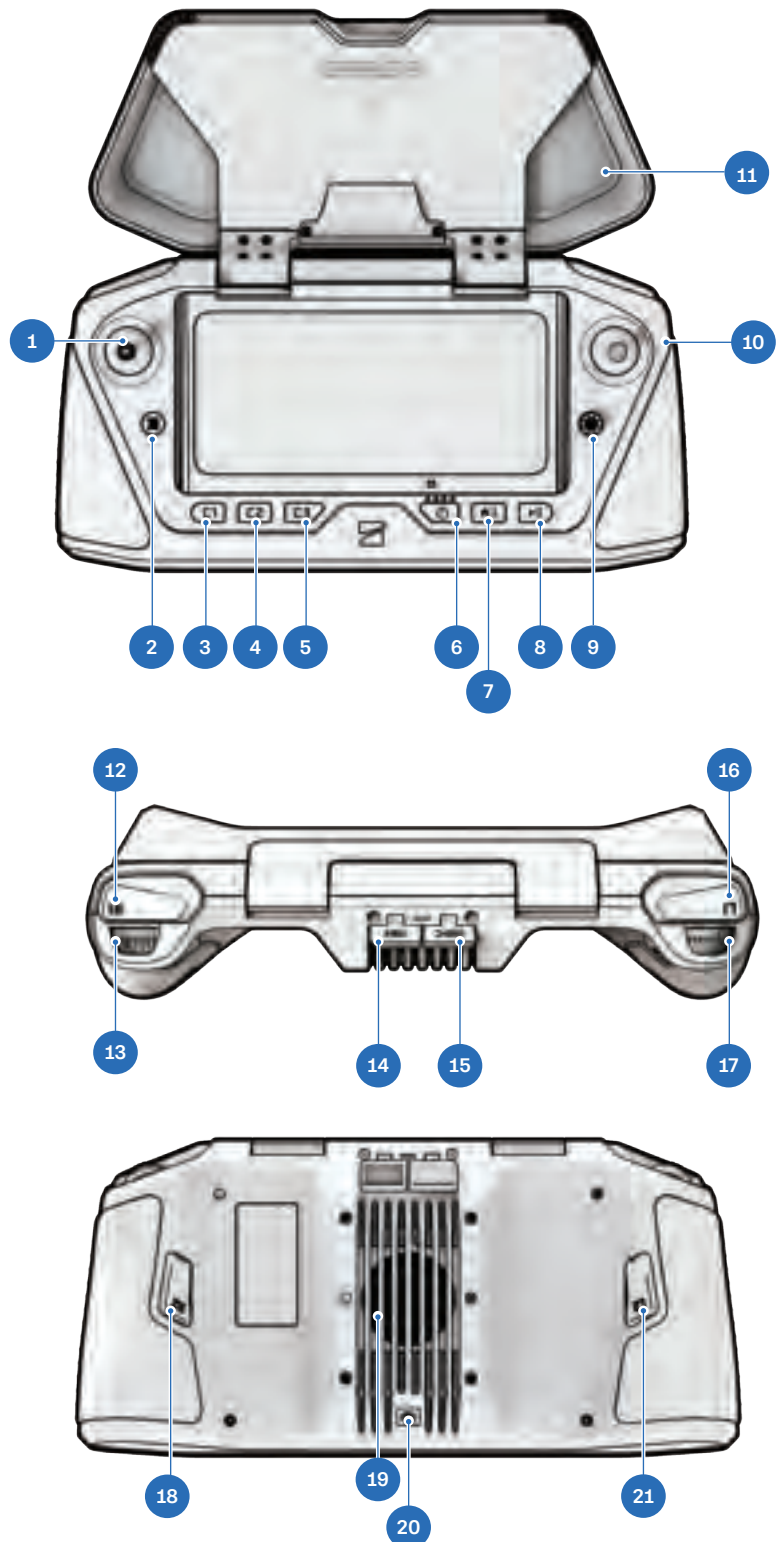
Scan for more information about the sensor packages available for purchase.

Skydio X10 Controller Hardware

1. Left joystick
2. Menu/Back button
3. C1 button¹
4. C2 button¹
5. C3 button¹
6. Power button
7. Launch/Return/Land button
8. Pause button
9. Directional pad (D-pad)
10. Right joystick
11. Controller cover/antennas
12. R1 button (Shutter)
13. Right wheel
14. HDMI port
15. USB-C charge port
16. L1 button (Boost)
17. Left wheel¹
18. R2 button¹
19. Cooling fan
20. Neck strap² and tripod mount
21. L2 button¹

¹Customizable

²Neck strap sold separately



Skydio X10 Overview

Skydio X10 delivers the performance of a heavy-duty drone in a portable, backpack-friendly design. Weighing **under 4.7 pounds**, and measuring just 13.8 inches when folded, it's the most compact drone in its class – easily carried, set up, and flown by a single pilot. With a deployment time of **less than 40 seconds**, an **IP55 rating**, and a maximum flight speed of **45 mph**, the X10 is built for rapid, reliable performance in any environment.

As the first drone to integrate the **Teledyne FLIR Boson+ sensor**, the X10 achieves thermal resolutions of 640x512 with sensitivity down to $\leq 30\text{mK}$. Powered by an onboard **NVIDIA Jetson Orin GPU**, it offers unmatched computing power to process complex tasks in real time. Six custom-designed navigation lenses provide 360° visibility, eliminating blind spots and enabling confident flight – even in **GPS-denied** or **high-EMI** environments.

Packing more megapixels and superior optics than any drone its size, the **modular X10 sensor packages** combine high-resolution visual and radiometric cameras. Four **attachment ports support tools** such as NightSense modules, Speaker, or Spotlight, with a maximum payload of 12 oz. Skydio X10 adapts swiftly to changing mission demands, making it the ultimate tool for any operation.

Software

Skydio X10 comes equipped with the following software features:

- 360° Obstacle Avoidance
- Low Light Flight
- Manual Flight
- Map Capture
- Motion Planning
- Object/Scene Recognition
- Offline Maps/Map Importing
- Point of Interest Orbit
- Real-time 3D mapping
- Skydio Visual Navigator
- Subject Detection
- Thermal Tools
- Track in Place (subject tracking)
- Visual Return-to-Home
- Waypoint Missions
- Zoom

In addition to these built-in capabilities, Skydio offers additional software applications—such as Remote Flight Deck for collaborative flight operations and 3D Scan for high-resolution, automated scanning—to further expand the functionality of Skydio X10.

Attachments and Sensor Packages

To adapt the X10 for specialized tasks, Skydio provides a range of [attachments](#) and [sensor packages](#).

Attachments are optional accessories that can be added to enhance the functionality of the drone, such as advanced lighting or communication tools. These add-ons allow you to customize X10 for unique mission requirements.

Sensor packages are a combination of cameras and sensors mounted on the front of the drone, stabilized by a gimbal. These packages may feature thermal imaging or integrated flashlights to support various operational needs. Sensor packages are designed to capture precise data across various environments and applications.

Solution Bundles for Specific Use Cases

Skydio also offers pre-configured solution bundles tailored to specific applications. These bundles provide the ideal combination of hardware, software, and accessories for tasks like Drone as First Responder (DFR), bridge inspection, and utility management, ensuring you have the tools you need right out of the box.

Skydio Connect

Skydio Connect includes various radio connectivity options between Skydio X10, the Skydio X10 Controller, and Flight Deck controls, whether you're flying with the controller or via browser.

There are two Skydio Connect options when purchasing your Skydio X10:

Skydio Connect SL provides a proprietary, optimized point-to-point wireless link between X10 and the controller. With line-of-sight distances up to 7.5 miles or 12 kilometers, SL offers robust performance for most autonomous flight missions.

- Operating frequency: 2.4 GHz, 5GHz
- Range in ideal conditions: 7.5 mi (12 km)

Skydio Connect 5G allows you to fly Skydio X10 anywhere with a stable cellular connection. With the addition of Skydio Remote Flight Deck you can also operate your Skydio drones through an internet browser via Skydio Cloud. You will also be able to remotely operate your drones from any Skydio X10 Controller that is connected to your 5G network.



NOTE: When flying with Skydio Connect SL, always maintain a clear line of sight between the controller and drone.

Skydio Connect Fusion

If you purchased Skydio Connect 5G, you will have access to Connect SL along with **Skydio Fusion**.

Skydio Connect Fusion provides continuous, high-quality video by encoding separate streams for both Connect SL and Connect 5G, then dynamically selects the strongest feed in real-time.

Pilots do not need to manually manage radio links as Skydio X10 and the X10 Controller will optimize inflight connectivity by automatically switching between Connect 5G (cellular) and Connect SL (point-to-point), ensuring the best available connection is always in use.

- Fusion is set as the default **Flight Connection** within the **Radio** menu
- Select **FUS** in the telemetry bar to view detailed connection health status



Charging Skydio X10 Batteries

Skydio X10 batteries are shipped in a state of hibernation and will not power on your drone out of the box. Your batteries will automatically exit this state once they begin charging for the first time.

Using the Skydio X10 Dual Charger

The Skydio X10 Dual Charger sequentially charges two batteries. The Dual Charger will prioritize fully charging the battery with the highest charge level. If both batteries are depleted, it will prioritize the battery that is inserted first.

Step 1 - Remove battery from drone

Skydio X10 batteries are held in place using a magnetic connection.

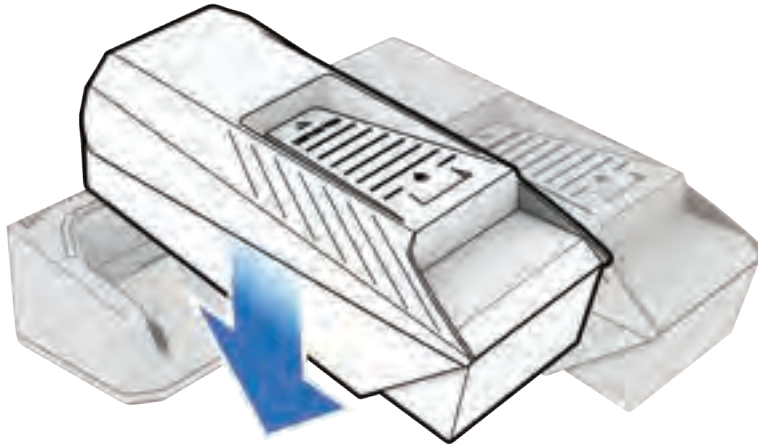
1. Firmly grip the drone chassis with one hand
2. Grip the battery with your other hand, placing your palm over the power button and wrapping your thumb under the battery
3. Using your fingers as leverage, press against the drone until the magnets disengage and slide the battery away from the sensor package



Charging

Step 2 - Place batteries into the X10 Dual Charger

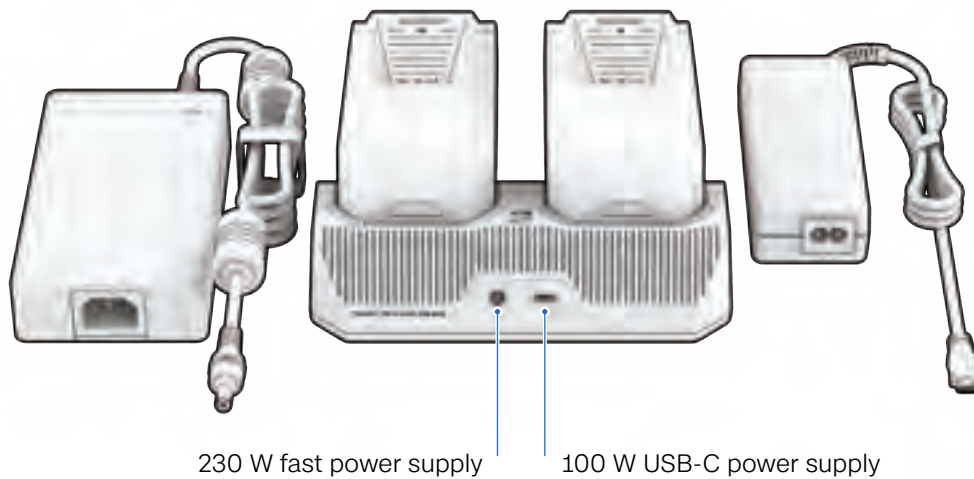
Ensure the battery and connection pins are free of debris and interference. Gently push down to ensure the batteries are properly seated.



Step 3 - Insert the power supply

Two charging ports are located on the back of the Skydio X10 Dual Charger. You may use either the 100 W USB-C power supply or the 230 W fast power supply.

Plug into a power source.



Charging

Battery Charge State	Light Behavior on X10 Dual Charger
Actively charging	Pulsing blue
Waiting to charge	Solid blue
Charging complete	Solid green

Power Supply	Input	Charge Time
230 W	20VDC, 11.5A	About 1 hour to charge a depleted battery
100 W	5-20VDC, 3A / 20VDC, 5A	About 1 hour 45 minutes to charge a depleted battery



CAUTION: *Third-party adapters and cables are not supported. Only use the Skydio-provided power supplies and cables to charge your batteries.*



NOTE: *Pass-through charging is not currently supported. This functionality will be enabled in a future software update.*

Using Skydio X10

Step 1 - Insert battery

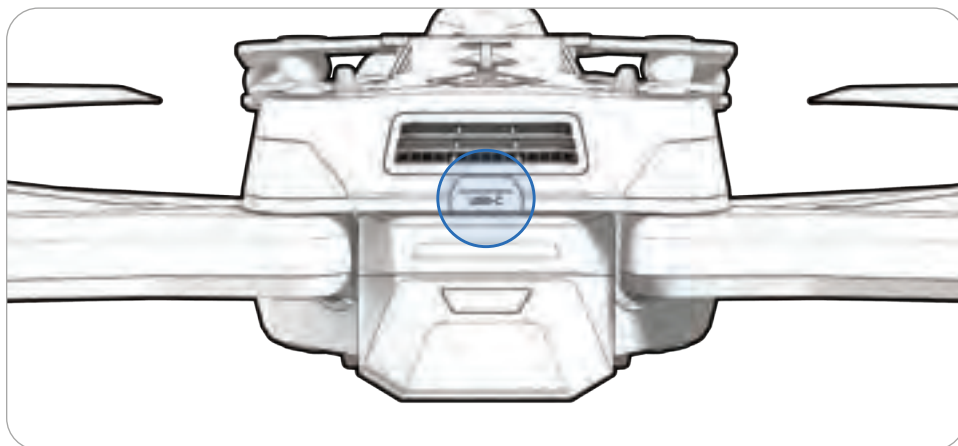
Align the battery with the rails and slide toward the sensor package until the magnets engage.

- Ensure the battery and rails are free of debris and interference
- Ensure the battery is completely seated before flying



Step 2 - Locate the charging USB-C port

The charging port is located on the back of the drone above the battery. This is the only USB-C port that supports charging.

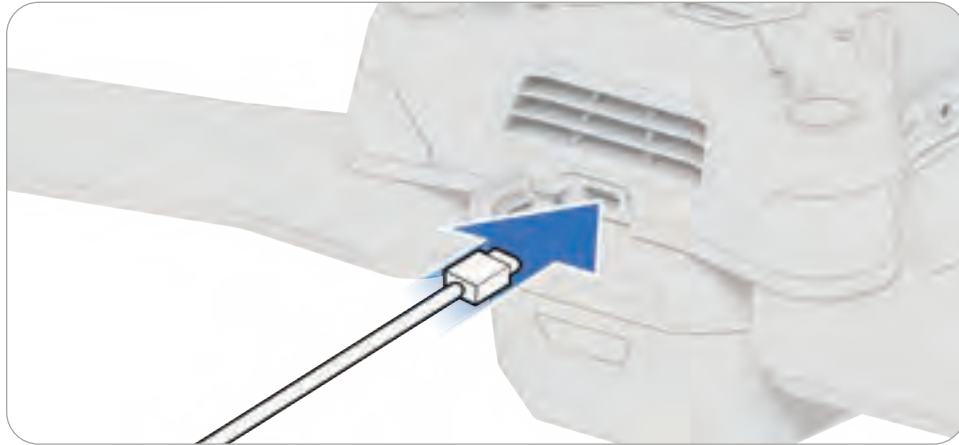


Charging

Step 3 - Insert the 100 W power supply

Plug into a power source.

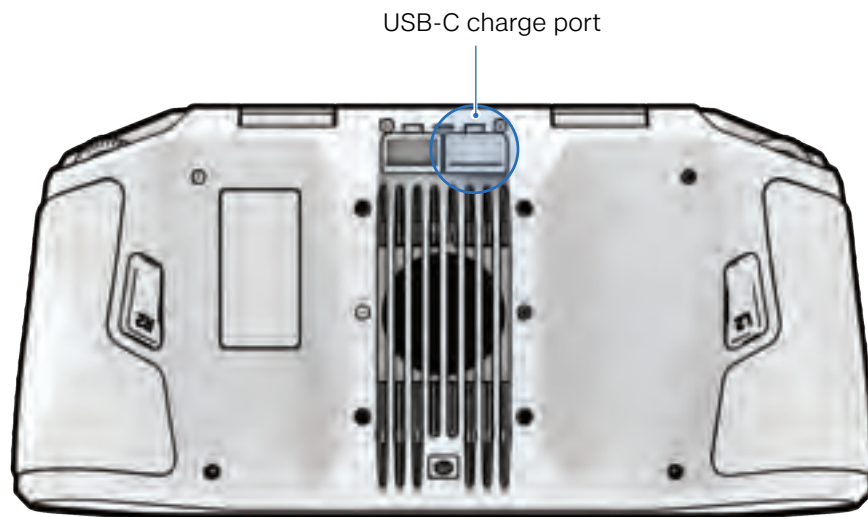
- It will take about 2 hours to fully charge a depleted battery using the 100 W power supply



Charging the Skydio X10 Controller

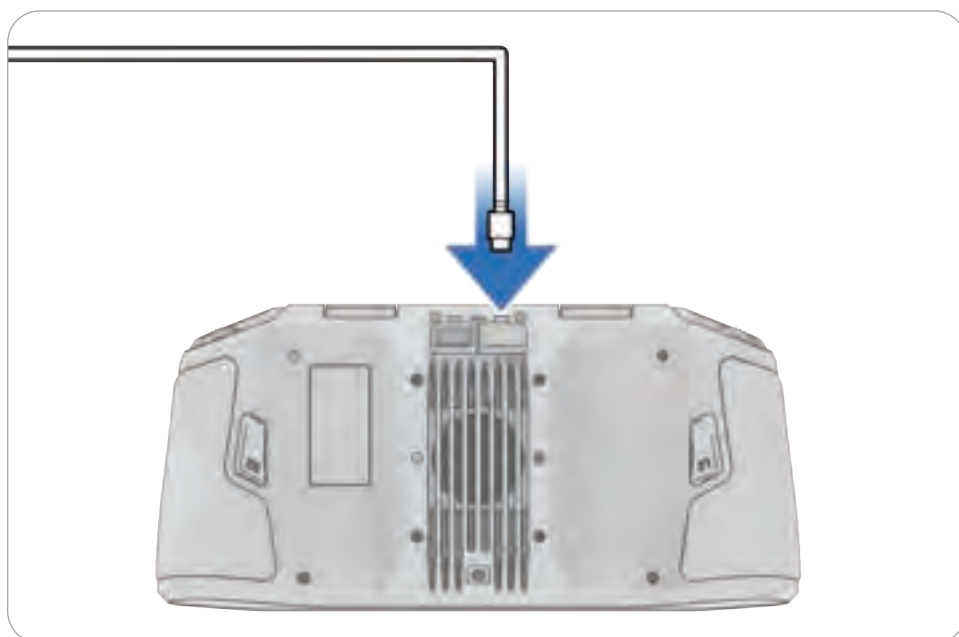
Step 1 - Locate the USB-C port

The charging port is located on the back of the controller.



Step 2 - Insert the 100 W power supply

Connect your Skydio X10 Controller to the 100 W power supply. Plug into a power source. The lights on the front of the controller will turn on and indicate the level of charge.



Skydio Cloud Setup

Before flying, you or an Organization Admin will need to configure your Skydio Cloud account in order to manage your fleet or sync your media. This includes configuring your organization settings, adding users, claiming your Skydio X10, and connecting to wireless networks.



NOTE: Skydio X10 and the X10 Controller must be claimed in Skydio Cloud to receive software updates and fly with Connect 5G.

As part of our onboarding process, Skydio will create the initial admin user for your organization. An admin level is required to set up your Skydio Cloud organization.



INFO: For more information about setting up Skydio Cloud, visit [Getting Started with Skydio Cloud](#).

Log in and configure settings

Step 1 - Log in

Visit cloud.skydio.com and enter your email address. Enter the verification code sent to your email address.



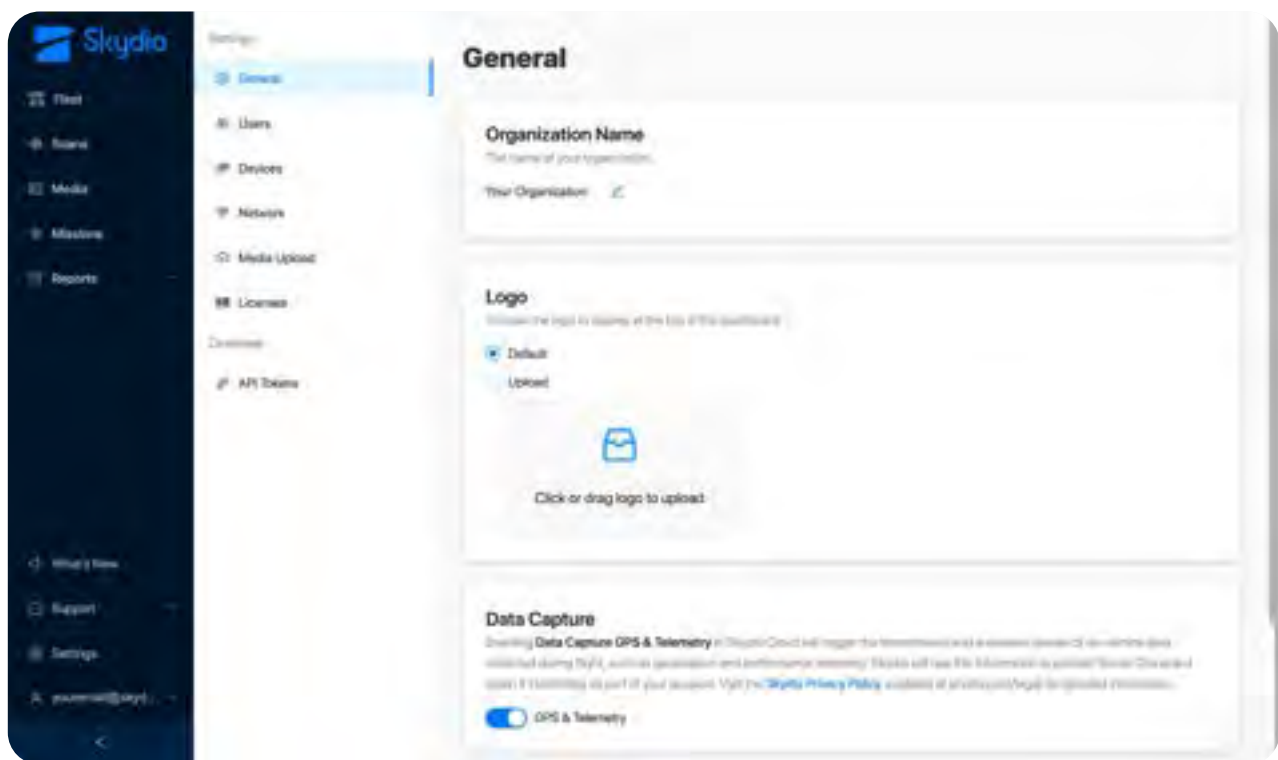
Skydio Cloud Setup

Step 2 - Select Settings > General

Upload a logo for your organization to replace the Skydio logo in the upper left corner (optional).

Enable Data Capture which allows GPS and telemetry data to automatically upload to the cloud after each flight.

If Data Capture is toggled off, the flight path and other telemetry data will not display on the flight screen.



Add Users

Step 1 - Select Settings

Step 2 - Select Users

Step 3 - Add members

Select **Add User**. Add an email address for the user and assign a role.

This is a crucial step to ensure pilots in your organization can access Skydio Flight Deck on their X10 controller.



NOTE: Only Admin level accounts can add users. A member email address can only be associated with one organization at a time. Use the **Login Methods** tab to configure settings such as [Single Sign-On \(SSO\)](#).

Claim Devices

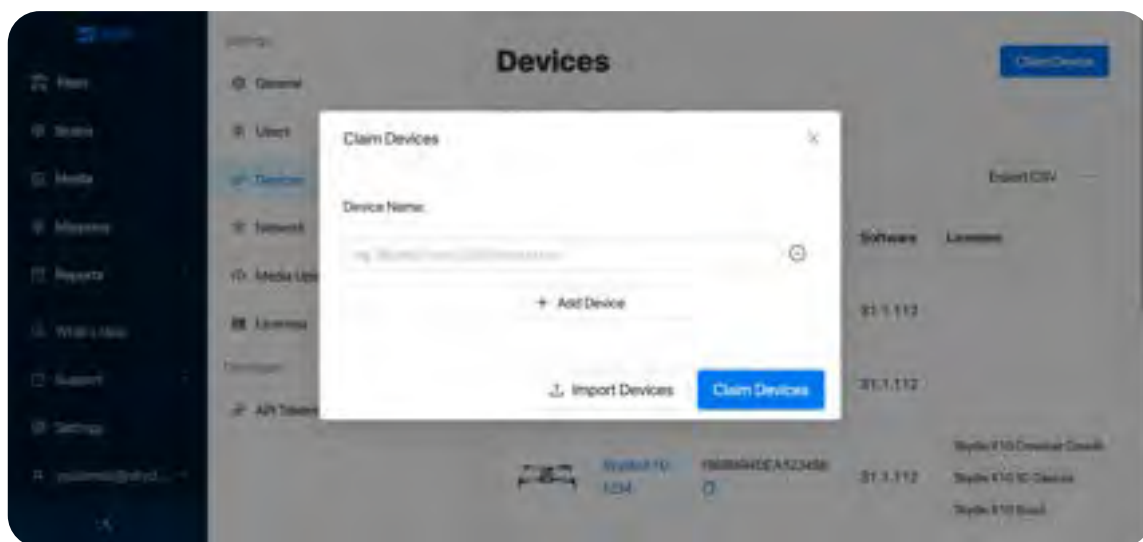
You will need to **claim your drone and controller in Skydio Cloud to receive software updates**. Claiming your drone, controller, and batteries also enables you to track usage metrics and assign additional software licenses. Skydio X10 is not automatically associated with an organization, meaning you will need to claim the devices that you want in your fleet.

Step 1 - Select Settings

Step 2 - Select Devices

Step 3 - Select Claim Device

Step 4 - Depending on the device you are claiming, enter either the UAV name, serial number, or battery serial number

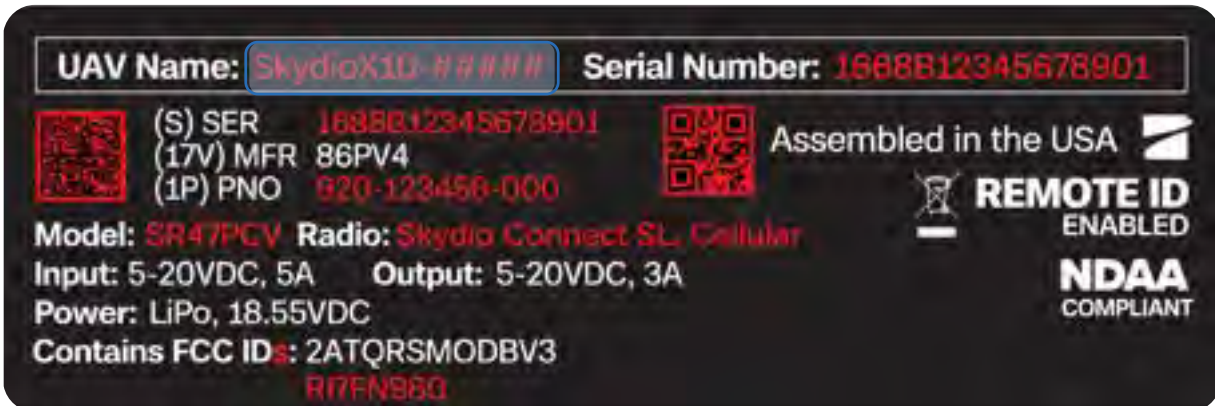


NOTE: For organizations with large fleets, your Customer Success Manager can assist in bulk claiming your new Skydio drones and batteries in Skydio Cloud.

Finding Drone Name, Controller Serial Number, and Battery Serial Number

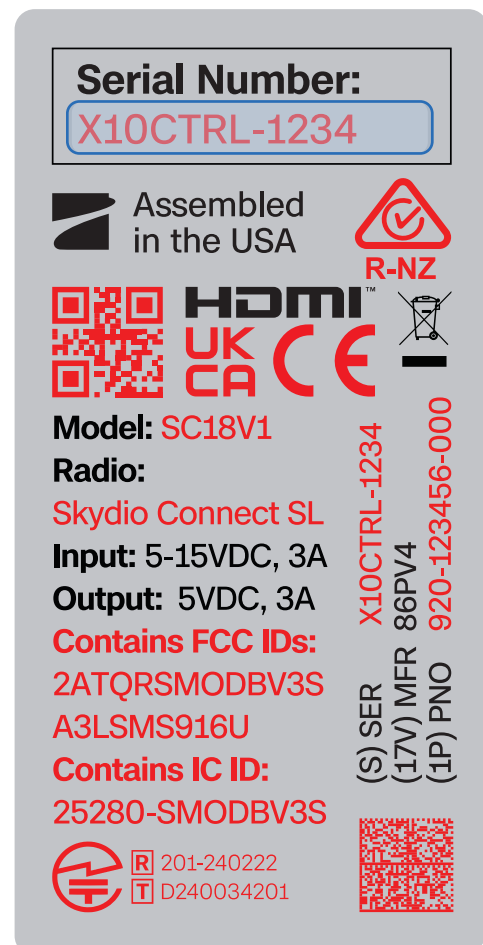
Skydio X10 UAV Name

This begins with **SkydioX10-** and can be found on the label inside the battery bay of the drone.



Skydio X10 Controller Serial Number

This begins with **X10CTRL-** and can be found on the label located on the back of the controller.



Skydio Cloud Setup

Skydio X10 Battery Serial Number

This 16-character number can be found on the battery label below the QR code.



INFO: After an initial flight, you can also find the battery serial number listed in the Battery column under **Reports > Flights**.

Assign Licenses

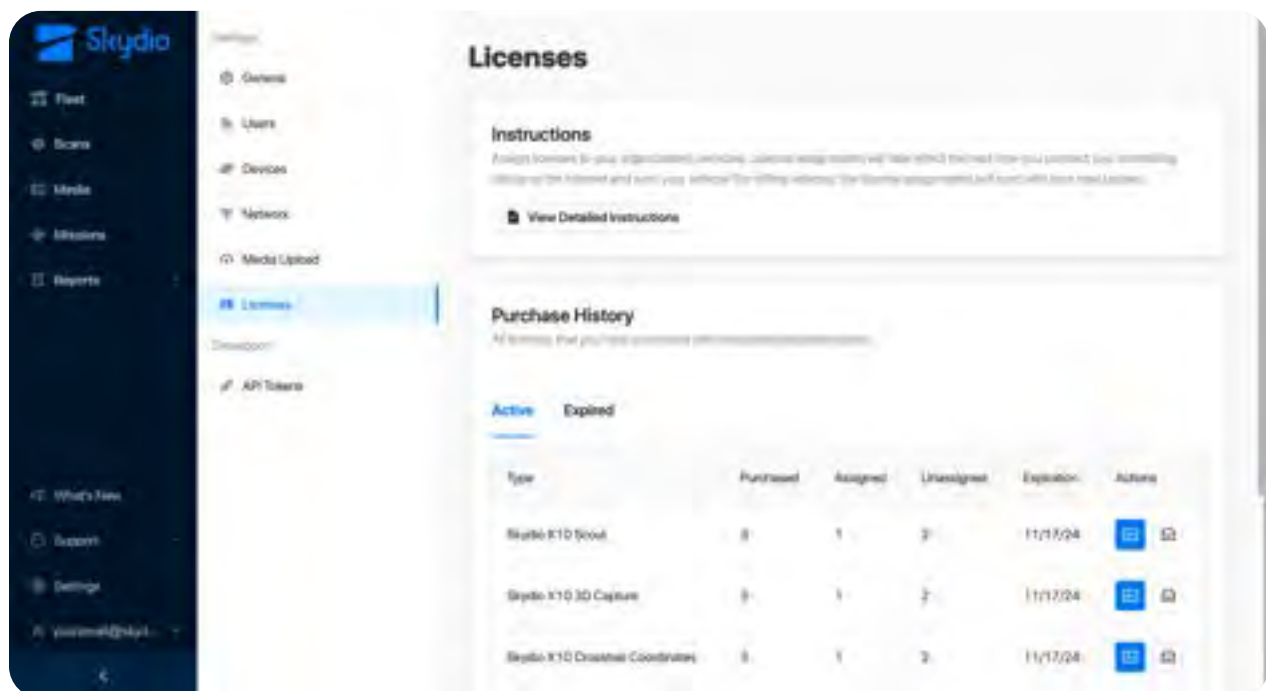
If you purchased the following add-on software packages, you will need to assign these software licenses to your drones:

- 3D Scan
- Tower Capture
- Scout
- Crosshair Coordinates

Step 1 - Select Settings

Step 2 - Select Licenses

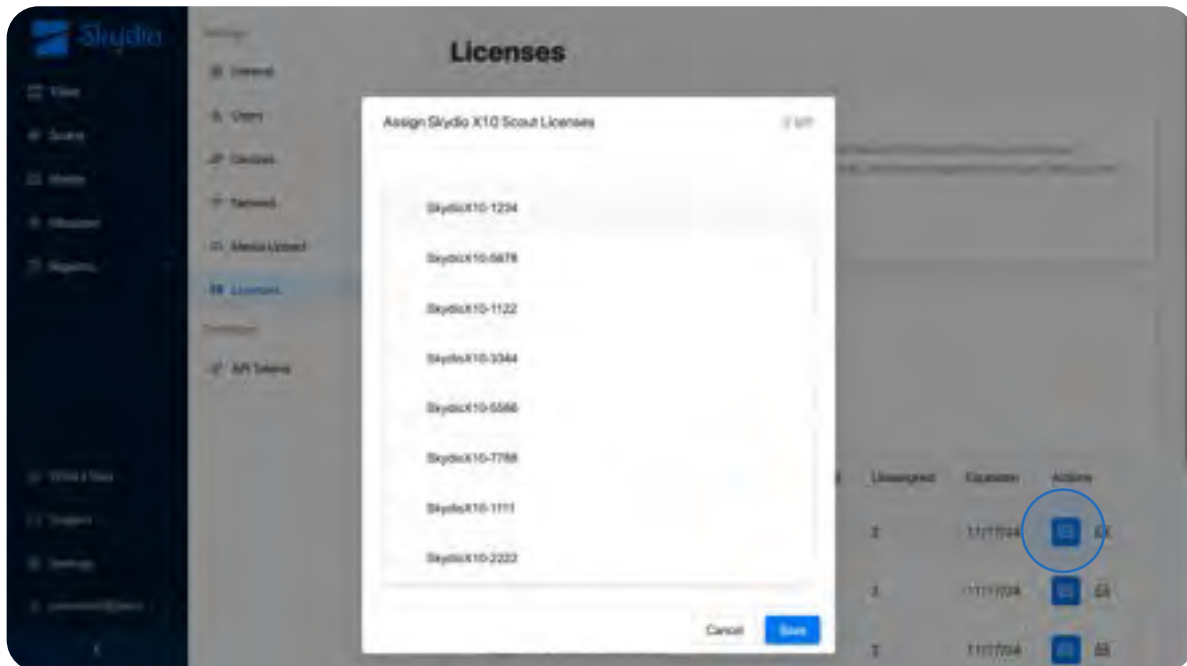
The Licenses page offers information about your license Purchase History including all active and expired licenses, how many have been assigned, and how many remain unassigned.



Skydio Cloud Setup

Step 3 - Assign Licenses

Select the blue icon to assign that license to an eligible vehicle.



INFO: For more information about software licenses, visit [How to assign Skydio advanced software licenses in Skydio Cloud](#).

Media Sync

If you have purchased Media Sync for your organization, you will need to add a wireless network and configure your media upload preferences.



INFO: For more information about setting up Media Sync, visit [How to use Media Sync in Skydio Cloud](#). You also have the ability to enable watermarks on your photos and videos. For more information, visit [How to enable watermark overlays in Skydio Cloud](#).



Preflight

Learn the basics of setting up before launching.

This section covers

Skydio X10 Setup

Skydio X10 Controller Setup

Updating the Skydio X10 Controller

Updating Skydio X10

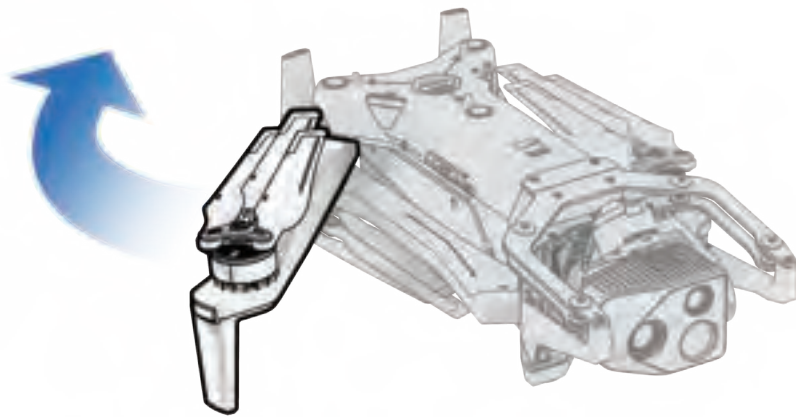
Skydio Connect 5G Setup

Battery Warming in Cold Environments

Skydio X10 Setup

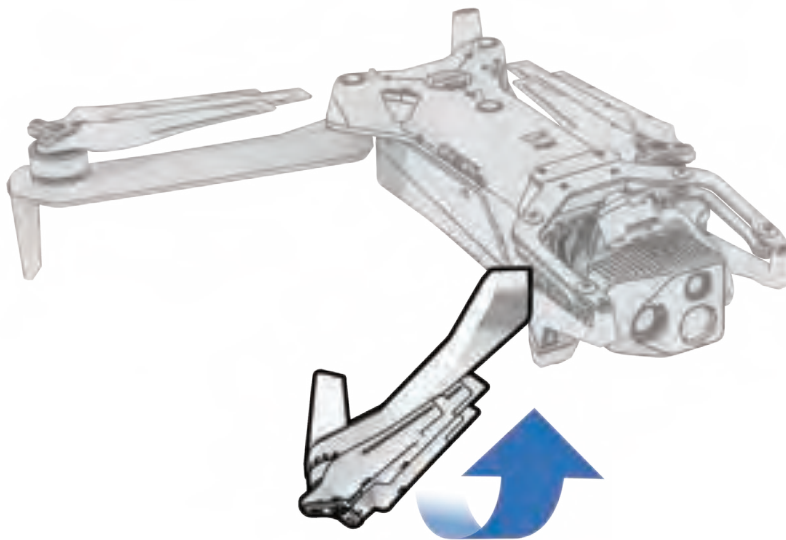
Step 1 - Unfold the rear arms

Hold the drone with the sensor package facing away from you. Pull **laterally** away from the chassis until you feel the arm seat into place.



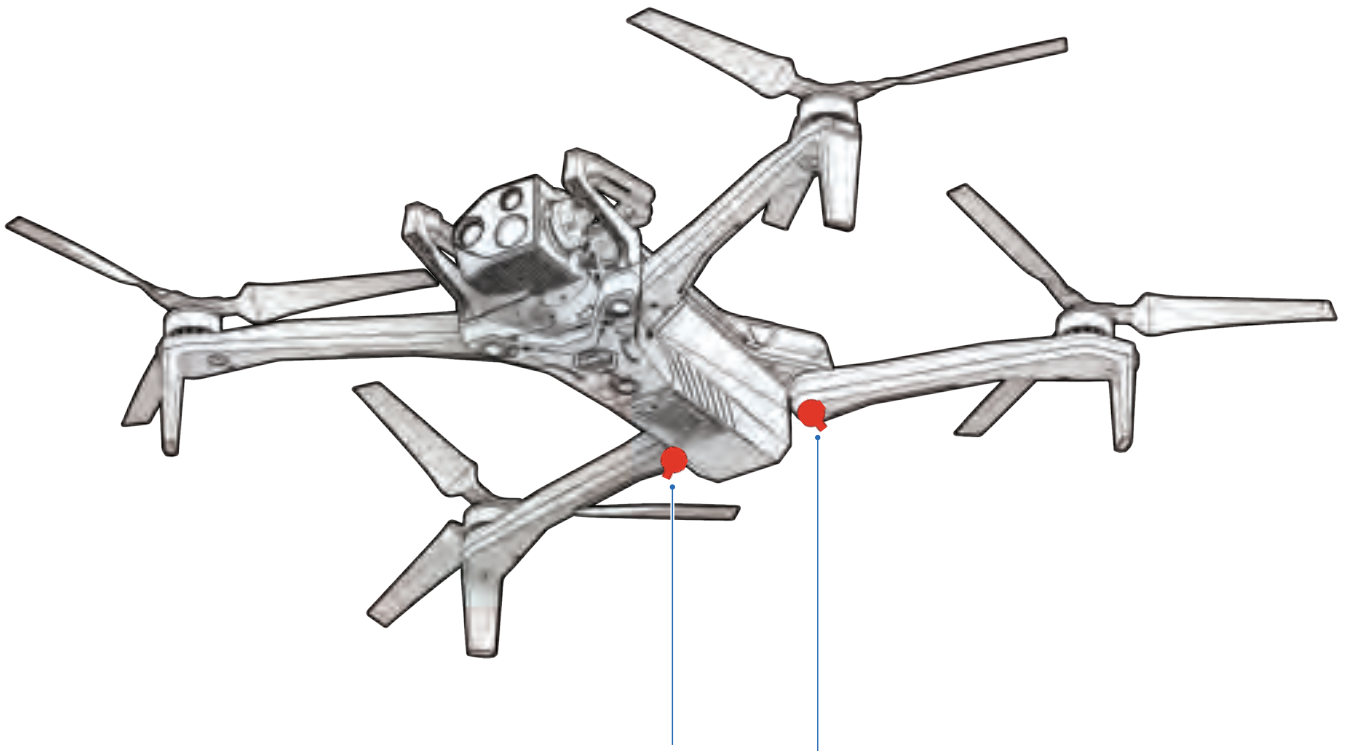
Step 2 - Unfold the front arms

Push **down** and **forward**. Gently continue until you feel the arm seat into place.



Step 3 - Remove the red packaging stickers

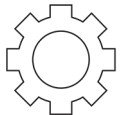
There are red stickers located on the underside of each rear arm. Use the tab to gently lift and pull them off before flying.



Step 4 - Verify and format microSD cards (pre-installed)

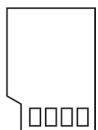
Ensure you have two UHS Speed Class 3 (or faster) microSD cards inserted in the slots on the side of the drone.

- Minimum 256 GB
- Format your cards before flying using **Manage Data** within the **Information** menu (**Global Settings** > **Information** > **Devices** > **Manage Data**)



Logs card

Supports software updates, scan data and records flight logs



Media card

Stores media captured during flight

Step 5 - Insert battery

Align the battery with the rails and slide toward the sensor package until the magnets engage.

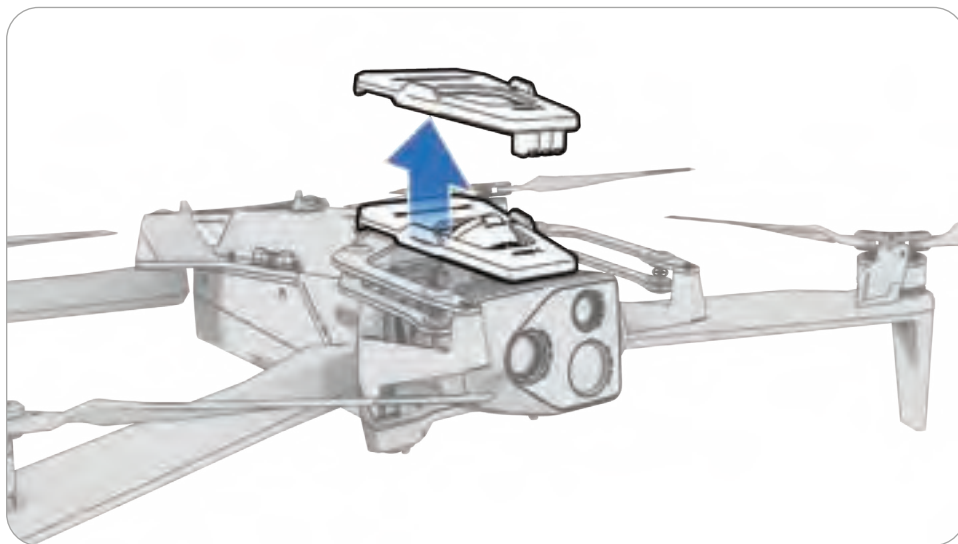
- Ensure the battery and rails are free of debris and interference
- Ensure the battery is completely seated before flying



Step 6 - Remove the sensor package lock

Hold the sensor package and gently pull to remove from the top of your drone.

- Save this piece to reattach when storing or transporting



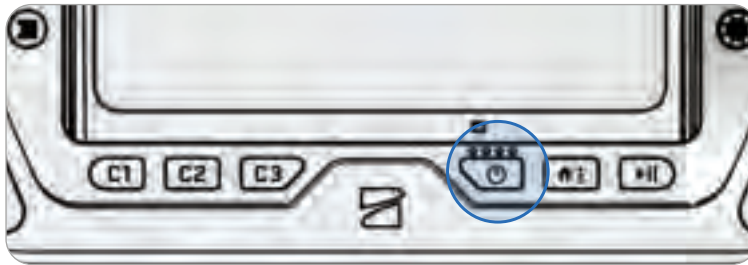
Skydio X10 Controller Setup



Scan for more information about Skydio Connect Fusion/SL/5G.

Step 1 - Power on the Skydio X10 Controller

Open the controller lid and hold the Power button for five seconds. The lights on the front of the controller will turn on and indicate the level of charge.



NOTE: While powered off, you can check the level of charge by pressing the Power button once.

Step 2 - Set up Skydio Flight Deck

Skydio Flight Deck is the dedicated flight software on your controller. Follow the on-screen prompts to begin setup.

- Connect to a WiFi network
- Provide the email address associated with your Skydio Cloud account and enter the activation code sent to your email
- If your Organization Admin has [configured Single Sign-On \(SSO\)](#) via Skydio Cloud, you have the ability to log in using your organization's Identity Provider (IdP)
- Enable **Shared Controller** to let multiple accounts sign in to the same controller (optional; an Org Admin can do this at any time)
- Set a password for your controller (optional)



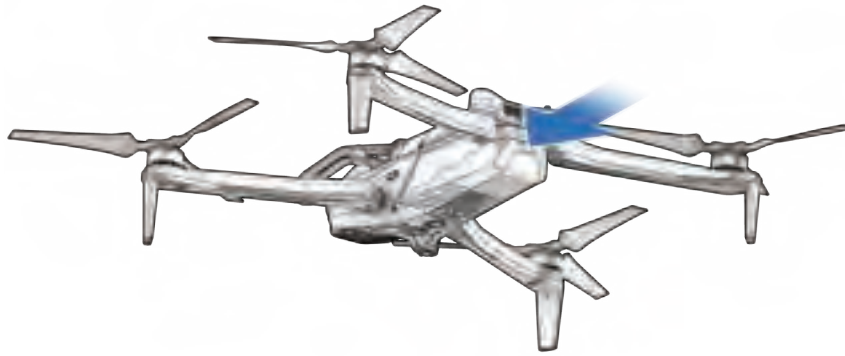
CAUTION: To maintain operational security and prevent conflicts with controller functionality, never sign in to personal or third-party accounts (such as Google Drive, Gmail, OneDrive, or other cloud services) on the X10 Controller. If you need to import maps or export logs, use the supported workflows provided by Skydio.



CAUTION: The password cannot be recovered or reset. Ensure that your password is entered correctly and is written down and stored in a safe location. If the password is lost, the controller will need to be replaced.

Step 3 - Power on Skydio X10

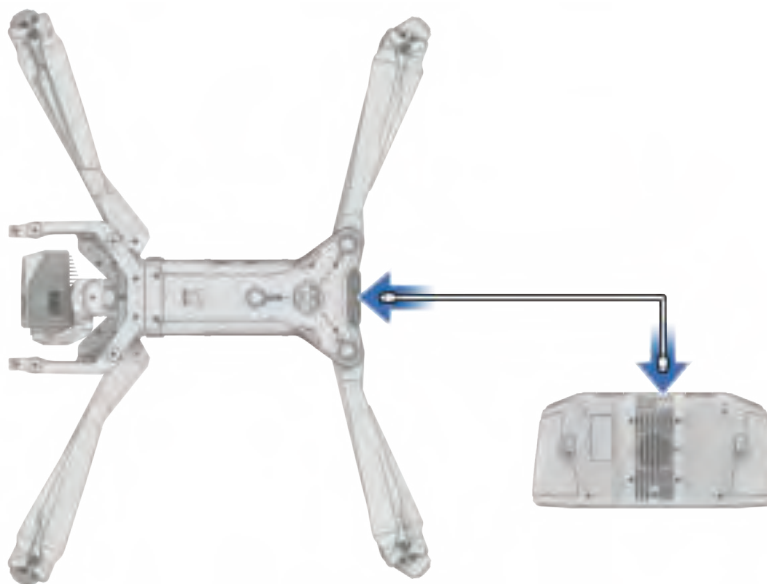
Press and hold the Power button on the battery for three seconds. The lights on the drone arms will turn blue as X10 powers on.



Step 4 - Pair the drone and controller

Use the USB-C pairing cable to connect your devices. Wait as pairing completes. The lights on the drone will turn solid blue and the name of your drone will appear on the screen when pairing is successful.

Once paired, the drone and controller will automatically connect before future flights.



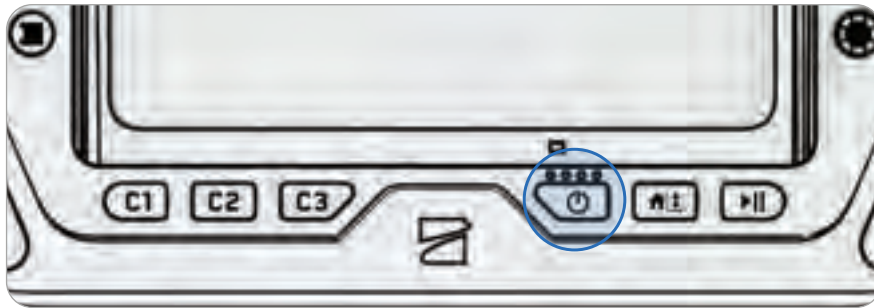
Updating the Skydio X10 Controller



NOTE: Check for available updates before flying. You must update the Skydio X10 Controller first before updating Skydio X10.

Step 1 - Power on the Skydio X10 Controller

Open the controller lid and hold the Power button for five seconds. The lights on the front of the controller will turn on and indicate the level of charge.

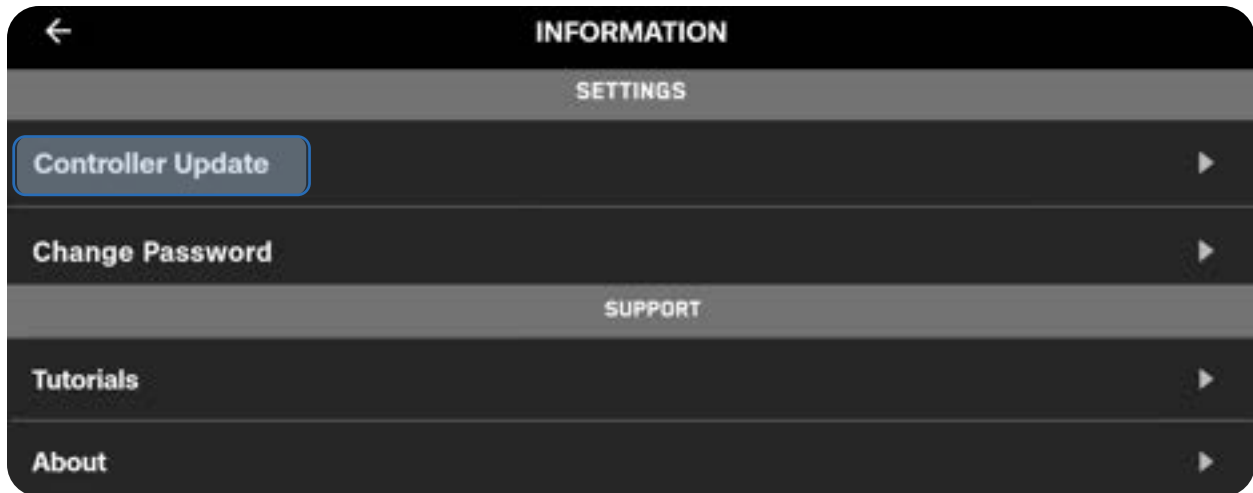


Step 2 - Navigate to the Information menu

Located within Global Settings.



Step 3 - Select Controller Update under Settings



Step 4 - Select Check for Update

Step 5 - Select Update

Follow the on-screen prompts to update your controller.



Updating Skydio X10

Skydio will not force an update for your system, however, for optimal performance, we recommend that you keep your Skydio system up-to-date. If an update is available, you will see a red notification icon in the **Information** menu.

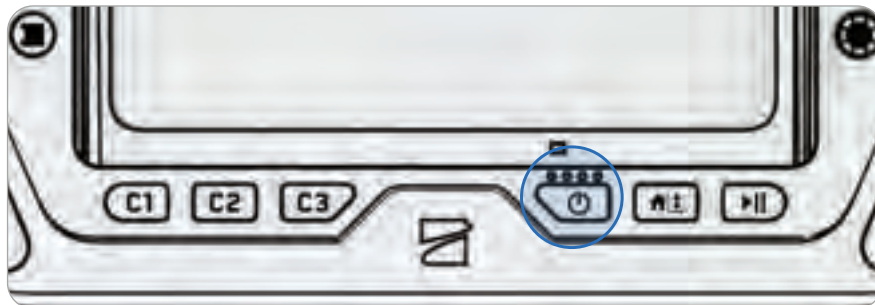
Step 1 - Power on Skydio X10

Press and hold the Power button on the battery for three seconds.



Step 2 - Power on the Skydio X10 Controller

Open the controller lid and hold the Power button for five seconds. The lights on the front of the controller will turn on and indicate the level of charge.



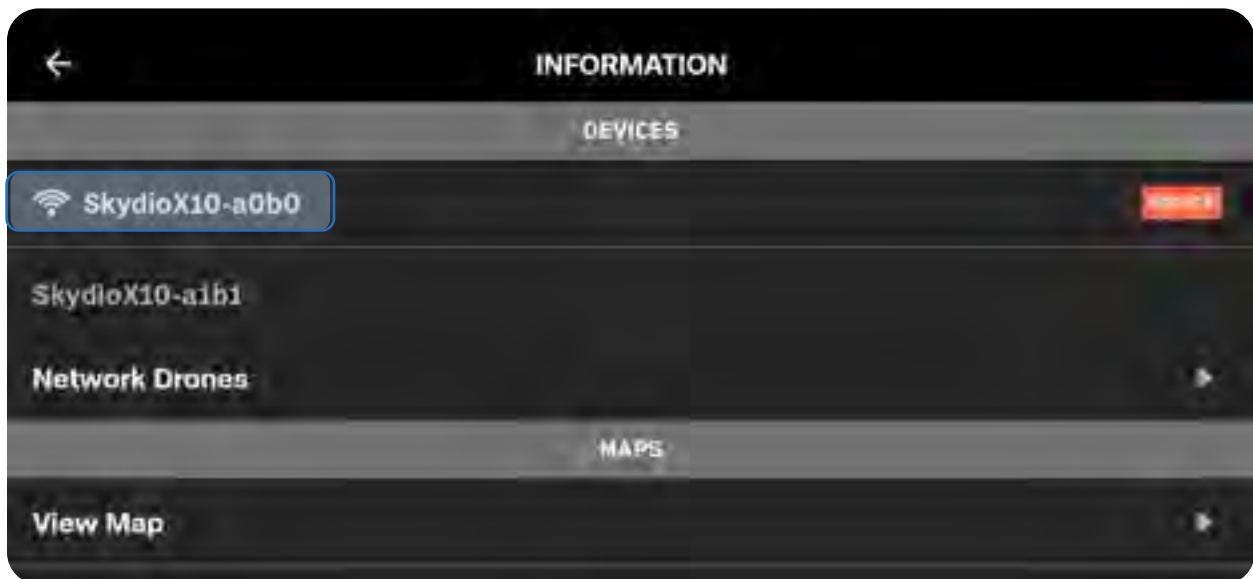
Preflight

Step 3 - Navigate to the Information menu

Located within Global Settings.



Step 4 - Select your Skydio X10 under Devices

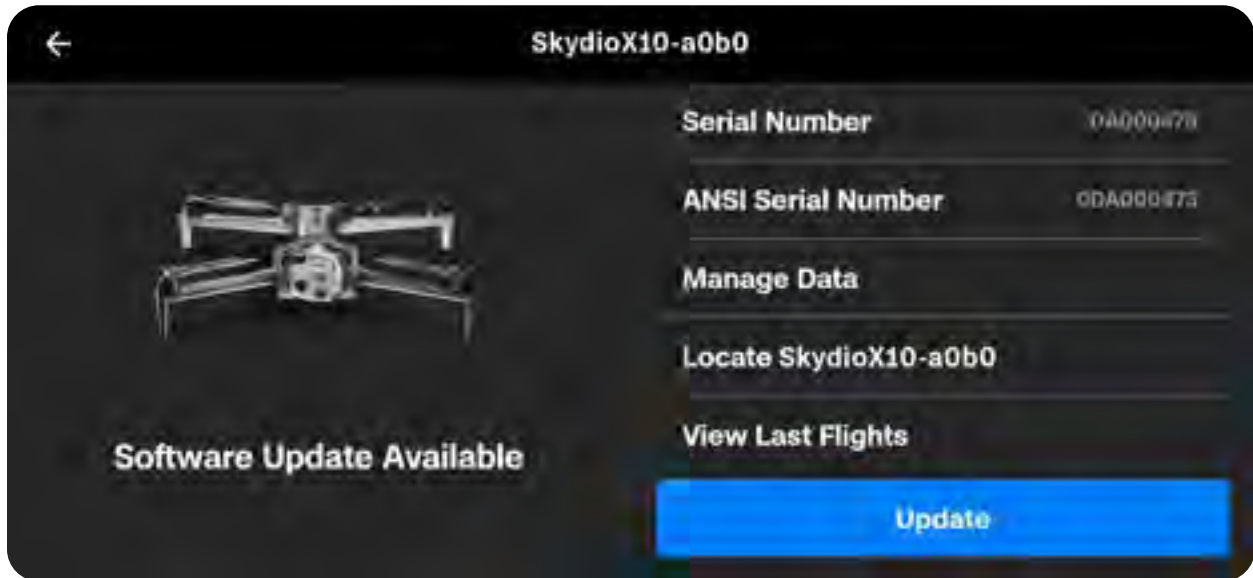


Preflight

Step 5 - Select Update

Follow the on-screen prompts to update your drone.

Select **Check for Updates** anytime to look for available updates.



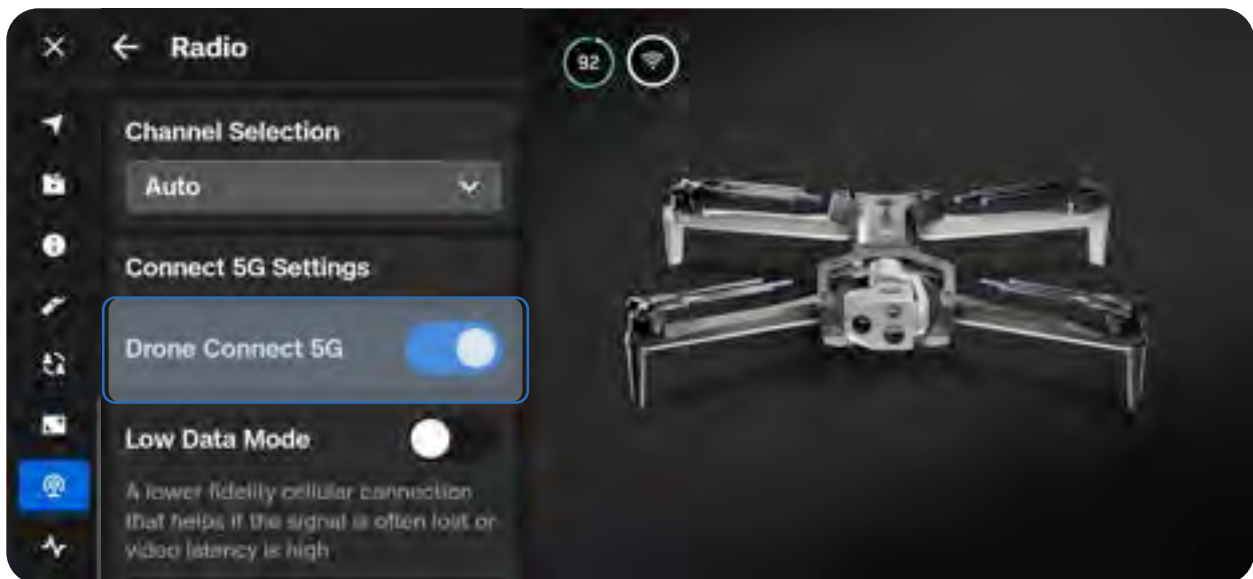
Skydio Connect 5G Setup



NOTE: Flight over 5G Cellular is only available if you've purchased an X10 system with Skydio Connect 5G and subscription to Skydio Remote Flight Deck with Skydio Connect 5G.

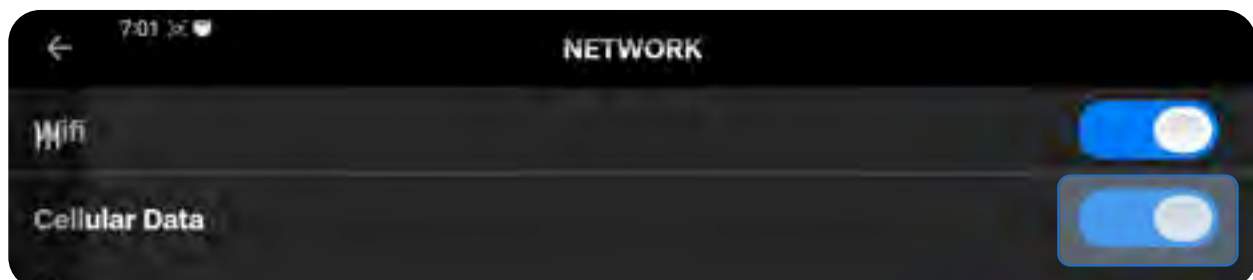
Step 1 - Enable 5G Cellular for your drone

On your X10 Controller, select the **Global Settings** menu and the **Radio** tab. Toggle on **Drone Connect 5G**.



Step 2 - Enable 5G Cellular for your controller

Select the **Global Settings** menu and the **Information** tab. Scroll down to **Device Settings**, select **Network**, then ensure **Cellular Data** is toggled ON.



Battery Warming in Cold Environments

For extremely cold environments, the Skydio X10 battery features self-warming technology to enable flight within 5 minutes.

If the X10 battery is below 32°F (0°C), Skydio X10 will be prevented from launching. You will need to pre-warm your batteries before you can launch.

- Battery self-warming is supported down to -4°F (-20°C).
- Skydio X10 will allow launching with reduced performance while continuing to operate the self-warmer until full performance is restored.
- A battery must have at least 30% charge to self-warm.
- A fully charged battery is strongly recommended when flying in cold weather.
- Flight time will be reduced in cold temperatures.
- Minimize aggressive maneuvering in extremely cold environments.



Pre-warming Skydio X10 Batteries

If Skydio X10 detects that a battery is too cold to launch, the battery will automatically begin self-warming. Launch will become unblocked when the battery is sufficiently warmed for the given state of charge.

You also have the option to prewarm a battery before inserting into the drone:

Step 1 - Triple tap the battery button to begin the self-warming process

- The lights on the battery will turn orange during the self-warming process
- Triple tapping again will end the self-warming process

Step 2 - If the battery is cold enough it will begin self-warming

- When self-warming is complete, the lights on the battery will briefly turn blue and then power off

Step 3 - Insert the battery into the drone and power on

- If the self-warming was still in progress prior to inserting, it will continue while the battery is in the drone
- Launch will become unblocked when the battery is sufficiently warmed for the given state of charge



Navigating Skydio Flight Deck

Skydio Flight Deck is the dedicated flight software on your controller. In this section you will learn about core menu locations and setting customizations.

This section covers

Gate Screen

Global Settings

Flight Controls

Return Behaviors

Sensing (Obstacle Avoidance)

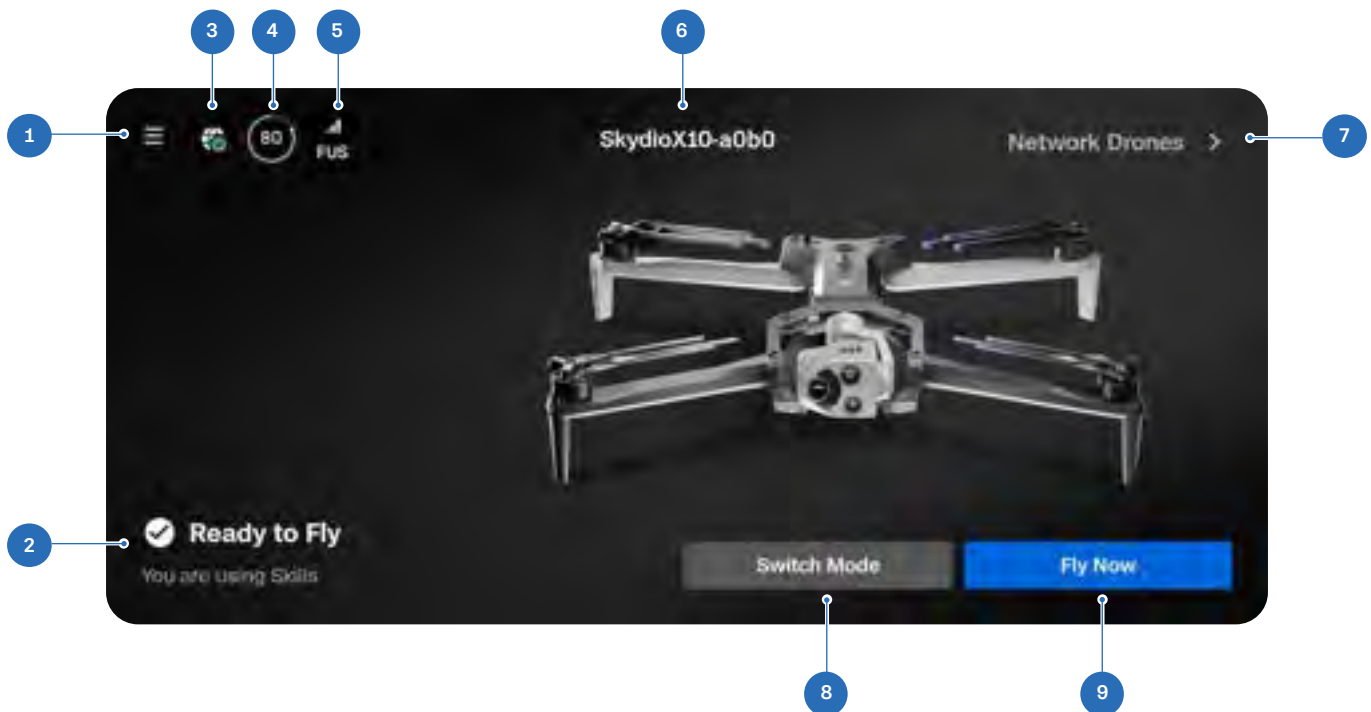
Display

System Status

Quick Actions

Gate Screen

After powering on and connecting to your drone, the first screen you will see is the **Gate Screen**. This screen is the first step to starting your flight, switching modes, or configuring preflight settings.



1. Global Settings
2. Flight Status
3. X10 Controller Network Status
4. Drone Battery
5. Signal Strength (select to use Connect Fusion, SL, or 5G)
6. Drone Name
7. Network Drones (displays active drones in your organization)
8. Flight Mode Selector (Skills or 3D Scan)
9. Fly Now (opens Flight Screen)



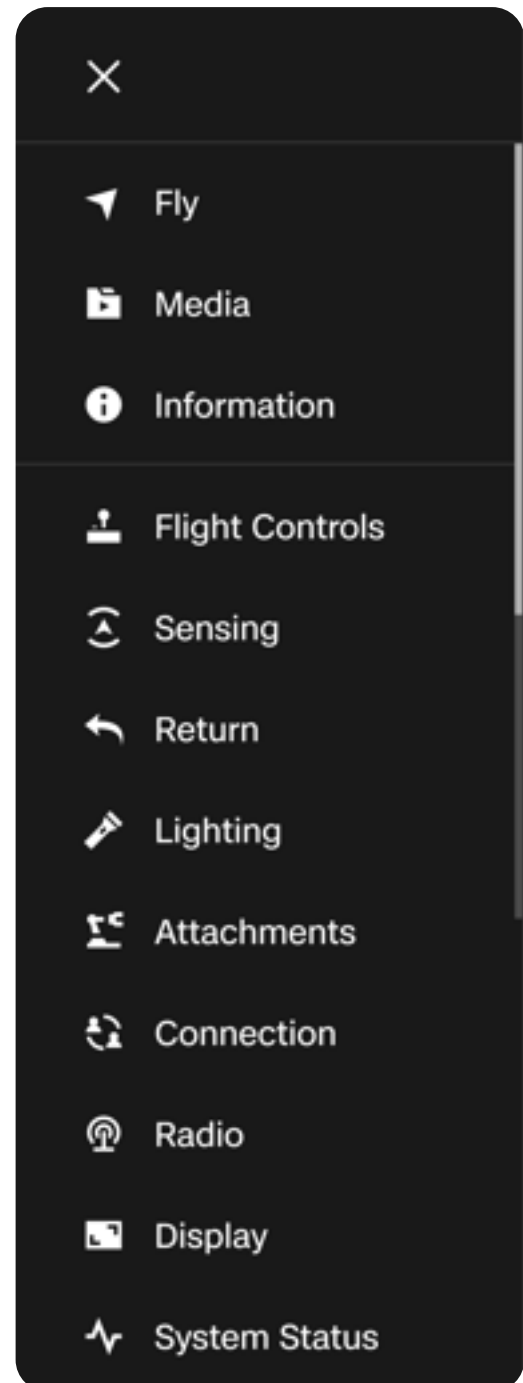
INFO: For more information about flying over cellular, visit [How to fly Skydio X10 over cellular connectivity](#).

Global Settings

The Global Settings menu is accessible before and during flight. You will use this menu to navigate to a variety of settings, such as low battery behaviors, stream layouts, and obstacle avoidance behavior.

Select the Global Settings icon in the top left of the screen to access the following menus:

- Media
- Information
- Flight Controls
- Sensing
- Return
- Lighting
- Attachments*
- Connection
- Radio
- Display
- System Status



**Only appears if you are using an attachment*

Media

Use this menu to view photos, videos, and scans from your recent flights.

- Select an image or video to view
- Press and hold on a thumbnail to select multiple or delete

If you capture photos using **Interval**, all photos captured will appear as a single stack. Selecting the stack will allow you to scroll through individual images one by one.

Only standard color and thermal JPGs will display in the Media menu. To access your DNG or RJPNG files, you must transfer the files from your drone.



NOTE: Media is not accessible inflight.

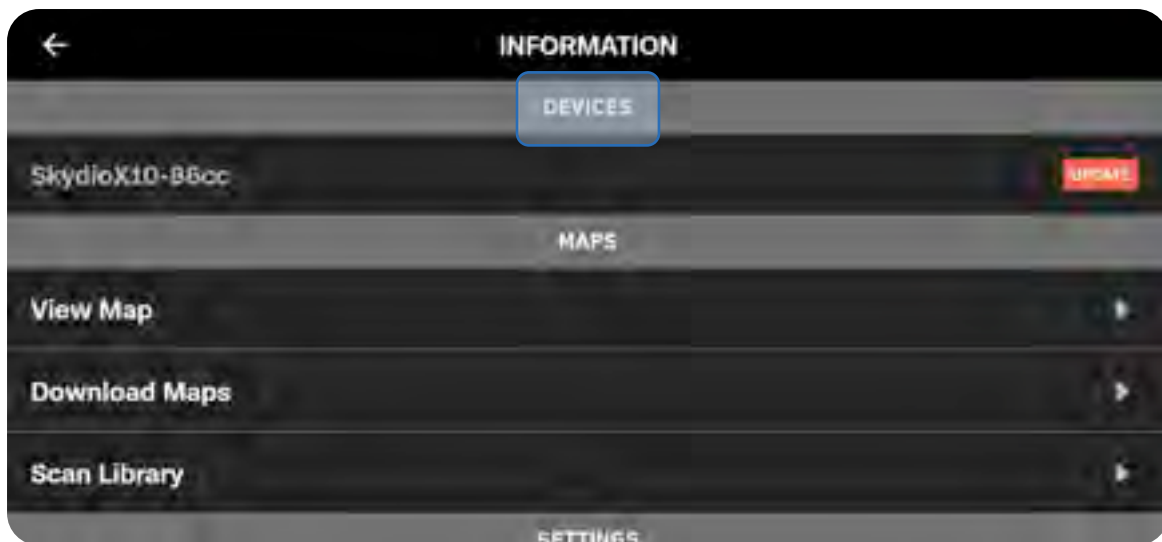
Information

While you are connected to Skydio X10, the Information menu provides access to settings such as drone and controller updates, radio frequency selections, the map library, and more.

Devices

Lists the name of the drone that is currently connected, as well as other X10 drones that have previously paired to the controller.

Check for drone updates and verify software versions by selecting the name of a drone.



Navigating Skydio Flight Deck

Manage Data

Select to format the Log and Media cards or Factory Reset your drone.

Locate Skydio X10

In the event that your Skydio X10 is lost, you may view its last known location. If the Coordinate setting is enabled, the latitude and longitude of the current or last known location will be displayed.

View Last Flights

Displays the feed that was last viewed from the Flight Screen, even if the drone is not connected. Designed to assist you with locating your drone in the event of a crash, emergency landing, or low battery landing in an unintended location.

Cloud Settings

Provides visibility into the licenses you have assigned to your drone, features, and the networks you have added in Skydio Cloud.

Upload Files

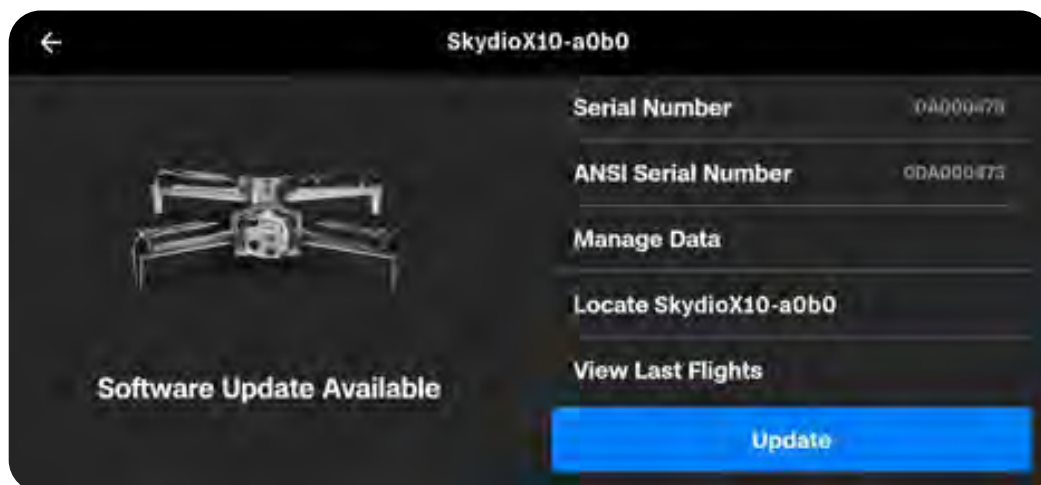
Monitor the progress of file uploads from the X10 Controller to the Cloud, including flight telemetry.

Overwrite Media

Manage your media storage by automatically deleting old media to ensure you always have enough storage space to start a new flight. Select Delete Oldest Media to automatically delete the oldest media stored on the microSD memory card.

Anti-flicker

Adjust anti-flicker settings if you experience flickering in your video. This setting is for users located outside of North America, in countries where the frequency of the alternating current in household electrical outlets is 50 Hz.



Maps

View Map

View your current location, search, and configure map settings.

The location of Skydio X10 (if connected), the controller, Launch Point, and Home Point (if set) are all indicated on the map. You will also be able to view alerts for nearby crewed aircraft via ADS-B alerts.



Skydio X10



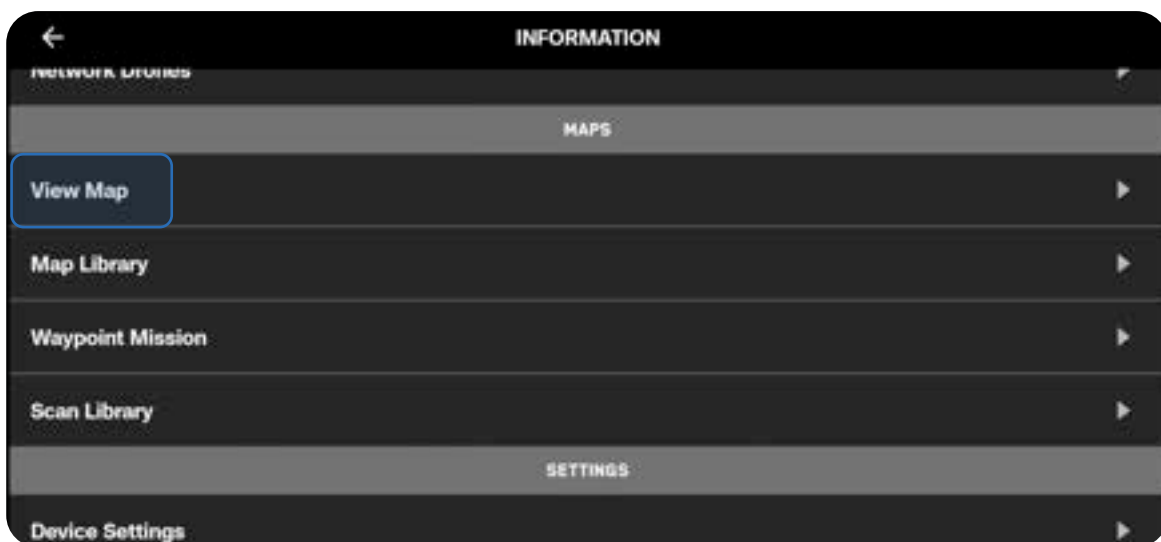
Controller



Launch Point



Home Point



Navigating Skydio Flight Deck

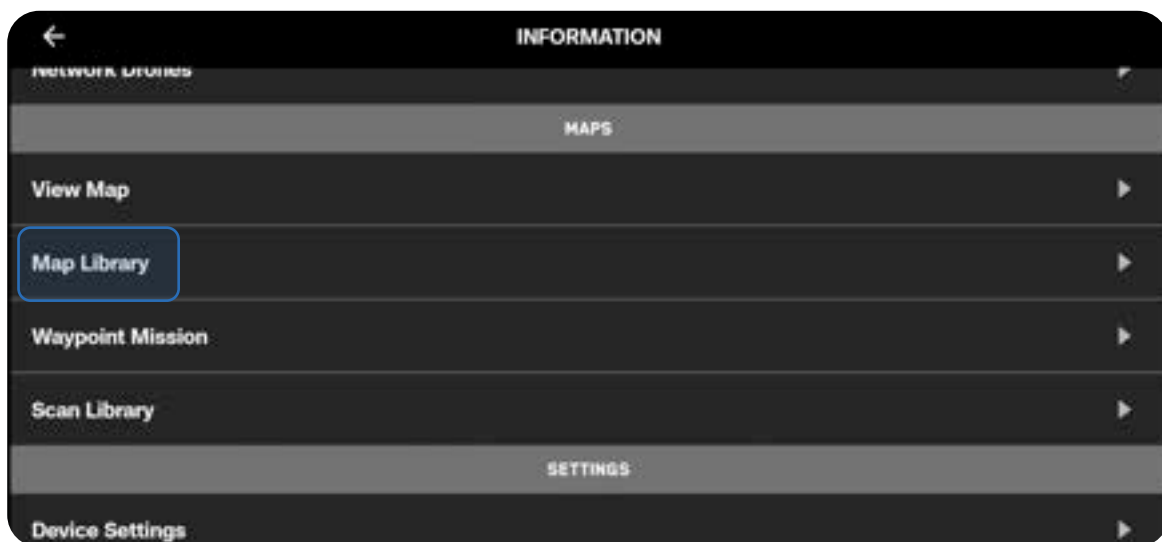
Map Library

In the Map Library, you will see two sections, **Base Maps** and **Map Layers**:

- **Base Maps** - Displays imported or downloaded maps (e.g., for use in GPS-denied environments)
- **Map Layers (Early Access Program)** - Displays imported map files that will be used as custom layers

Importing a **custom map layer** enables you to display critical information, such as mission boundaries, wildfire zones, or asset locations, as visual overlays on the Skydio X10 Controller. Supported map formats include:

- KML/KMZ
- GeoPDF
- GeoJSON
- MBTiles
- GeoTIFF

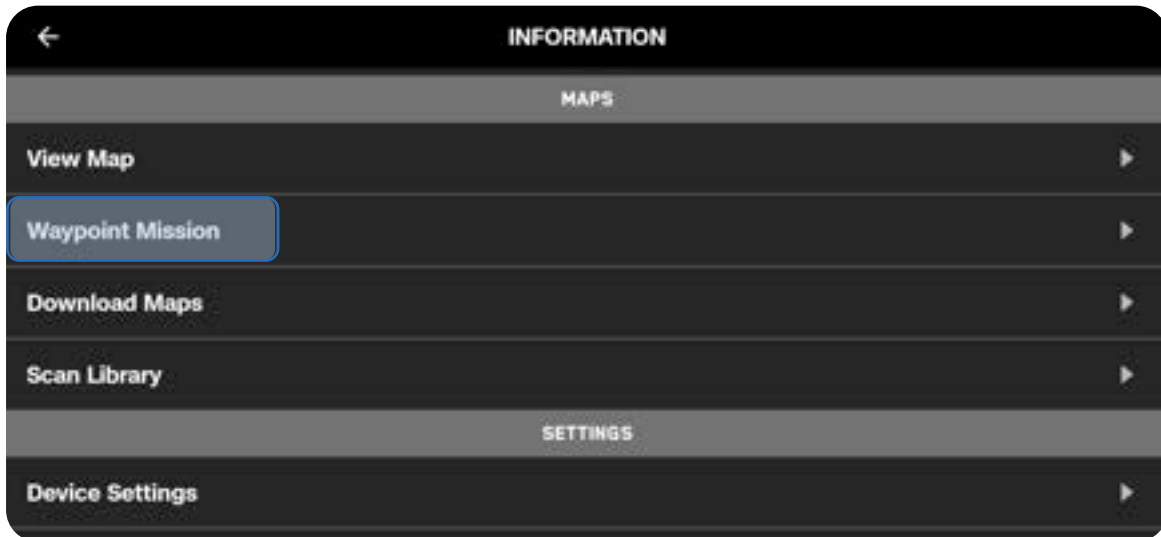


Navigating Skydio Flight Deck

Waypoint Mission

Stores the most recent Waypoint Mission. Press and hold on a waypoint to delete it. Press and hold on the map to add a waypoint or delete all waypoints.

Only one Waypoint Mission will be saved at a time.



Navigating Skydio Flight Deck

Download Maps

- Select the blue + icon under Add New Maps and a satellite view of your current location will display.
- Drag and pinch-to-zoom on the map until your desired location is centered on the screen
- Use the search bar to enter coordinates or type a location
- Select Download Map to save

The map will be 3.5 x 3.5 square miles, centered around the target point even if you are zoomed in. Your map will appear in this menu once the download is complete.



NOTE: The map library is able to store 10 maps at a time. Delete unused maps to make room for new maps.

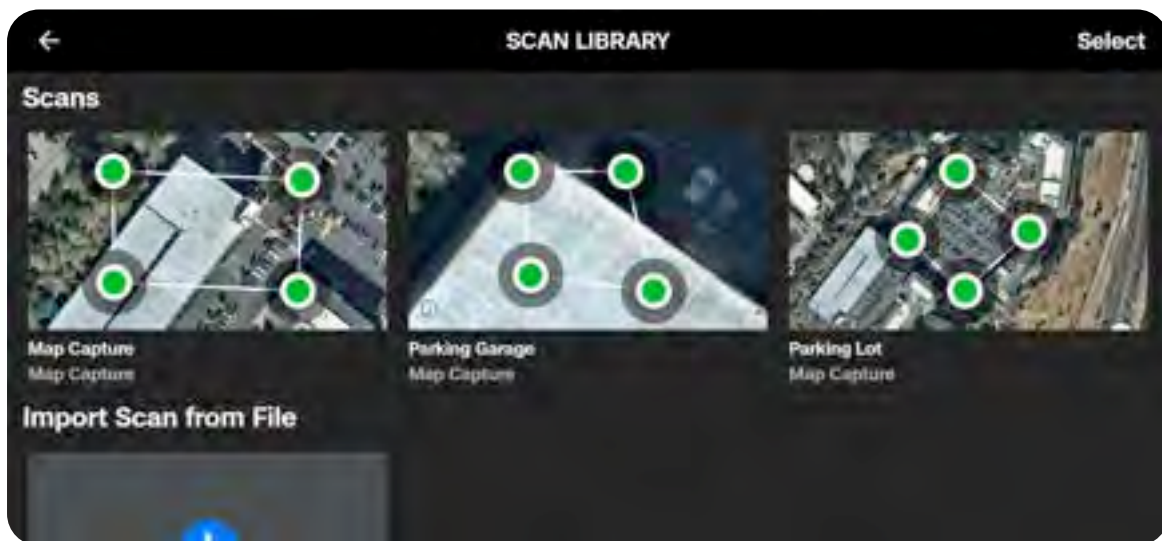
Navigating Skydio Flight Deck

Scan Library

View or repeat saved Map Capture scans. You also have the ability to import a previous Map Capture.



INFO: For more information, visit [How to use Map Capture](#).



NOTE: Only .mission files created from Skydio Map Capture are supported when importing.

Settings

Controller Update

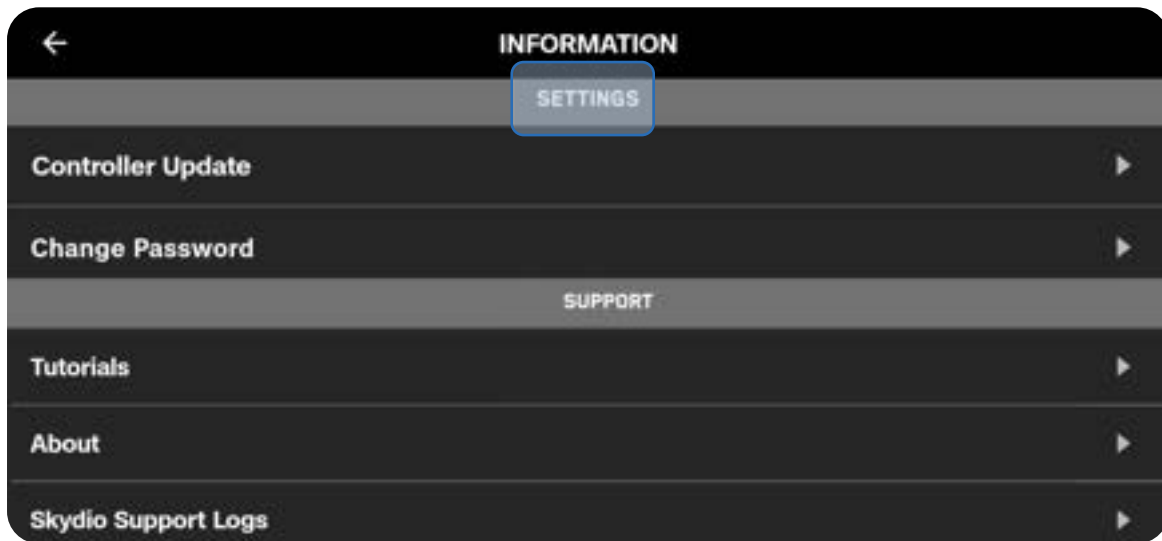
Use this menu to view the current software version of your controller and to check for or initiate updates.

Create/Change Password

Optionally add a password for your controller.



CAUTION: *The password cannot be recovered or reset. Ensure that your password is entered correctly and is written down and stored in a safe location. If the password is lost, the controller will need to be replaced.*



Support

Tutorials

Includes step-by-step instructions for actions such as calibration.

Hand Wave Calibration Guide

For use in environments with magnetic interference such as cars, metal bars, power lines, etc. You will need to calibrate before flying at night without NightSense.

About

View the current software version of the X10 Controller, the email associated with your account, and your organization.

Skydio Support Logs

To assist the support team and better troubleshoot any issues or questions you may have, we may require you to upload logs or other data from your drone to help us determine the root cause of any issues. Refer to the Sending Support Logs section for step-by-step instructions on uploading logs.

If you have any objection to this, please let the support team know. We will never review your videos or data without your permission. Do not reformat or factory reset your Skydio drone prior to contacting our support team.

Single Flight Log

Includes all logs from a specified flight. This option will show you the history of all flights, organized by date and time. Select which individual flight you wish to upload.

All Logs

Exports all logs saved on the controller from all flight history. This option allows you to sync logs whether you are connected to the drone or not.

Legal

View legal documentation such as the Skydio Safety and Operating Guide.

Flight Controls

Use this menu to customize your joystick controls, input mapping, and altitude limits.



Moving the joysticks allows you to adjust the roll, pitch, yaw, and throttle of the drone.

Roll - Controls left and right movement

Pitch - Controls forward and backward movement

Yaw - Changes rotation around the vertical axis

Throttle - Controls altitude

Sensitivity

Gimbal Pitch

Controls how quickly the camera sensor package moves up and down.

- Default - 18%

Flight

Allows you to customize the maximum allowed speed for roll, pitch, yaw, and throttle.

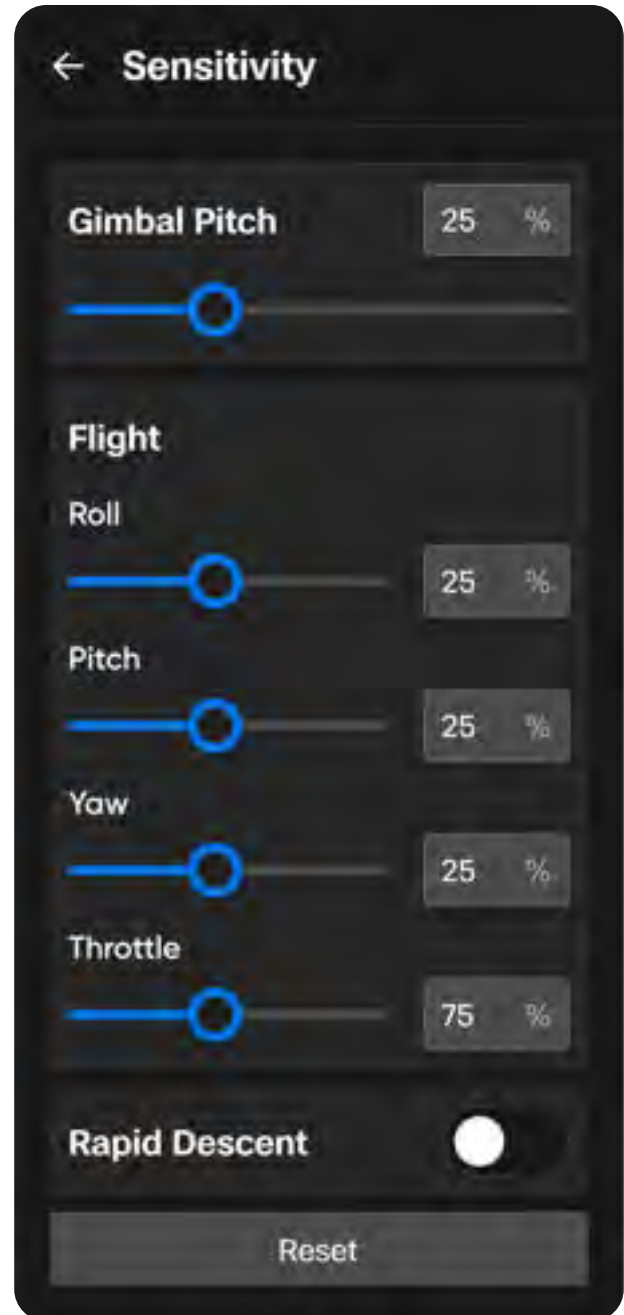
Default sensitivity:

- Roll - 35%
- Pitch - 35%
- Yaw - 45%
- Throttle - 100%

Rapid Descent

Allows you to increase the speed to the maximum descent rate when pressing and holding Boost (L1 button).

- Descent speed: 27 mph (12 m/s)



NOTE: Increase pitch sensitivity to increase the maximum speed of the drone.

Controls

Battery

Displays Skydio X10 Controller battery level.

Control Mode

Determines how your controller joysticks will maneuver X10. Select between Mode 1, 2 (default), and 3.

Input Mapping

Allows you to customize buttons and wheels with the following options:

- Obstacle Avoidance (Close, Minimal, Disabled)
- Cycle Display Layout
- Cycle Thermal Palettes
- Cycle Full Screen View
- Reset Sensor Package
- Toggle Strobe Lights
- Toggle RGB Lights
- Stop at Structure
- Exposure Compensation (+/-)
- No Function

Customizable buttons include: C1, C2, C3, L2, R2.

You can also invert your wheel directions, assign Exposure Compensation to your Right Wheel (instead of Zoom), or assign No Function to your wheels.



Navigating Skydio Flight Deck

Action Wheel

Assign functions to your controller D-pad and quickly preview and select functions in flight.

Once functions are assigned, hold the Back button on the controller to bring up the Action Wheel menu. While holding the Back button, press the D-pad up, left, or right to execute that function.

Camera Dragging

Drag your finger on the screen to pitch the sensor package and yaw the drone to look around.

Joystick/Roller Input

Enabled by default. This toggle functions as a safety precaution in the event that you experience a hardware issue with the joysticks.

When disabled, you will still have the ability to use touchscreen inputs and buttons on the controller.

Only disable this toggle if you experience errors with the joystick controls.



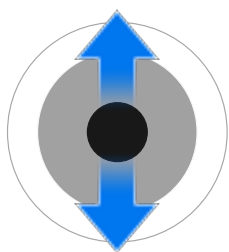
WARNING: Disabling the Joystick/Roller Input toggle turns off joystick functionality, which may result in serious bodily injury and/or damage. You may only disable this toggle if you experience errors with the joystick controls and proceed with extreme caution. Any injury and/or damage resulting from disabling this toggle is not covered under Skydio's Limited Warranty.

Control Mode

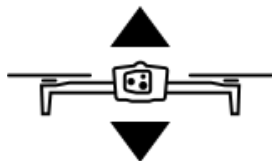
By default, flight controls are set to Mode 2.

In Mode 2, the left joystick controls the elevation and horizontal rotation of the drone, and the right joystick controls the forward, backward, and lateral movements of the drone.

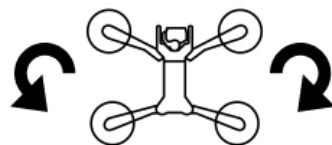
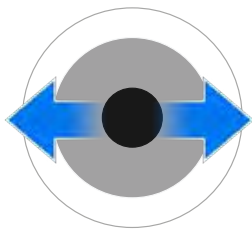
Left Stick



Up



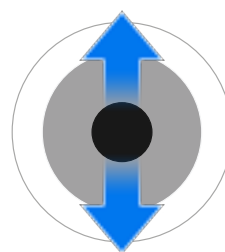
Down



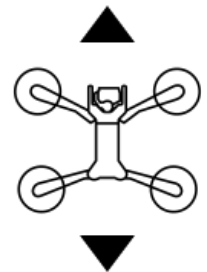
Rotate
left

Rotate
right

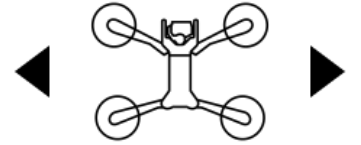
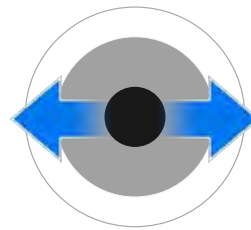
Right Stick



Forward



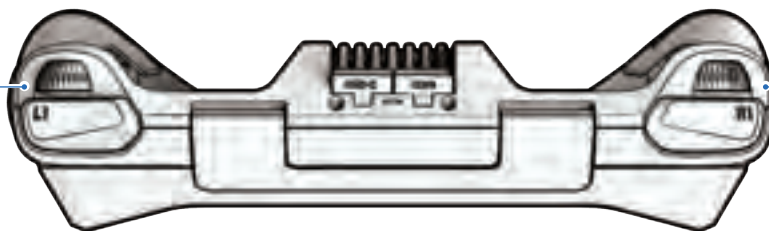
Backward



Left

Right

Camera gimbal
tilt



Zoom

Limits

Height Ceiling

When enabled, allows you to set the maximum allowed drone altitude above the Launch Point.

- Minimum: 30 ft (9 m)
- Maximum: 1500 ft (457 m)

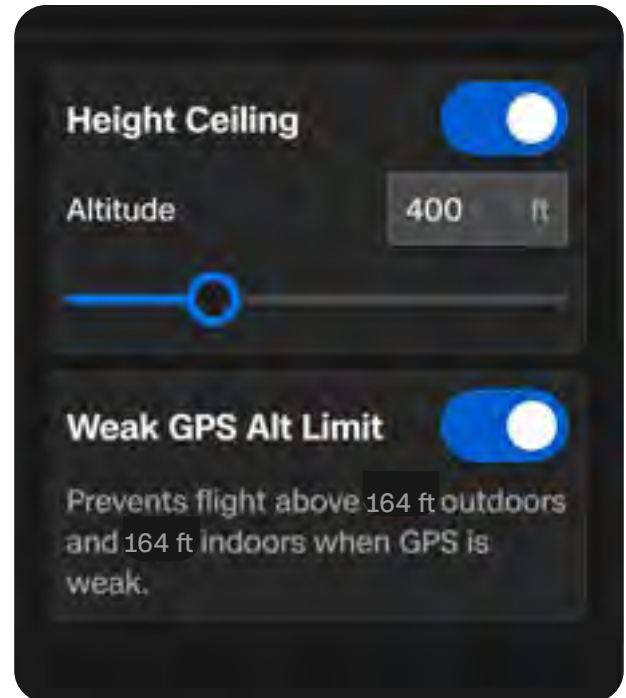
Height Ceiling settings persist across flights and power cycles.

Weak GPS Alt Limit

When enabled, prevents flight above 164 ft (50 m) when GPS is weak.

If GPS is lost, Skydio X10 maintains continuous navigation using Visual Navigation (VIO), which operates alongside GPS and can fully take over all navigation functions.

The drone can maintain stable VIO at altitudes up to 984 ft (300 m).



WARNING: Toggling Weak GPS Alt Limit OFF disables the altitude limit and your drone will fly using only visual navigation. To reduce the risk of an emergency landing, maintain a flight path near surfaces and objects.



NOTE: If you are using the NightSense attachment, the Weak GPS Alt Limit is 16 ft (5 m).

Sensing

Use this menu to adjust autonomous flight behaviors.



Navigating Skydio Flight Deck

Flight Environment

Select either Standard or Low Light to best represent your flight environment

Standard (default) - flight in normal daytime or in brightly lit conditions (i.e. indoors)

Low Light - flight at night or in low-light conditions with poor visibility.

Obstacle Avoidance

When flying near obstacles your drone will follow your selected distance setting. Choose between Standard, Close, and Minimal.

Standard (default) - Drone stays 24 in (60 cm) away from obstacles (15 in, 39 cm in narrow spaces)

- Top ground speed: ~36 mph (16 m/s)

Close - Drone stays 6 in (15 cm) away from obstacles (5 in, 13 cm in narrow spaces)

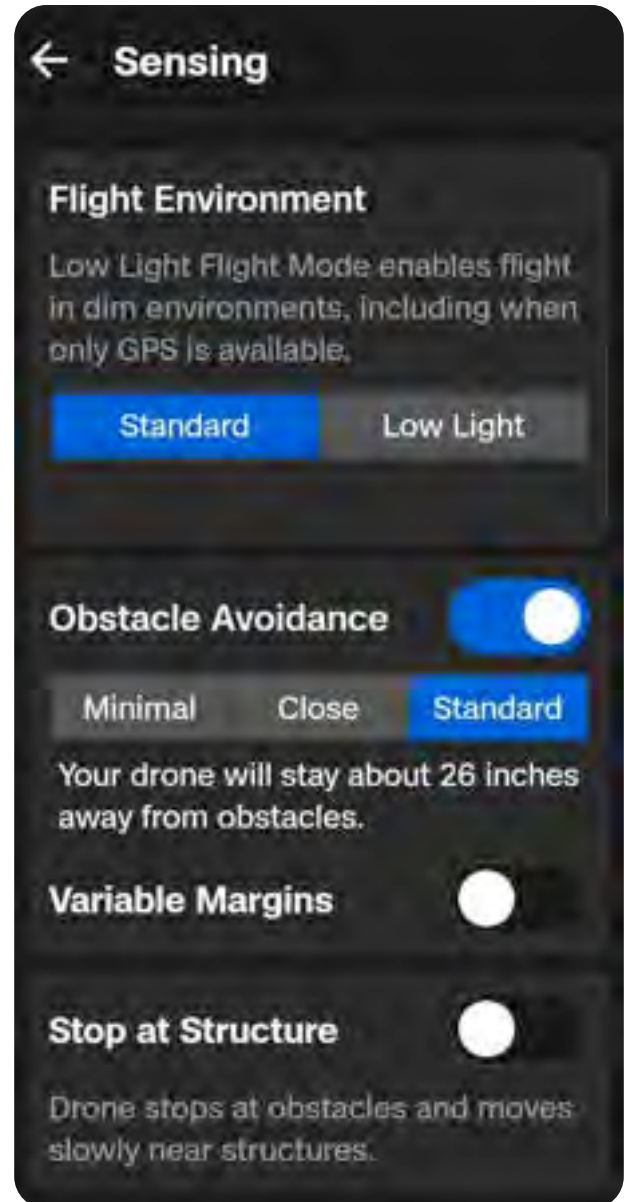
- Top ground speed: ~18 mph (8 m/s)

Minimal - Slight course corrections to avoid obstacles, but primarily relies on the pilot to avoid collisions.

- Top ground speed: ~18 mph (8 m/s)

Disabled (toggle off) - Skydio X10 will not avoid obstacles and there is a high risk of collision

- Top ground speed: ~45 mph (20 m/s)



Navigating Skydio Flight Deck

Variable Margins

Skydio X10 uses AI and visual navigation to dynamically, and temporarily, reduce obstacle avoidance margins when moving through narrow spaces. Margins will also dynamically expand if the drone detects environmental dangers, such as wind.

Enabled by default. Disable to turn off the dynamic margin behavior.

Stop at Structure

Perform finer, more controlled inspections on structures such as bridges or building facades.

When enabled, your drone will not deviate from its course when it is within 8 ft (2.5 m) of a structure.

The drone will reduce speed and maintain position, allowing for more precise maneuvering in the immediate vicinity of the structure.

- Adjust the maximum speed using the Speed Near Obstacles slider
- Maximum controller speed settings apply when no structure is present
- Stop at Structure is active during manual flight, including when paused during a 3D Scan



CAUTION: Flying with Close, Minimal or Disabled settings greatly increases the risk of collision. Minimal or Disabled settings are used to navigate tight spaces and should only be used if you are an experienced pilot. Skydio recommends turning down controller throttle, roll, and pitch sensitivity to the lowest setting and proceeding at a maximum speed of 2 mph (1 m/s).

Return

Configure your standard return behaviors as well as how Skydio X10 will return if connection is lost.



WARNING: Before flying, ensure you have set your Lost Connection Return Behaviors. This is a critical step that ensures your drone returns safely and lands in an accessible location.

Return

Height Behavior

Customize the altitude behavior of Skydio X10 when returning.

Use Return Height to set the altitude at which the drone will ascend to before returning.

Absolute means your drone will ascend to the specified Return Height above the Launch Point before returning

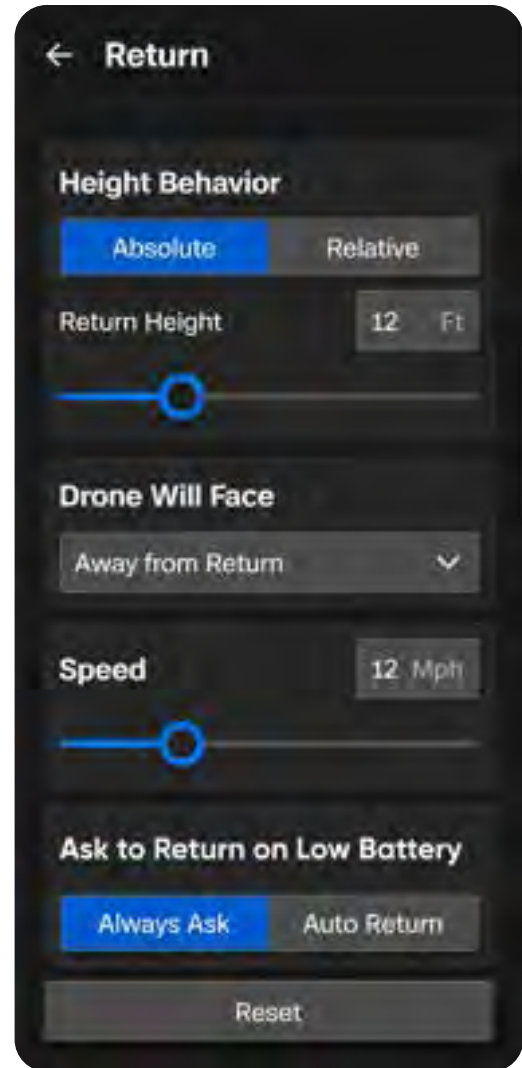
- For example, if the Return Height is 32 ft and the drone is at 20 ft at the time the return is commanded, Skydio will ascend 12 ft before returning

Relative means your drone will ascend to the specified Return Height above the current position before returning

- For example, if the Return Height is 32 ft and the drone is at 20 ft at the time the return is commanded, Skydio will ascend 32 ft and then return at a height of 52 ft

Drone Will Face

Set Skydio X10 to either look toward or away from the return destination while flying.



Speed

Set the speed at which Skydio X10 returns.

- Vision return: 1 - 35 mph (0.5 - 16 m/s)
- GPS return: 1 - 45 mph (0.5 - 20 m/s)

Ask to Return on Low Battery

When the battery is only sufficient to return and land, choose between a prompt or an automatic return.

- **Always Ask (default)** - you will be asked to select the return location each time the battery level is low
- **Auto Return** - the drone will automatically return to either the Launch Point or the Home Point (if set) when the battery is low (default in Remote Flight Deck)

Lost Connection

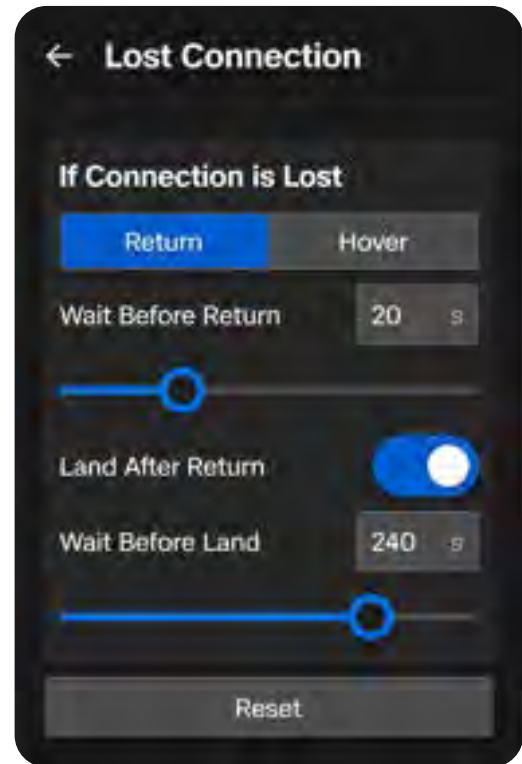
If connection is lost, Skydio X10 will default to the **Lost Connection** settings. Select between **Return** and **Hover** upon lost connection.

Return

Wait Before Return - set the amount of time you want Skydio X10 to wait before it initiates a return flight, allowing time to reconnect

Land After Return - when enabled, your drone will return, hover for a specified amount of time, then land.

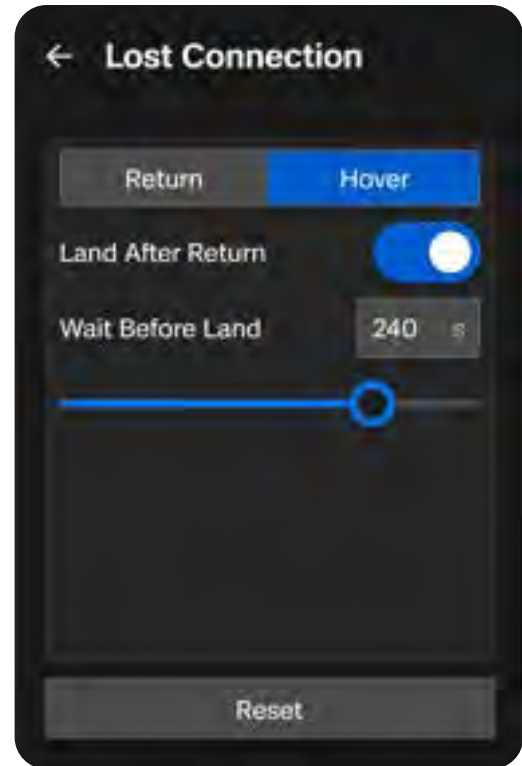
Wait Before Land - the amount of time between 0 to 300 seconds (default is 240 seconds) that you want your drone to wait above the landing location before landing. This setting is only enabled when Land After Return is toggled on.



Hover

Land After Hover - when enabled, Skydio X10 will hover for a specified amount of time, then use visual navigation to find a safe area to land.

Wait Before Land - the amount of time between 0 to 300 seconds (default is 240 seconds) that you want your drone to wait before landing. This setting is only enabled when Land After Hover is toggled on.

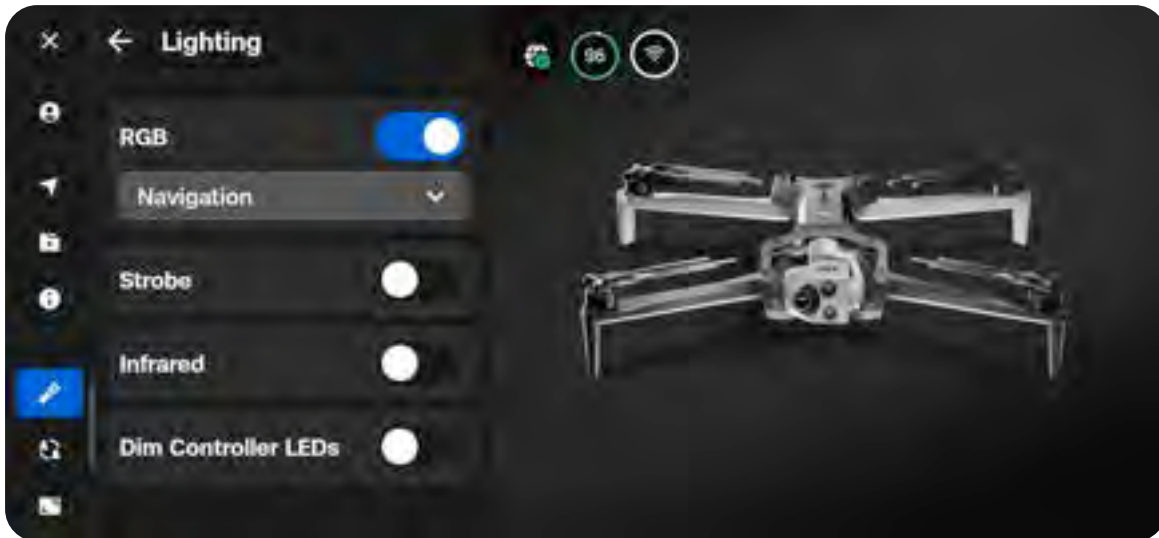


Skydio X10 will continue hovering as it tries to regain connection. If it fails to reconnect and reaches low battery:

- If you have an automatic return set, your drone will return to either the Launch Point or Home Point (if set)
- If you do not have an automatic return set, your drone will use visual navigation to find a safe area to land
- If VIO is degraded (flying in Low Light without NightSense), your drone will be unable to use visual navigation and will descend vertically and land

Lighting

Customize the inflight behavior of the RGB/strobe lights that are located at the end of the arms.



RGB

Choose between Navigation, Police, or Emergency lights.

- **Navigation (default)** - When enabled, the lights on the end of the arms will appear red and green while flying. When the drone is powered on and grounded, the lights will appear blue.
- **Police** - When enabled, the front and back RGB lights will flash red and blue and alternate colors between the right and left arms.
- **Emergency** - When enabled, the front and back RGB lights will flash red and white and alternate colors between the right and left arms.

Strobe

Enable to visually track the drone in low-light conditions. Skydio X10 strobe lights meet the FAA requirement of being visible at a distance of 3 statute miles (4.8 km).

Infrared

Cannot be seen by the naked eye. These broadcast an IR light that can only be detected with an infrared lens. Assists with navigation in low-light.

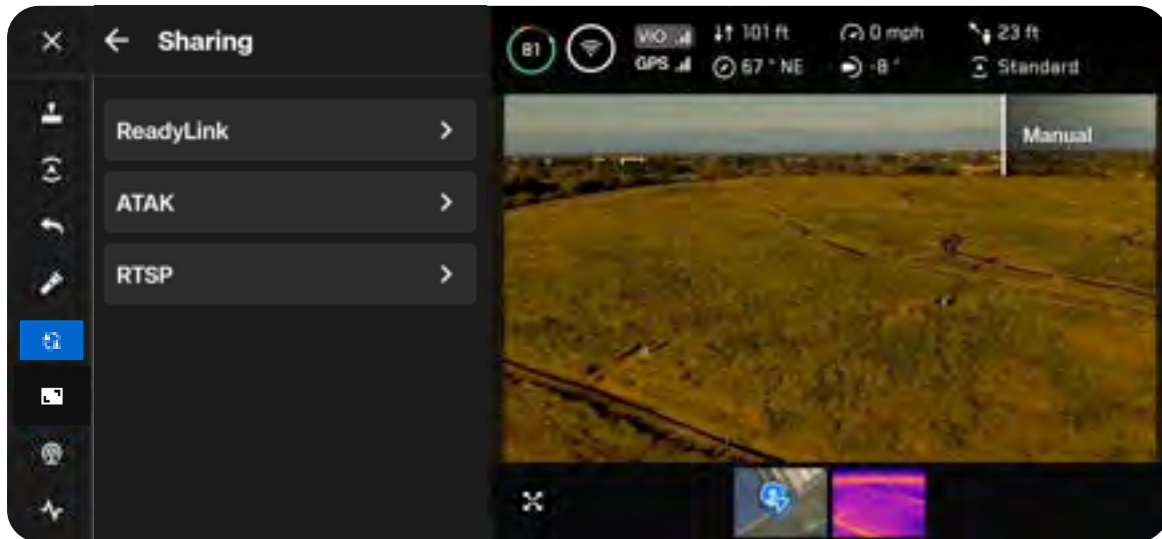


CAUTION: Police and Emergency lights option is only for strict use by authorized users comprising public safety first responders and emergency personnel performing a public safety mission.

Sharing



Use this menu to share a live stream of your flight or configure an ATAK or RTSP stream.



Sharing a Live Stream with ReadyLink

ReadyLink allows you to share a live stream of your flight with others via a QR Code.

- **Connect SL** - You will see the ReadyLink option appear in the Sharing menu **after launching**
- **Connect Fusion or Connect 5G** - A ReadyLink will automatically generate after launching

Use the **ReadyLink Quick Action** in the Flight Screen to access the QR Code at any time.



NOTE: Skydio Live Streaming is an optional software add-on available for purchase. You must have Live Streaming to use ReadyLink. When flying over Connect 5G or Fusion, Skydio X10 will live stream automatically.

Step 1 - Launch and open the ReadyLink menu

Navigate to **Global Settings > Sharing > ReadyLink**



Navigating Skydio Flight Deck

Step 2 - Select Start ReadyLink

Wait a moment as the QR Code is generated.

Select the QR Code to share your stream. While streaming, you will be able to see how many viewers are watching your flight.



Navigating Skydio Flight Deck

Use the **ReadyLink Quick Action** in the Flight Screen to access the QR Code at any time.

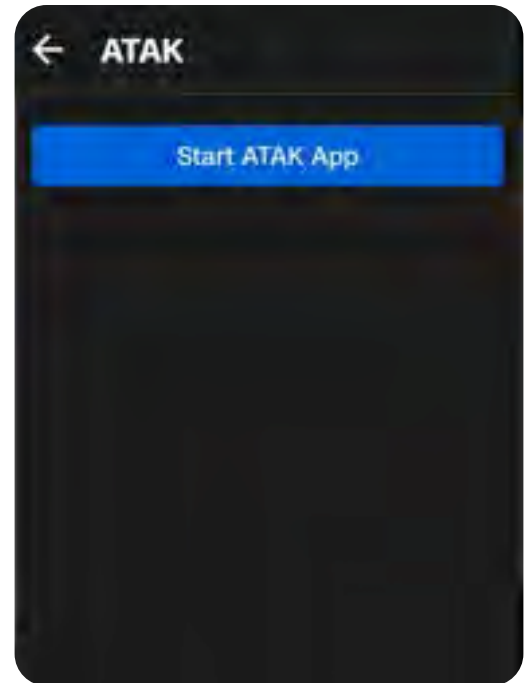


The controller will display ReadyLinks associated with your drone, including PIN codes when applicable. Active ReadyLinks appear based on your Skydio Cloud permissions, with the default ReadyLink shown first.



ATAK

The integration of the Skydio X10 Controller and the Android Team Awareness Kit (ATAK) app allows you to seamlessly monitor your Skydio fleet. ATAK provides geospatial information about your drones and controllers for increased situational awareness in the field.



INFO: For more information about setting up ATAK, visit [How to set up ATAK with the Skydio X10 Controller](#).

RTSP (Real-Time Streaming Protocol)

RTSP enables live video streaming from your Skydio drone directly to any compatible media player or system. This protocol allows for flexible, real-time monitoring of video feeds, whether you are using it for surveillance, situational awareness, or integrating with third-party systems.

DroneSense Integration: Missions and Call Signs

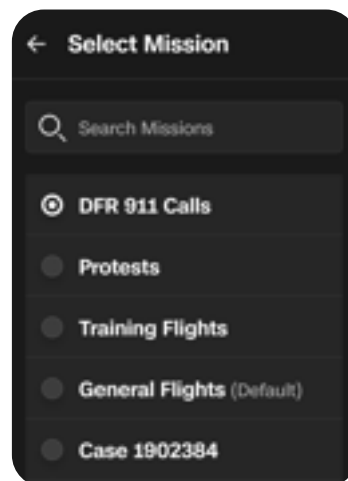
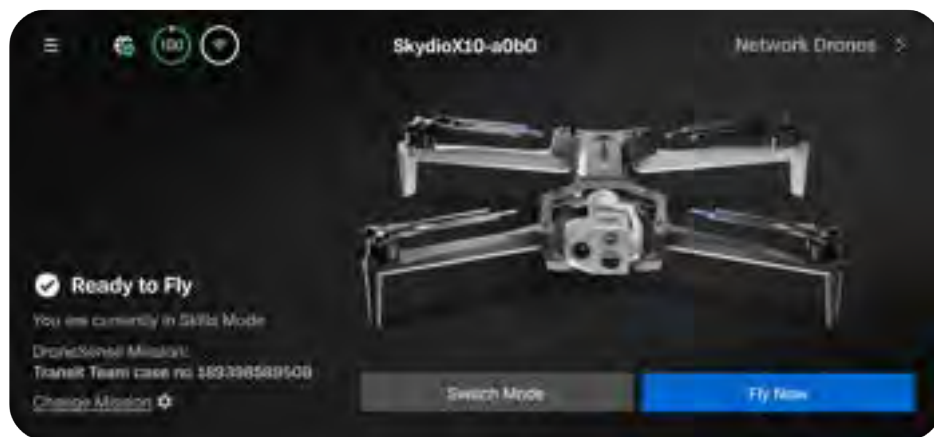
Skydio's integration with DroneSense allows pilots to link flights to active missions directly from the Skydio Controller. This streamlines mission tracking and removes the need for the DroneSense CoPilot app.

If your organization has DroneSense enabled, a DroneSense option will appear under **Global Settings > Sharing** on the controller. From there, pilots can select or update an active mission either before launch or mid-flight.

Assigning a mission ensures that each flight is properly categorized in DroneSense, improving mission documentation and real-time awareness across your team. This is especially useful for DFR operations, training sessions, or incident-based response, where mission coordination and call sign accuracy are critical.



NOTE: This integration must be enabled by an Organization Admin in Skydio Cloud before it will appear on the controller. For more information, visit the [Integrations section of our Support Site](#).



Display

Customize your Flight Screen display, including telemetry metrics, unit type, and depth style.

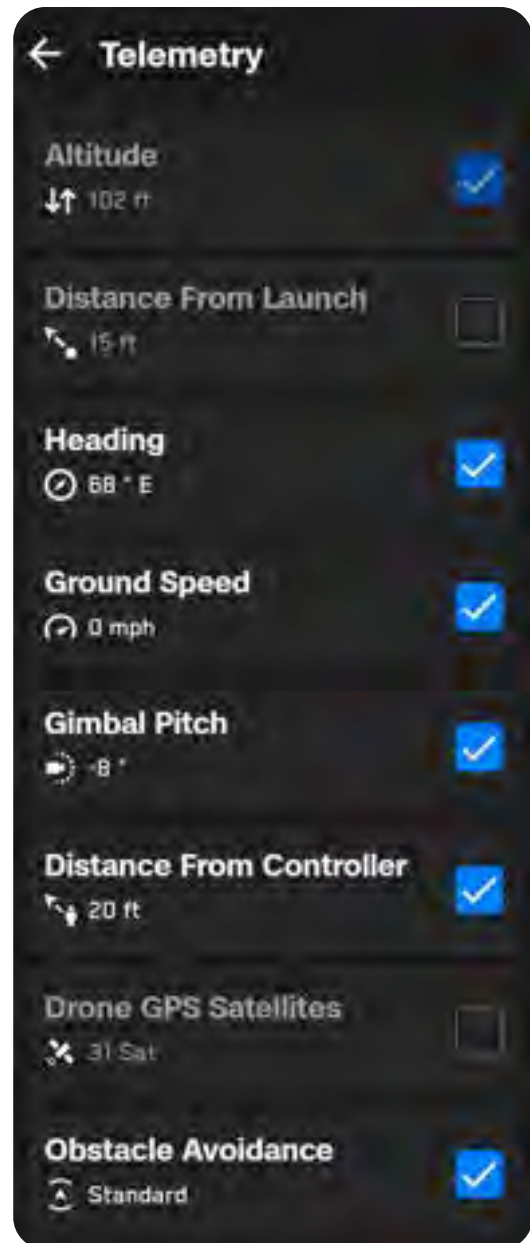


Navigating Skydio Flight Deck

Telemetry

Customize the telemetry metrics you want to display while flying. Select the blue checkmark to enable or disable the corresponding telemetry information. Up to 6 telemetry metrics may be displayed while flying.

Altitude is required.



Navigating Skydio Flight Deck

Air Traffic

Use this menu to configure ADS-B settings for the inflight map.

ADS-B Alert Range - Use the adjustable alerting range to customize the detection sensitivity, where **V** is the vertical distance from the drone before the alert is triggered, and **H** is the horizontal distance.

When an aircraft enters your alert range, its icon will turn red and you will hear an audible alert through the controller speakers.

The default range is set to **Small**.

- Small - H: 0.5 NM; V: +/- 500 ft (152 m)
- Medium - H: 1 NM; V: +/- 500 ft (152 m)
- Large - H: 1.5 NM; V: +/- 500 ft (152 m)

ADS-B Exclusion Range - Provides information about the distance at which aircraft are removed from the map.

Aircraft are excluded:

- Outside 6.0 NM
- Above 3,000 ft (914 m)



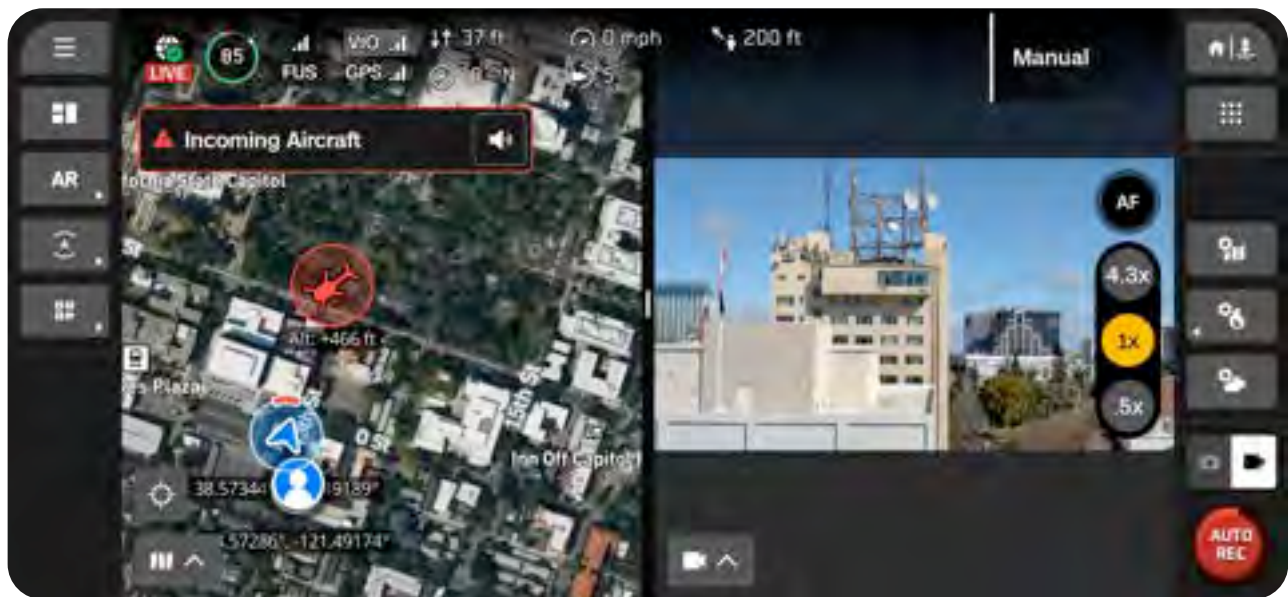
ADS-B Alerts

Automatic Dependent Surveillance–Broadcast (ADS-B) is a surveillance technology that enables situational awareness of nearby crewed aircraft. Skydio X10 is equipped with an onboard ADS-B In receiver, which detects ADS-B Out signals broadcast by crewed aircraft. This allows the X10 Controller to display nearby air traffic in real-time and issue alerts when traffic enters a pilot-defined proximity.

ADS-B traffic awareness is required for Beyond Visual Line of Sight (BVLOS) operations from the controller. This capability **supports waiver compliance by providing both visual and audible alerts when aircraft are nearby**, helping you avoid conflicts and maintain a safe distance.

Use ADS-B to:

- **Check the airspace before launching** – View real-time positions of other aircraft on the map before flight
- **Maintain situational awareness during flight** – Monitor air traffic as it approaches your area of operation
- **Respond to audible alerts** –If a crewed aircraft enters your defined alerting range, take appropriate action (e.g., descend to a safe altitude)



Setting ADS-B Alert Ranges

When an aircraft enters the alert radius, a notification will appear on the controller:

Aircraft icon – Displays either a fixed wing aircraft or rotorcraft

- Shows the altitude **relative** to your drone
- Select this icon to view more flight details (e.g., vertical/horizontal speed, ground distance relative to your drone)

Visual notification – An alert appears in the top left of the screen and displays **Incoming Aircraft**

Audible alert – When an aircraft enters your alert radius, the X10 Controller will play an audible beep, along with a spoken “Traffic”

- Mute this alert within the notification, or mute all notifications using the **Sound** menu (within Global Settings)

Step 1 - Navigate to the Display menu

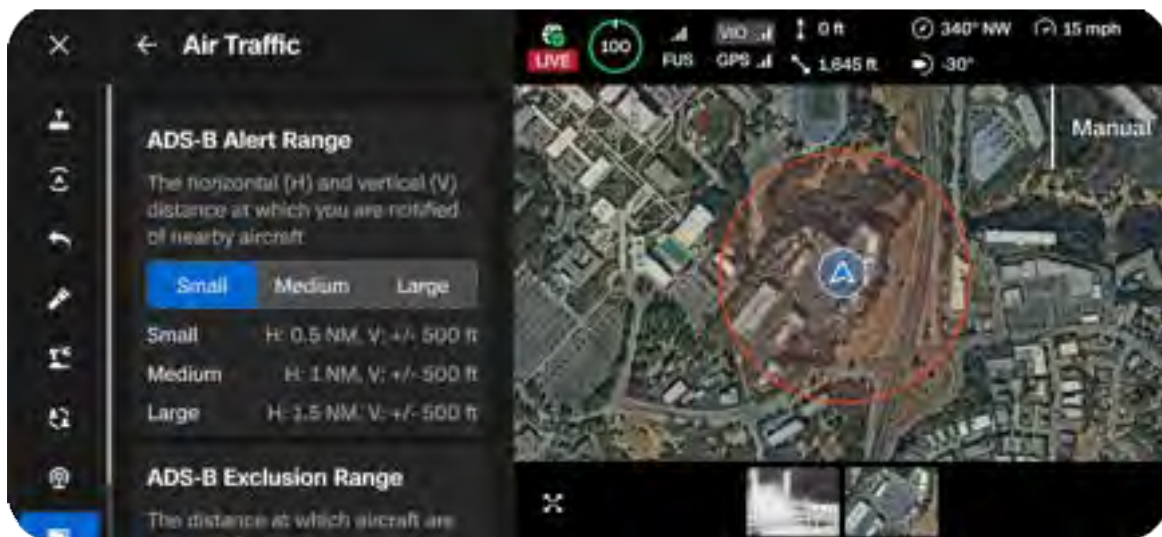
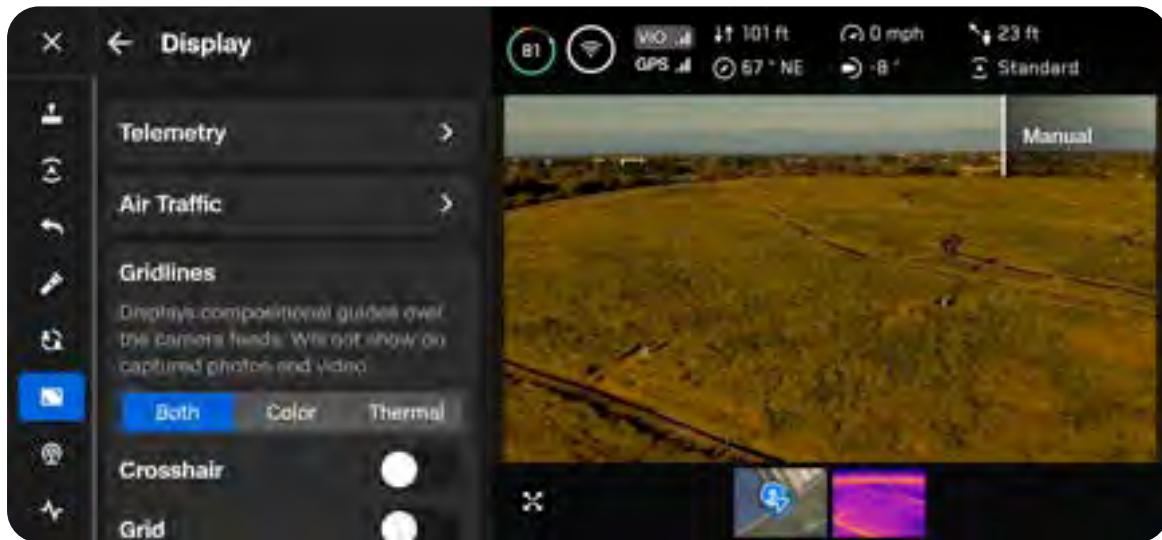
Located within Global Settings.



Navigating Skydio Flight Deck

Step 2 - Select Air Traffic

Use this menu to configure your ADS-B alert range.



Navigating Skydio Flight Deck

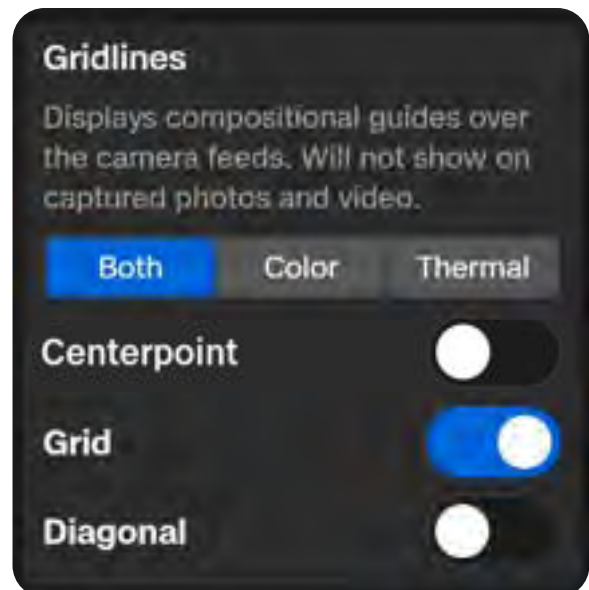
Gridlines

Choose from **Centerpoint**, **Grid**, or **Diagonal** to enable on-screen framing guides. Assists with centering and framing subjects.

Each tool adjusts for day and night visibility. Available in **Photo** and **Video** modes for both color and thermal cameras.

Use the toggles to enable one, two, or all three overlays. When enabled, they display on-screen only, without appearing in captured media.

- **Centerpoint** - Displays a crosshair in the center of the screen to align subjects
- **Gridlines** - Adds a rule of thirds overlay for balanced framing
- **Diagonals** - Displays two intersecting diagonal lines from opposite corners, aiding in angular alignment



Navigating Skydio Flight Deck



Centerpoint



Grid



Diagonal

Navigating Skydio Flight Deck

Display Brightness

Adjust the brightness of your X10 Controller screen.

Units

Choose between Imperial or Metric units.

Depth Style

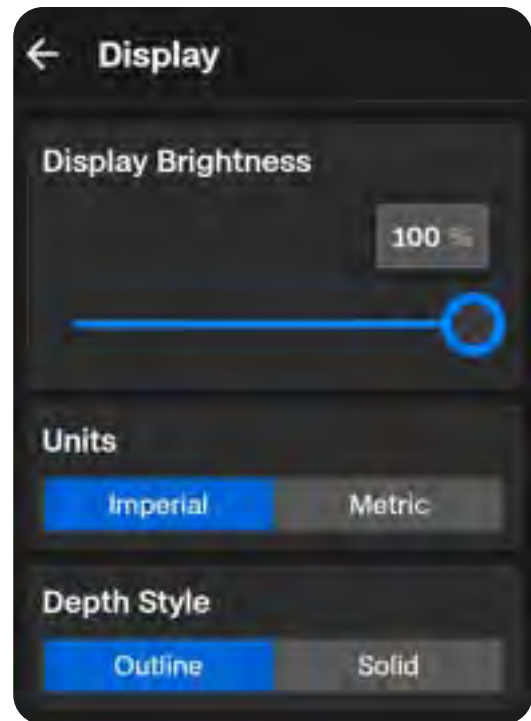
Only applies if you have enabled Depth View within the AR Quick Actions (located on the left side of the Flight Screen).

Select **Solid** or **Outline** when displaying visual information about what obstacles the drone sees.

- Solid displays boxes filled with color
- Outline displays wireframed or unfilled boxes

The **AR Quick Actions** button on the Flight Screen cycles between the distances from objects at which the visual information will start showing on screen.

- Off
- 6 ft (2 m)
- 13 ft (4 m)



Display Layouts

While flying, you have the option to use a Single, Split or Grid layout to set the number of streams that appear while flying. Available feeds include:

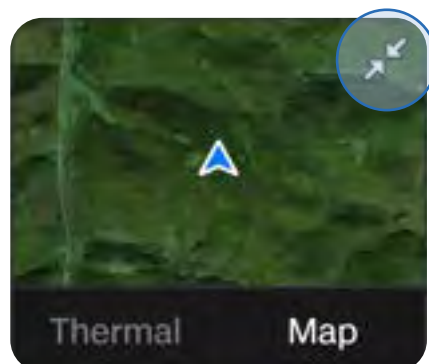
- Color
- Thermal
- Map

Single Layout

Displays one stream at a time.

In this layout, you will see a Picture-in-Picture (PiP) in the bottom left of the screen.

- Minimize using the two arrows in the top right
- Use the buttons on the bottom to select whether the Color, Thermal or Map feed displays in the PiP



Navigating Skydio Flight Deck

Split Layout

Choose two streams to display. Drag the middle handlebar to resize streams.

The primary feed displays on the right.



Grid Layout

Choose three streams to display. Press and drag the middle handlebar to resize streams.

The primary feed displays on the right.



Customizing Display Layout

Step 1 - Select the Display Layout icon in the left sidebar

Use this button to cycle through the various layout options. The icon reflects the next layout in the queue rather than the layout you are currently using.



Navigating Skydio Flight Deck

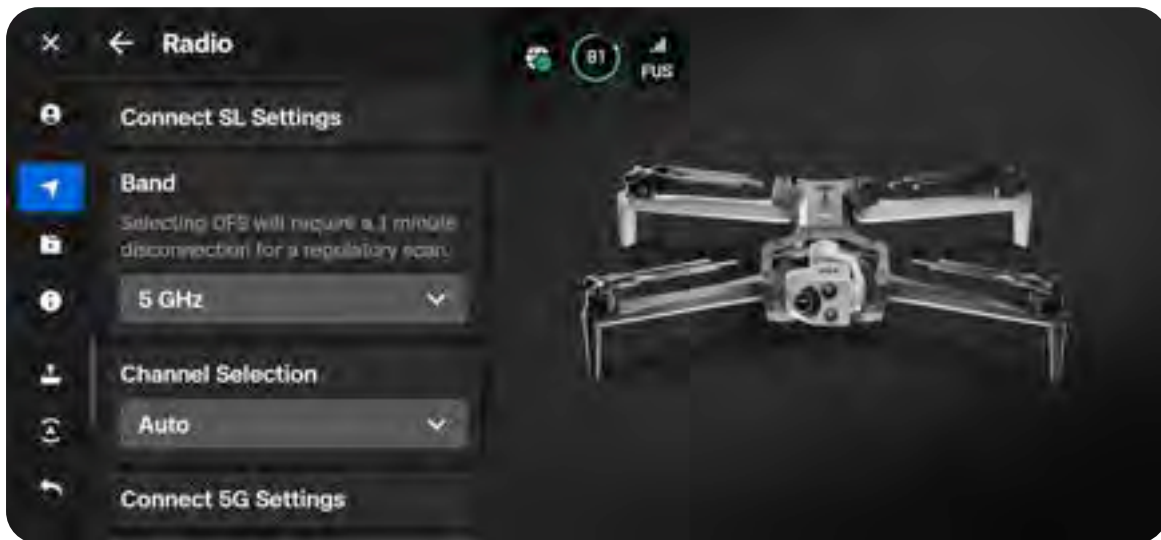
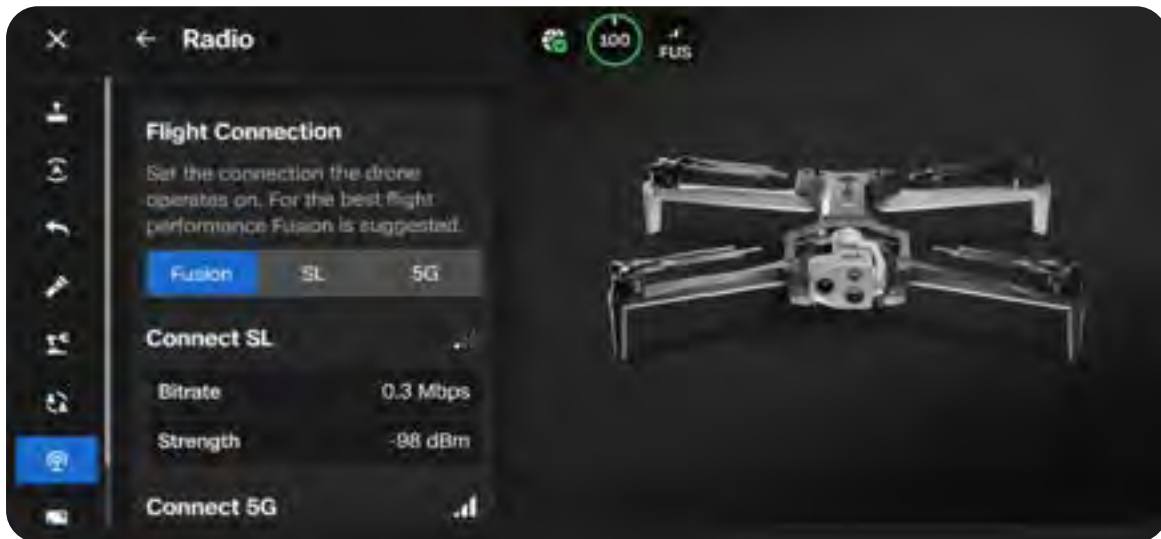
Step 2 - Use the View Selector to select which feeds display

A menu will appear with the stream options. Drag the middle handlebar to resize streams.



Radio

Use this menu to configure your radio settings.



Navigating Skydio Flight Deck

Flight Connection

Set the connection X10 operates on. Displays the current link quality between the drone and controller, including signal strength (in dBm) and bitrate.

- **Fusion (default)** - Optimizes inflight connectivity by automatically selecting the strongest signal between Connect SL and Connect 5G, eliminating the need for manual switching
- **SL** - The drone will only use Connect SL (point-to-point)
- **5G** - The drone will only use Connect 5G (cellular)

Band (Connect SL)

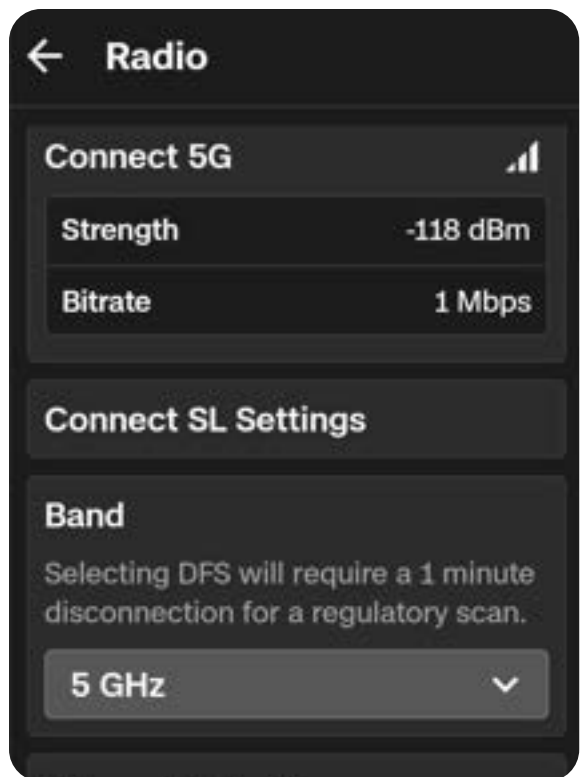
Select your radio bandwidth:

- 2.4 GHz (coming soon)
- 5 GHz
- DFS (Dynamic Frequency Selection)

Channel

Select your radio frequency channel to avoid congestion from other signals.

Set to **Auto** by default.



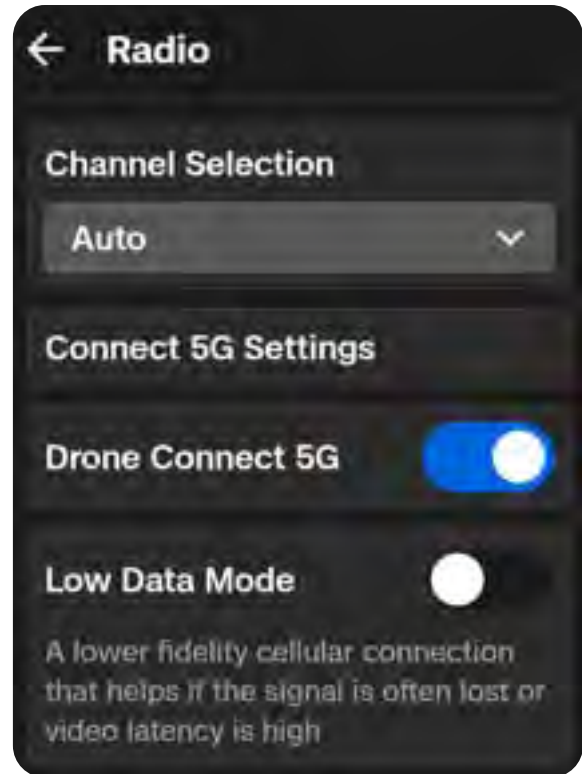
Navigating Skydio Flight Deck

Drone Connect 5G

Enables 5G cellular connection to the drone. For more information about setting up and flying over 5G cellular, visit: [How to fly Skydio X10 over cellular connectivity](#).

Low Data Mode

Helps reduce instances of lost connection or high video latency when flying in locations with poor cellular coverage. Enable this toggle if you're experiencing frequent lost connections or high video latency when Drone Connect 5G is enabled.



Navigating Skydio Flight Deck

Skydio Connect SL Frequencies

2.4 GHz (coming soon)

Auto

- 1: 2401-2423 MHz
- 2: 2406-2428 MHz
- 3: 2411-2433 MHz
- 4: 2416-2438 MHz
- 5: 2421-2443 MHz
- 6: 2426-2448 MHz
- 7: 2431-2453 MHz
- 8: 2436-2458 MHz
- 9: 2441-2463 MHz
- 10: 2446-2468 MHz
- 11: 2451-2473 MHz

5 GHz

Auto

- 36: 5170-5190 MHz
- 40: 5190-5210 MHz
- 44: 5210-5230 MHz
- 48: 5230-5250 MHz
- 149: 5735-5755 MHz
- 153: 5755-5775 MHz
- 157: 5775-5795 MHz
- 161: 5795-5815 MHz
- 165: 5815-5835 MHz

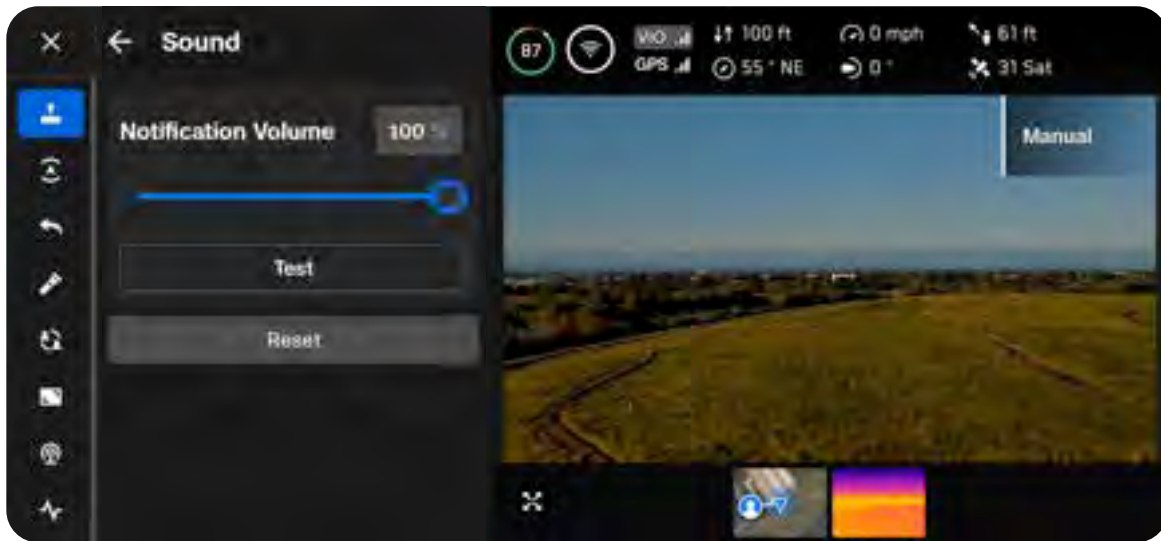
DFS (Dynamic Frequency Selection)

Auto

- 52: 5250-5270 MHz
 - 56: 5270-5290 MHz
 - 60: 5290-5310 MHz
 - 64: 5310-5330 MHz
 - 68: 5330-5350 MHz
 - 96: 5470-5490 MHz
 - 100: 5490-5510 MHz
 - 104: 5510-5530 MHz
 - 108: 5530-5550 MHz
 - 112: 5550-5570 MHz
 - 116: 5570-5590 MHz
 - 120: 5590-5610 MHz
 - 124: 5610-5630 MHz
 - 128: 5630-5650 MHz
 - 132: 5650-5670 MHz
 - 136: 5670-5690 MHz
 - 140: 5690-5710 MHz
 - 144: 5710-5730 MHz
-

Sound

Allows you to control the notification volume that plays through the X10 Controller speakers. Set the slider to zero to mute all sounds.



System Status

Provides a comprehensive overview of the system at a glance.

Displays:

- Drone connection quality
- GPS signal quality and position accuracy
- Live Stream status
- Remote ID broadcast status



Navigating Skydio Flight Deck

Drone Connection

Skydio Connect SL

Refers to the connection quality between the drone and controller.

Skydio Connect 5G*

Refers to the cellular connection quality of the drone to cellular towers.

**You must have purchased a Skydio X10 Controller with Connect 5G on it*

GPS

Satellites

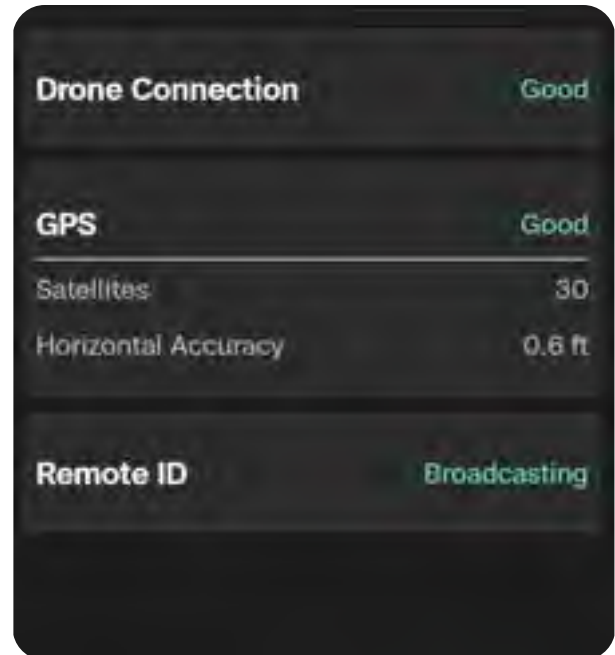
Represents the count of GPS satellites that the drone is currently receiving signals from. A greater number of satellites results in a more accurate position as well as greater safety and reliability during flight.

It is important to establish a strong GPS connection, especially before flying over water.

- Connection to 13 or more satellites is considered a Good connection
- We recommend 18 or more satellites before flying over water

Horizontal Accuracy

Quantifies the position of Skydio X10 on a two-dimensional plane. Especially useful when mapping or surveying.



Remote ID

Displays the status of Remote ID broadcasts.

Remote ID requires drones operating in U.S. airspace to identify themselves by broadcasting information that will enable the authorities to identify pilots who are not following the rules. Only authorized individuals from public safety organizations may request the identity of the owner of the drone from the FAA.

Remote ID broadcast includes:

- Launch location
- Drone ID - your Skydio ANSI serial number
- Drone location and altitude
- Drone velocity
- Control station location and elevation
- Time of operation
- Emergency status

Quick Actions

The left sidebar of your Flight Screen contains a variety of Quick Action menus. With Quick Actions, you have the ability to quickly **toggle** a setting or **cycle** through setting options.

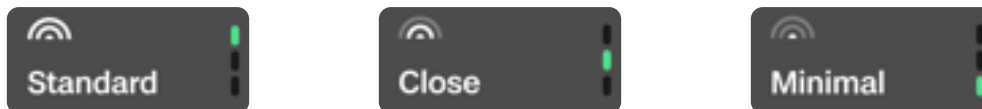
Toggle

Full green bar indicates a setting is ON.



Cycle

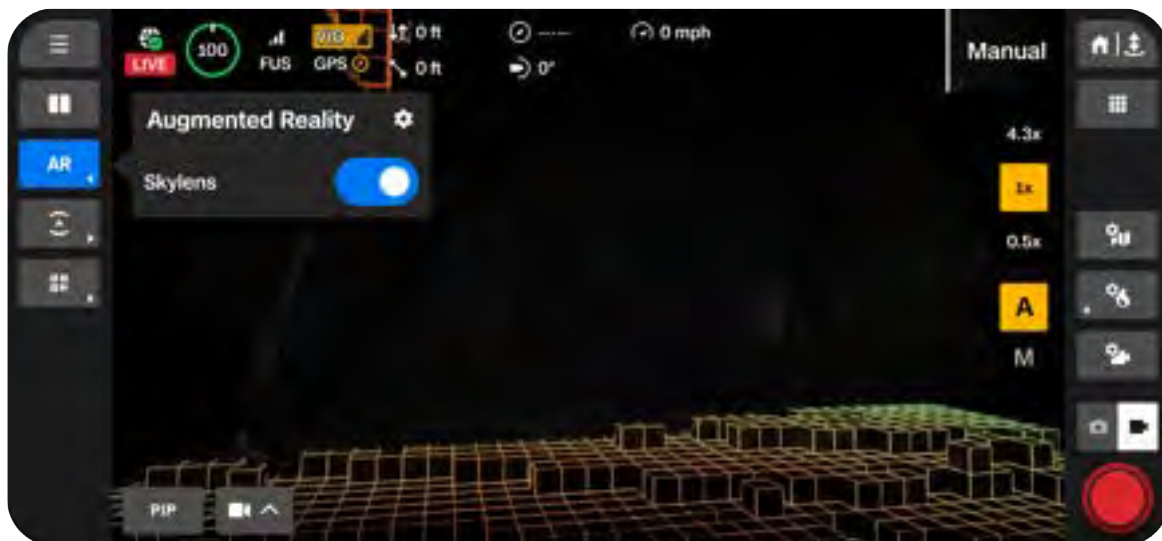
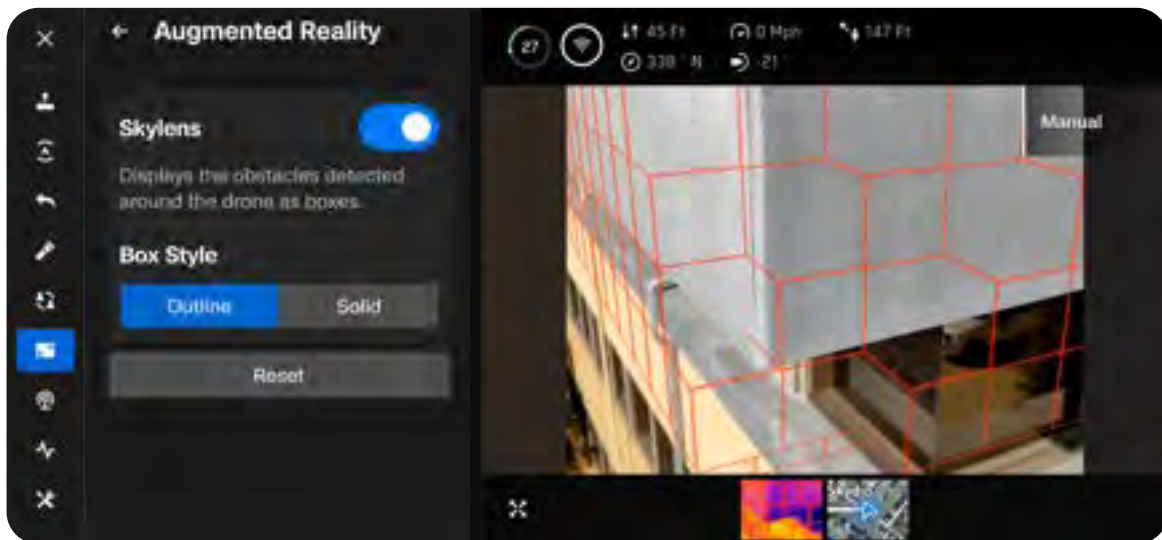
A menu will pop out and label the current setting. The green bars indicate the number of available settings.



Skylens (Augmented Reality) Quick Actions

Enable to visually display where Skydio X10 detects obstacles in the environment. Appears as either solid or outlined augmented reality (AR) boxes, depending on your selection within the **Display** menu.

Use the **AR Quick Action** button to enable or disable Skylens. Objects that are closer appear red.



Obstacle Avoidance Quick Actions

Select the **Sensing Quick Action** to toggle obstacle avoidance **ON** or **OFF**. Use the drop-down menu to set the **Range** as Standard, Close, or Minimal. Select the **gear icon** to enter the **Sensing settings menu** (within Global Settings).

A yellow border appears when in Close or Minimal obstacle avoidance mode, or if obstacle avoidance is disabled.



Obstacle avoidance disabled

Inflight Map

View your current location, search, set a Home Point, and configure map settings.

- The location of Skydio X10, the controller, Launch Point, and Home Point (if set) are indicated on the map
- Press and hold on a location to set a **Home Point**



Skydio X10



Controller



Launch Point



Home Point

When an aircraft enters your ADS-B alert radius (Display > Air Traffic), a series of notifications appear:

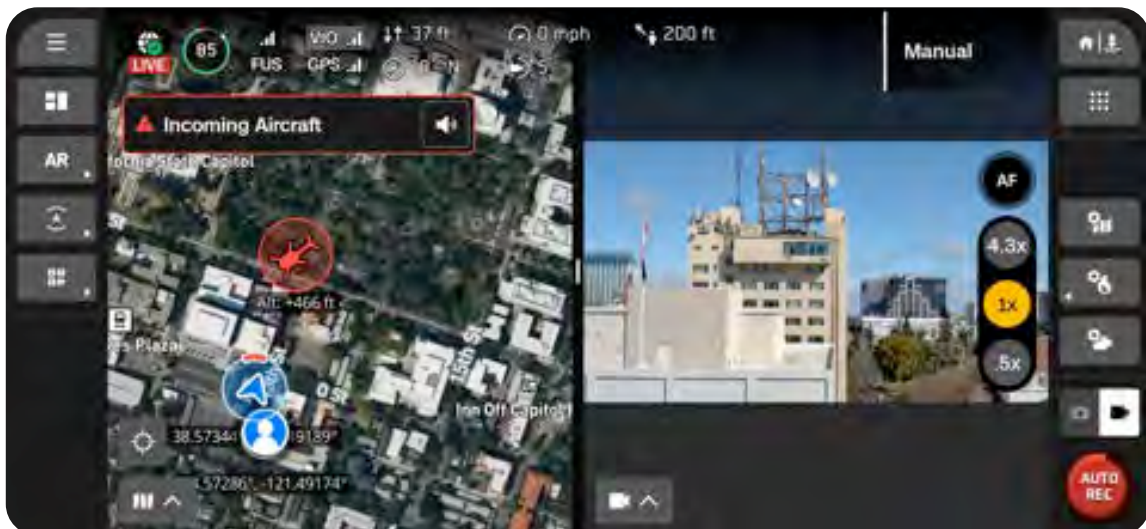
Aircraft icon – Displays either a fixed wing aircraft or rotorcraft

- Shows the altitude **relative** to your drone
- Select this icon to view more flight details (e.g., vertical/horizontal speed, ground distance relative to your drone)

Visual notification – An alert appears in the top left of the screen and displays **Incoming Aircraft**

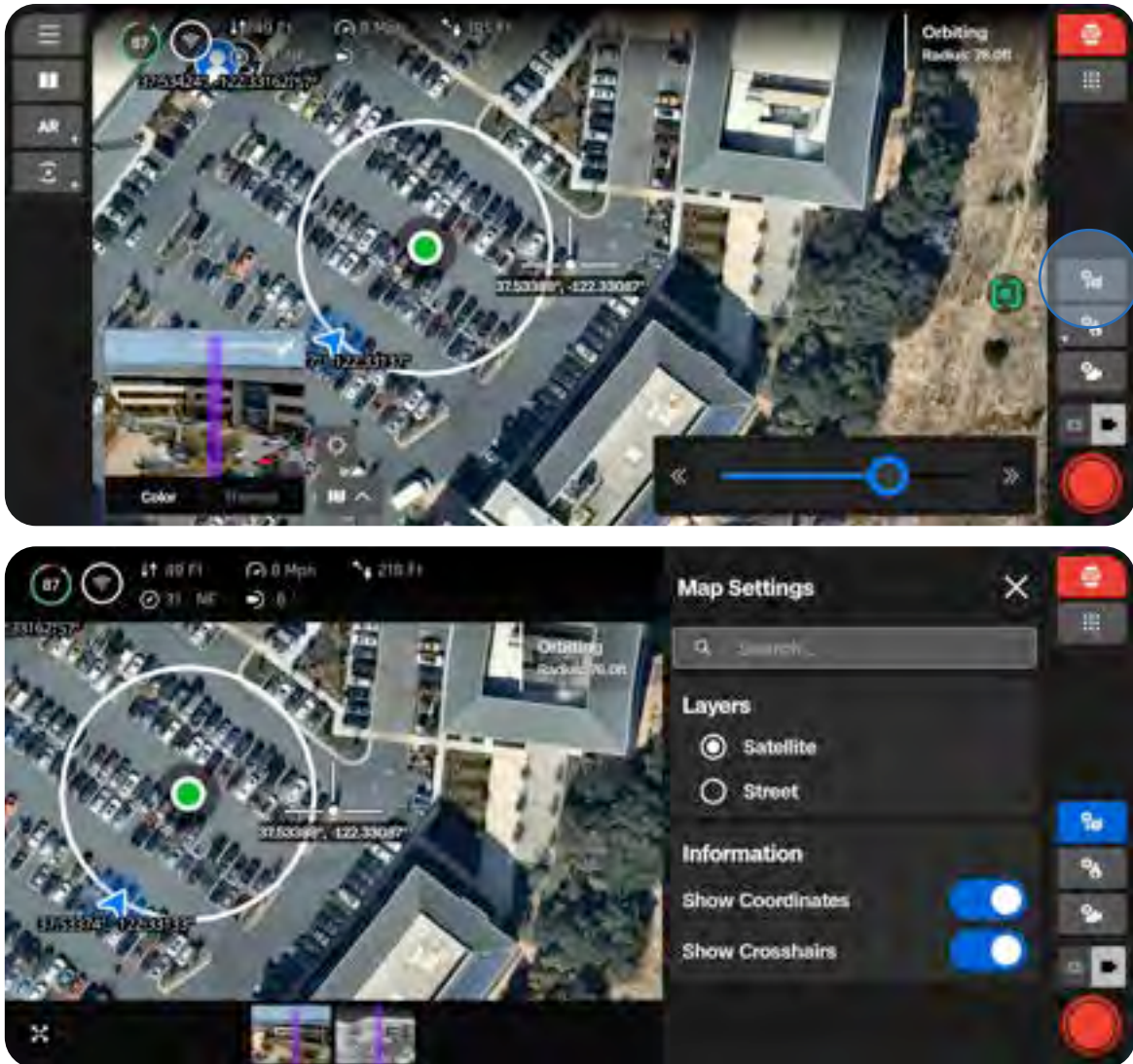
Audible alert – The X10 Controller will play an audible beep, along with a spoken “Traffic”

- Mute this alert using the **Sound** menu (within Global Settings)



Navigating Skydio Flight Deck

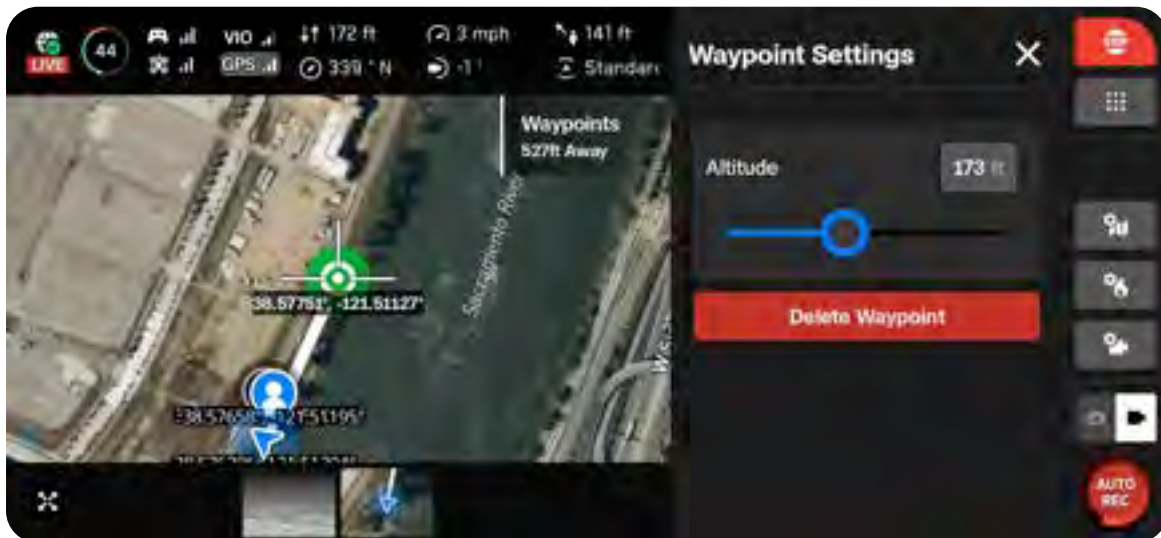
Adjust your map settings during flight using the **Map Settings** icon.



Fly Here Now

Press and hold on the map and select **Fly Here Now** to immediately fly to that location.

- Select anywhere along the flight path between the drone and the Fly Here Now location and add **transit points**
- Edit the altitude of the transit points or destination using the menu on the right
- Both the Fly Here Now point or transit points are draggable



NOTE: The drone will fly at maximum speed while executing a Fly Here Now mission.



Camera Settings

Learn how to adjust photo and video settings such as zoom, exposure, ISO, and resolution.

This section covers

Overview

Capture Settings (Photo and Video)

Focus and Exposure

Zoom Settings (Photo and Video)

Shutter Indicators

Photo Settings

Video Settings

Sensor Package Flashlights

Overview

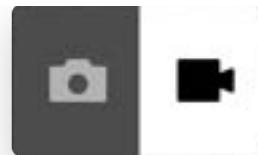
When your drone captures a photo or video, it will save one image file with the color camera. Two files will save if you have JPG and DNG enabled. If you would like Skydio X10 to also capture an image file with the thermal camera, enable **Thermal Capture** within the **Thermal Settings**.

Use **Camera Mode** on the right sidebar to switch between photo or video. Skydio X10 can capture photos or videos but not both at the same time. Your color and thermal cameras will always be in the same Camera Mode.

Access your photos and videos using the **Media** menu located in **Global Settings**.



Photo enabled



Video enabled



NOTE: Photo and Video mode settings are independent of each other and persist through mode changes, but not power cycles.

Capture Settings (Photo and Video)

Brightness Exposure Value (EV)

Refers to the amount of light the camera allows in.

Negative numbers result in darker images (less exposure) while positive numbers result in brighter images (more exposure).

- Brightness is set to Auto by default

White Balance

Balances the color temperature in your photo. If the whites in your picture are too orange, for example, adding the opposite color (blue) will balance them out.

Lower values result in a cooler (blues) image while higher values result in a warmer (yellows) image.

- Auto (default) means Skydio X10 will automatically adjust the White Balance for its environment

ISO

Brightens or darkens your photo. When in low-light conditions, raising the ISO value will brighten the image, however you may see some graininess.

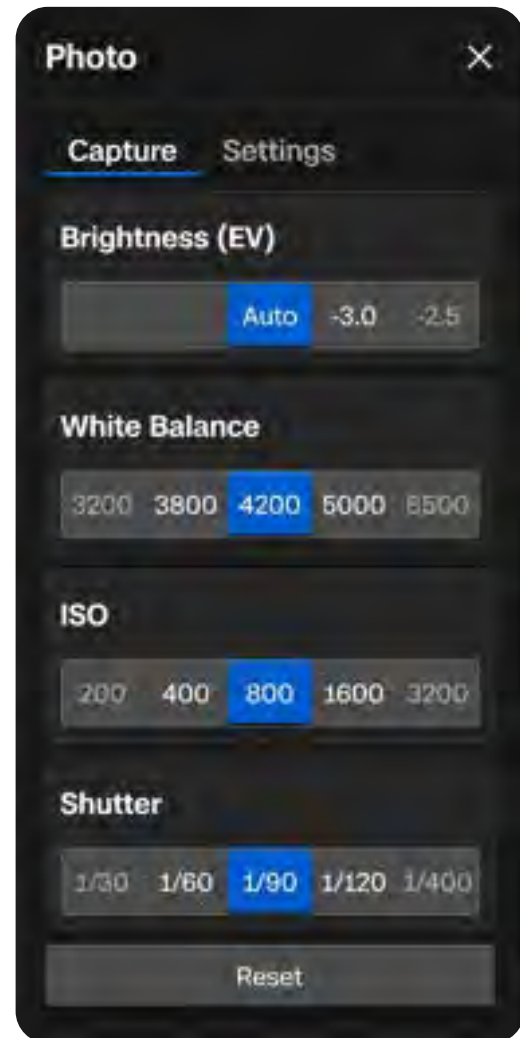
- Auto means Skydio X10 will automatically adjust the ISO for its environment

Shutter

Refers to the length of time a photo is exposed.

Slower shutter speed means greater exposure, while faster shutter speed means less exposure.

- Auto means Skydio X10 will automatically adjust the Shutter speed based on the available light



Focus and Exposure

To view the various focus options, select the Focus Control button on the right side of the Flight Screen.



AutoFocus (AF)

By default, your camera will be set to automatically adjust focus and exposure. In this focus mode, focus will generally prioritize objects in the center of the screen.



Manual Focus (MF)

After selecting the Manual Focus icon, a focus slider will appear:

- Drag the slider to adjust the plane of focus
- Select outside the slider to return to the full parameter view
- To go back to Auto focus, select the Focus, Manual box again



Tap to Focus

Touch anywhere on the screen to focus on an area of interest, or select the icon to center. The Focus Indicator will turn green once the image is sharp and clear.



Manual Capture Controls

The Manual Focus option provides you with greater control over image settings. Switch between AutoFocus and Manual Focus at any time.

Switching Modes

Select the Focus Control button to switch between Auto and Manual focus. In Manual Focus, you control:

- Exposure Value (EV)
- ISO
- White Balance (WB)
- Shutter Speed

Touch Controls

- Select the parameter box you wish to control (EV, ISO, WB, Shutter Speed)
- A value selector will appear; drag up or down to set your value
- Select outside of the selector to confirm and return to the main control view
- A **lock icon** indicates a manual value, select the lock to return the parameter to Auto

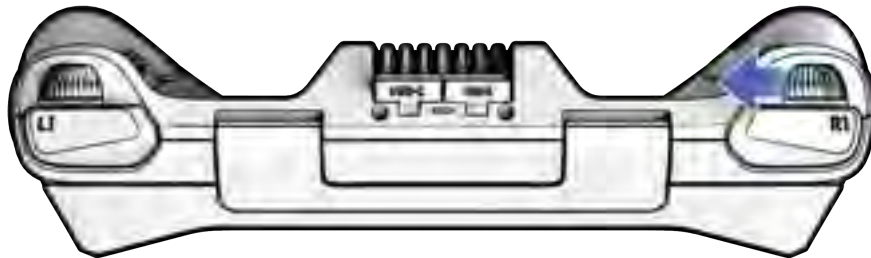
D-Pad Controls

From Auto mode, press down on the D-Pad to toggle into Manual mode. The active parameter is highlighted with a yellow border. Use the D-Pad to:

- Press down to open a parameter's value menu
- Press up or down to scroll through available values
- Press down again to confirm
- Press down repeatedly to cycle focus between parameters

Zoom Settings (Photo and Video)

To zoom in digitally, place your finger on the right controller wheel and push it to the left. Customize this using Input Mapping (Flight Controls > Controls).



Quickly snap to a zoom level using the Zoom buttons on the right side of the screen.

VT300-Z Sensor

- 4.3x - transitions between narrow and telephoto lens, 128x max system zoom
- 1x - default zoom level of narrow lens
- .5x - Surround Vision

VT300-L Sensor

- 2x - transitions between wide and narrow lens, 64x max system zoom
- 1x - default zoom level of wide lens
- .5x - Surround Vision



NOTE: Images and videos taken while zoomed in will be saved at that zoom level.

Camera Settings

Surround Vision

Surround Vision uses Skydio X10 navigation cameras to generate an ultra-wide view of your environment for situational awareness.



NOTE: Photos and videos captured when in Surround Vision will be saved at 1x zoom.



Thermal Zoom

The thermal camera will zoom up to 16x, but you may continue to zoom using the color camera. If any Tools are enabled, such as Region of Interest, they will dynamically adjust to fit the screen as you zoom.

Shutter Indicators

The Shutter is located in the bottom right of the Flight Screen and indicates the current state of Photo or Video mode.



Photo

Ready for Capture



Pressed



Disabled



Interval



Video

Manual Record



Manual Record (Pressed)



Recording



Recording (Pressed)



Auto Recording



Auto Recording (Paused)



NOTE: You have the ability to enable Shutter Sounds when capturing photos within Photo Settings.

Photo Settings

Save to Custom Folder

You have the ability to create a custom folder preflight or inflight. This folder will be saved on the **Media card** under DCIM > [Custom Folder Name].

All photos and videos captured **after** creating a custom folder will automatically save to this new folder. To revert to default, select **Revert to Media**.

- If you do not enter a custom folder name, media will be saved in DCIM > 100XSKYDO
- Does not apply to photos captured in 3D Scan

File Type

Choose whether you want Skydio to capture JPG images only, or both JPG and DNG files.

- **JPG** - Digital image format containing compressed image data.
- **DNG** - RAW image format file, meaning it is not compressed and retains all original photo data. A DNG file is larger than a JPG file since it stores image data.

Resolution

Refers to the amount of detail in your video. Measured in megapixels.

- **Full** - Images are captured at the highest quality, providing more detail and clarity.
- **1/4** - Images are captured at one-fourth of the full resolution, resulting in smaller file sizes. Best for conserving storage space or transmitting images faster.



Camera Mode

- **Standard** - Designed for typical, everyday lighting conditions. Provides a balanced, standard level of exposure, image processing, and contrast.
- **Low Light** - Designed for environments with dim lighting, such as indoors or evening. Settings are adjusted to capture more light, reduce noise, and improve visibility. Only available with 1/4 Resolution.
- **HDR** - Designed to capture environments with a wide range of brightness levels. Only available with 1/4 Resolution.

Interval

When enabled, Skydio X10 will continuously capture photos at the specified time interval until the setting is disabled or the flight ends.

Shutter Sound

When enabled, you will hear a shutter sound when capturing photos, providing instant confirmation that an image has been taken.

- Enabled by default
- Sound only plays through the controller speakers, not through connected headsets

Video Settings

Auto Start Recording

When enabled, Skydio X10 will record video automatically.

When disabled, tap the on-screen Shutter button or on the controller (R1 button) to start/stop video recording.

File Type

Select between H.264 and H.265 compression formats, depending on your preferences for video quality, file size, and playback compatibility.

- **H.264** - Provides manageable file sizes without sacrificing video quality. Recommended for standard video recording, and compatible with most devices and video editing software.
- **H.265** - Ideal for high-quality video capture and maintains efficient compression.

Resolution

Select between 4K and Full HD, which refers to the amount of detail in your video. Measured in pixels.

- More pixels result in a high-resolution video
- Fewer pixels result in a lower resolution video



Camera Settings

Camera Mode

- **Standard** - Designed for typical, everyday lighting conditions. Provides a balanced, standard level of exposure, image processing, and contrast.
- **Low Light** - Designed for environments with dim lighting, such as indoors or evening. Settings are adjusted to capture more light, reduce noise, and improve visibility.
- **HDR** - Designed to capture environments with a wide range of brightness levels.

Aspect Ratio

Sets the shape and framing of your video.

- **16:9** - Provides a wider, broader field of view.
- **4:3** - Provides greater FOV in the vertical axis, resulting in more square-shaped framing. Images are taller, as opposed to wider.

Save to Custom Folder

You have the ability to create a custom folder preflight or inflight. This folder will be saved on the **Media card** under DCIM > [Custom Folder Name].

All photos and videos captured **after** creating a custom folder will automatically save to this new folder. To revert to default, select **Revert to Media**.

- If you do not enter a custom folder name, media will be saved in DCIM > 100XSKYDO
- Does not apply to photos captured in 3D Scan

Using the Flashlight on VT300-L or V100-L Sensor Packages

The VT300-L and V100-L sensor packages comes equipped with an onboard flashlight, providing effective illumination up to 10 ft (3 m) and enabling inspections in low-light environments.

Select the on-screen flashlight icon to turn the flashlight on or off in flight.



NOTE: The flashlight can only be controlled during flight and will not be operational on the ground nor during launch and landing.



WARNING: When using the flashlight on the VT300-L or V100-L sensor packages, do not stare directly into the light at any range for any extended period of time.



WARNING: After prolonged use of the flashlight, your sensor package may be hot to the touch and could present a serious burn risk. After landing, wait for your sensor package to cool down before handling.



Thermal Camera and Tools

The thermal camera with Skydio X10 includes a powerful suite of tools, such as radiometric capabilities, to assist you in a variety of use cases.

This section covers

Accessing Thermal Options

Flat Field Correction (FFC)

Thermal Tools

Thermal Settings

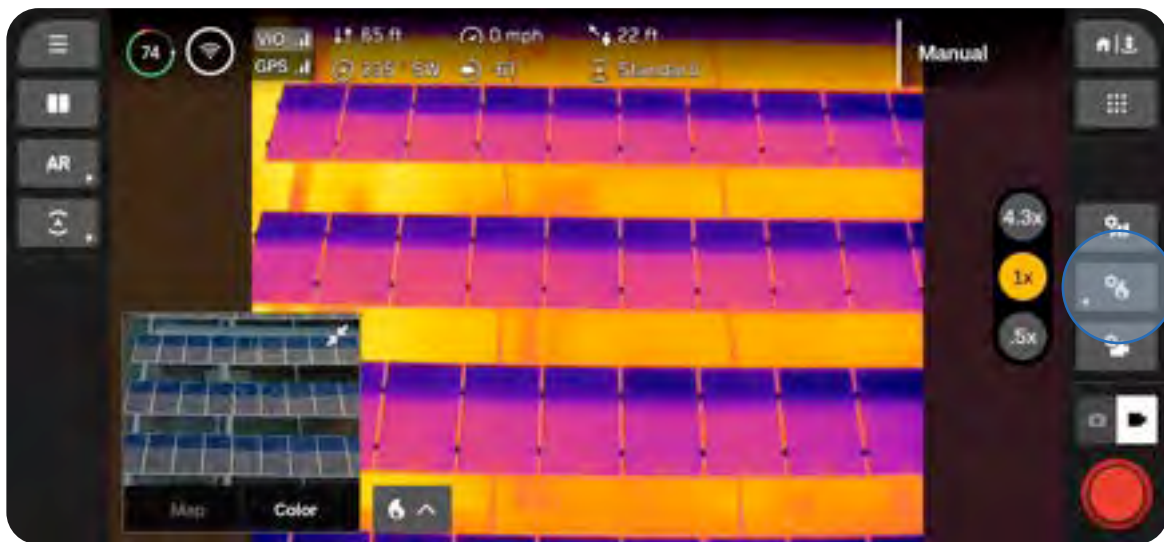
Thermal Parameters

Accessing Thermal Options

Quickly access your Thermal Tools and Settings using the quick action button located on the right side of the Flight Screen.

Step 1 - Select Thermal Settings

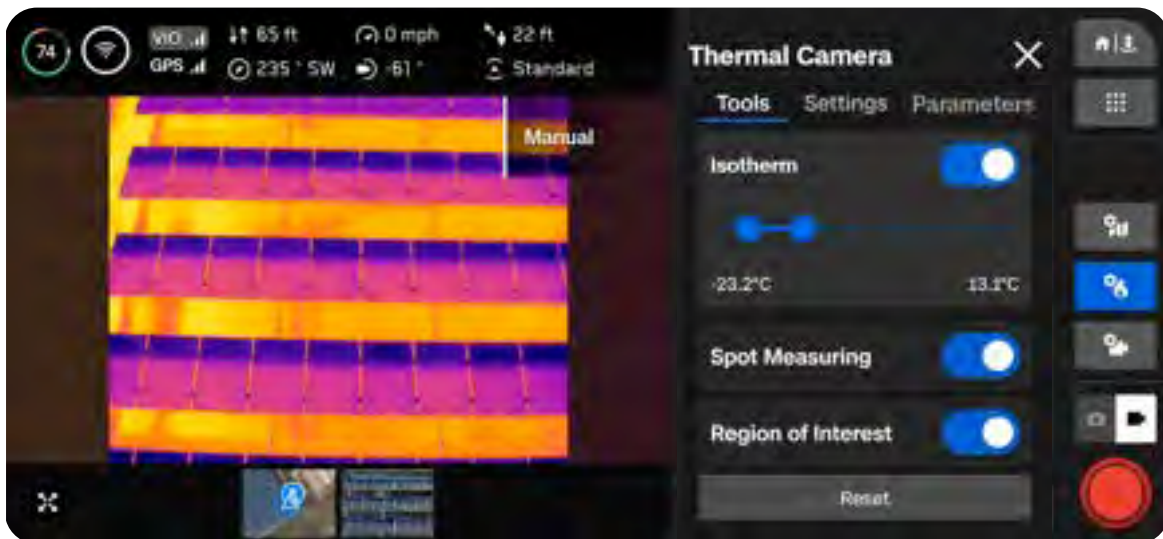
Easily enable some settings using the quick actions that appear, or access the full menu.



Thermal Camera

Step 2 - Select the gear icon to adjust settings

Use the tabs at the top to customize your Tools, Settings, and Parameters.



Flat Field Correction (FFC)

Flat Field Correction (FFC) mitigates and compensates for errors that build up over time during the thermal camera operation. This is performed in the background automatically at lower zoom levels, however, you can use the Thermal Settings to manually perform FFC at any time.

Before launching, your drone will complete an automatic FFC. As a result, you may hear a light clicking sound coming from the sensor package as this completes.



NOTE: If you manually perform FFC, the actuation of the shutter may be noticeable on the image at higher zoom levels.



Thermal Tools

Isotherm

Allows you to set a range of temperatures to detect. Use this setting to omit unwanted data outside of the specified range.

The defined range will appear as the currently selected palette.

- Temperatures outside of the defined range will appear as the default White Hot or Black Hot palette
- If White Hot or Black Hot are currently selected, the temperatures in the Isotherm range will default to Rainbow

Spot Measuring

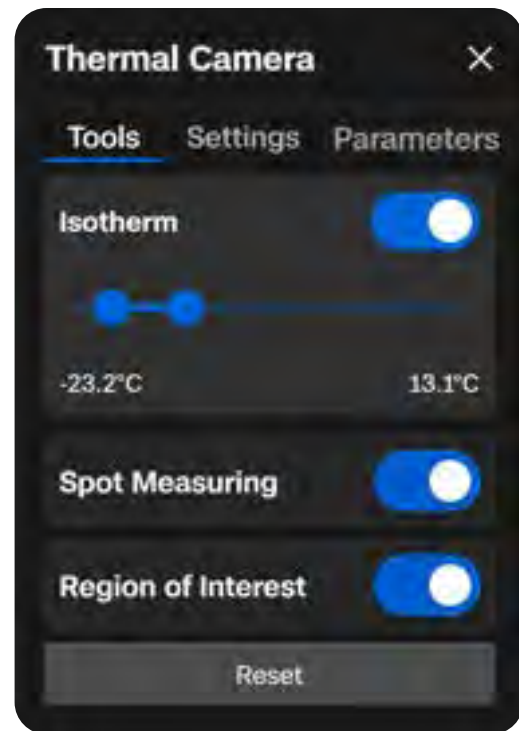
Enable to display the specific temperature value of an object on-screen as you drag your finger on-screen.

- Tap or drag your finger across the screen to view temperatures

Region of Interest

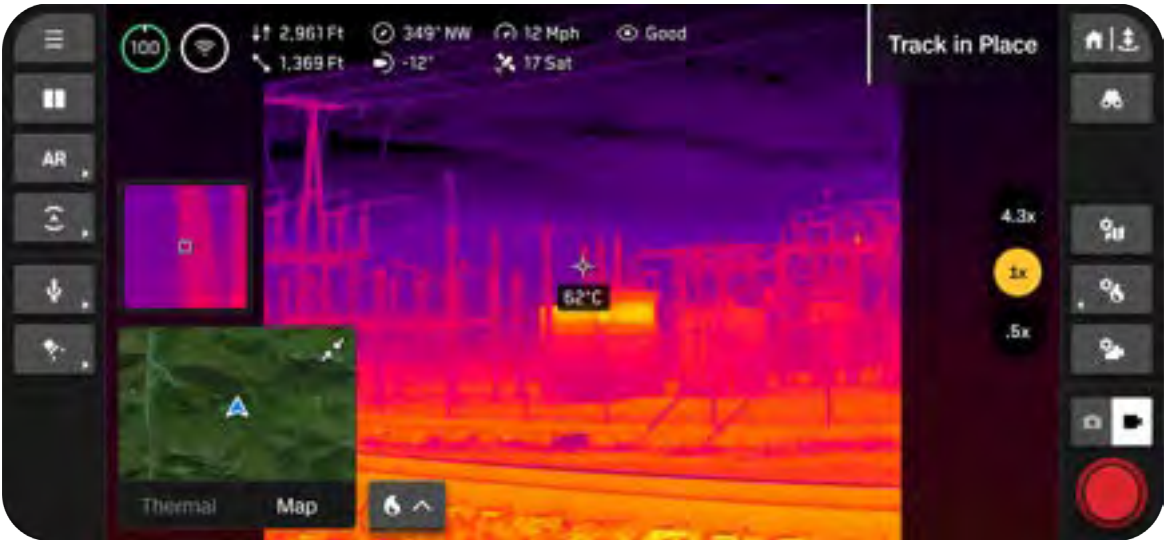
Enable to display an on-screen box that detects the minimum, maximum, and average temperatures within the outlined area

- **H** represents the highest temperature detected
- **L** represents the lowest temperature detected
- **A** indicates the average temperature detected
- Select the edges of the box to resize, or use the arrows to drag the box to a different location

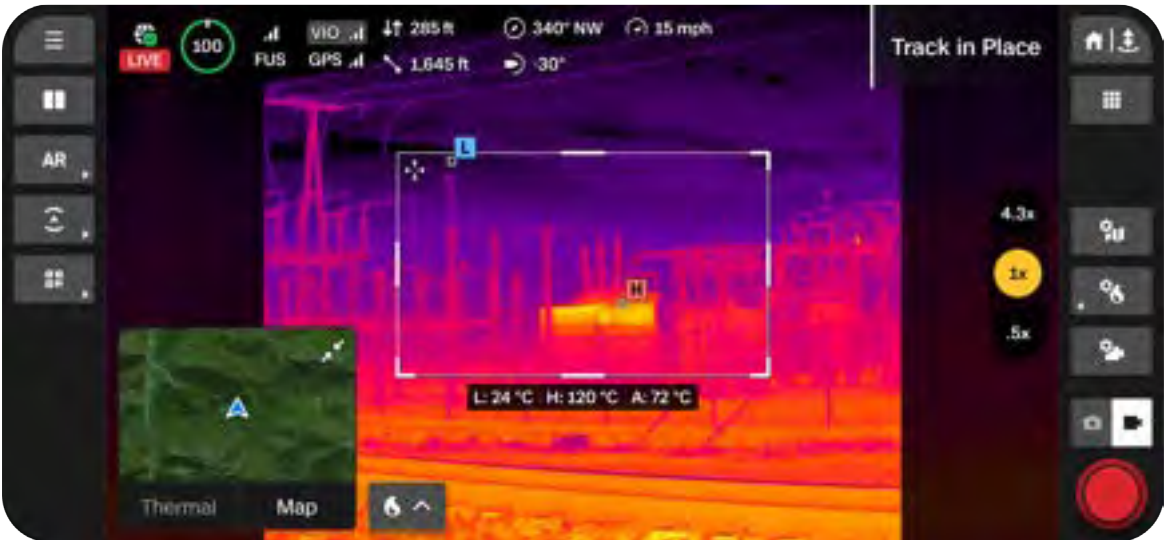


Thermal Camera

Spot Measuring



Region of Interest



Thermal Settings

Color Palette

Visual representations of temperature variations captured by a thermal camera

Ironbow - Quickly identify varying temperatures and spot thermal anomalies. Displays a specific range of colors, from blues to reds, which indicate different temperature levels.

- Warmer objects are presented in lighter colors and colder objects in darker colors

Rainbow - Uses the colors of a traditional rainbow to distinguish between subtle variations in temperature levels.

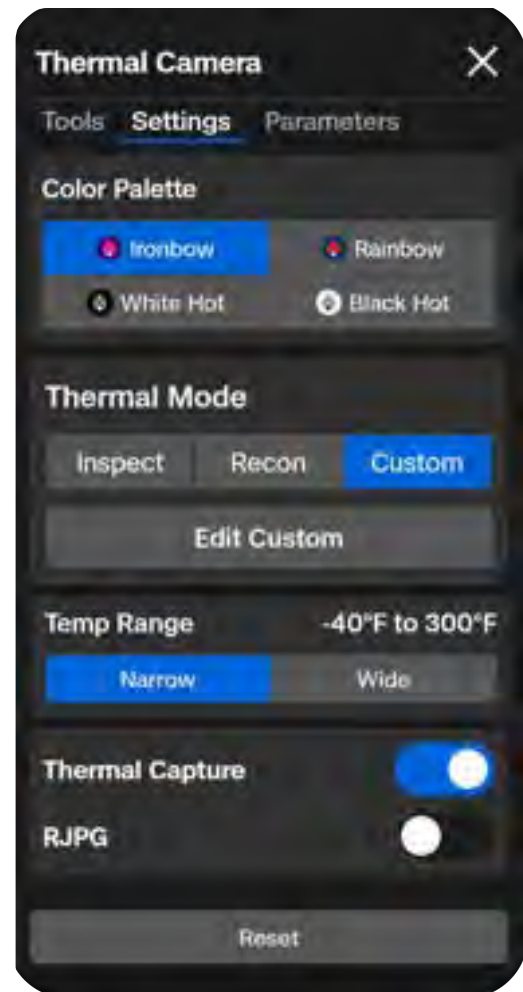
- Covers a broader range of colors without emphasizing specific temperature ranges

White Hot - Provides a clear visualization of temperature variations without a variety of colors.

- Brighter, whiter colors indicate warmer temperatures
- Darker colors represent cooler temperatures

Black Hot - Displays the inverse of a White Hot palette.

- Brighter, white colors indicate cooler temperatures
- Darker, black colors represent warmer temperatures



Thermal Camera

Thermal Mode

Adjust the signal amplification from the camera sensor to enhance temperature differences in an image.

Recon - Tuned to increase the contrast between the overall scene and targets. Best for search and rescue or situational awareness use cases. Recon helps differentiate the scene from things like people, vehicles, or animals.

Inspect - Tuned to decrease the overall contrast so that temperature anomalies are easier to identify in inspection use cases. This makes it easier to not only see the anomaly, but help track the source from the heat signature.

Custom - Allows you to fine-tune your Gain, High Tail, and Low Tail settings. Select Inspect or Recon as your Base Mode, then adjust your Gain, High Tail, and Low Tail settings from there. These settings will persist across flights and will be applied each time you select Custom as your Thermal Mode.

- Gain - Artificially increases contrast of areas with similar temperatures
- High Tail - Adjust saturation of the hotter items in the scene
- Low Tail - Adjust saturation of colder items in the scene

Temp Range

Select the range of temperatures Skydio X10 will detect.

Narrow - Detects temperatures ranging from -40°F to 302°F (-40°C to 150°C)

Wide - Detects temperatures ranging from -40°F to 662°F (-40°C to 350°C)

Thermal Capture

Enable to capture thermal images as JPG files.

RJPG - When enabled, your drone will capture a Radiometric JPG in addition to a standard JPG. A Radiometric JPG includes the radiometric data within the photo file.

Thermal Parameters

Emissivity

The measure of how efficiently an object emits thermal radiation. Adjust to match the camera readings to the true temperature of the object.

- Higher values means the camera is more sensitive to temperature variations
- Lower values means the camera is less sensitive to temperature variations

Surfaces that are better emitters (higher emissivity) provide more reliable temperature readings. For example, black electrical tape, rusted or oxidized surfaces, bodies of water, or human skin all absorb and emit energy. Set your emissivity high for these types of surfaces.

Reflective surfaces are not good emitters (low emissivity) and therefore not as reliable to measure. Stainless steel, shiny surfaces, or reflective windows tend to have a low emissivity. Set your emissivity low for these types of surfaces, but we recommend gathering your reading from a higher emissivity surface whenever possible.

Humidity

Set your humidity to accurately reflect the environment you are flying in. This should be the humidity between the sensor and the target.

Humidity is an important parameter to set because your thermal sensor will detect atmospheric conditions like humidity (and distance) which can affect the accuracy of your reading.

Distance

Set your distance to accurately reflect the range from your thermal sensor to the target.

The further away the target is, the more atmosphere your thermal camera will pick up along the way to the target which can affect the accuracy of your reading.





Flight

Review safe flight practices and important information you need to know while in flight.

This section covers

Preflight Inspection

Flight Screen

Changing Flight Modes

Launching

Battery Indicator

Flight Skills

Flying at Night and Flying in Precipitation

Returning and Landing

Preflight Inspection



WARNING: Carefully inspect your drone and environment before launching to ensure a safe flight.

- **Inspect the chassis** to ensure it is free of damage.
- **Inspect drone arms** and verify they are fully extended and free of damage.
- **Inspect the battery** and confirm it is securely seated prior to launching. Skydio X10 uses magnets to seat the battery which may attract metallic debris. Ensure the connector pins are free of debris or damage.
- **Clean the camera lenses and time of flight sensor** with a clean microfiber cloth. Cameras should be dust and smudge-free before flight.
- **Fan out the propellers and inspect** to verify they are firmly attached and properly seated in the motors and spin freely. Propellers should be free of cracks or damage. Do NOT fly with damaged propellers.
- **Inspect the sensor package** before powering on and ensure it moves freely and is not damaged, and **remove the Gimbal Stabilizer Clip** before flying.
- **Ensure all USB-C and microSD card seals are secured** over the ports.
- **If using an attachment, inspect the connections** and confirm cables are fully seated and thumb screws are secure. Loose or partially connected cables may detach during flight and cause damage.
- **Check your surroundings** before launching to ensure a safe environment for flight.
- **Point the controller cover/antennas toward the drone** for maximum wireless performance.
- **Verify batteries and the controller are fully charged** before flying.
- **Check for drone and controller updates** before flying.

Maximum Wireless Performance (Skydio Connect SL)

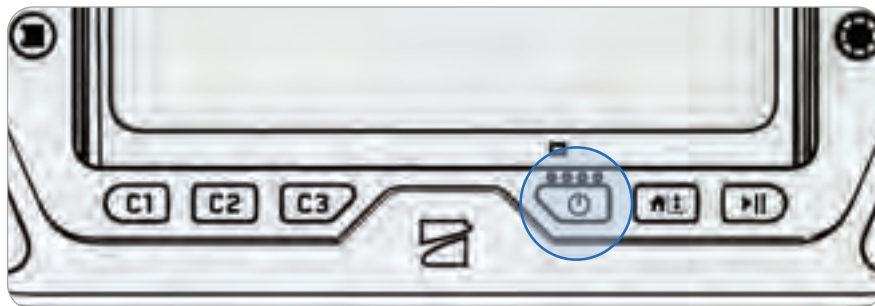
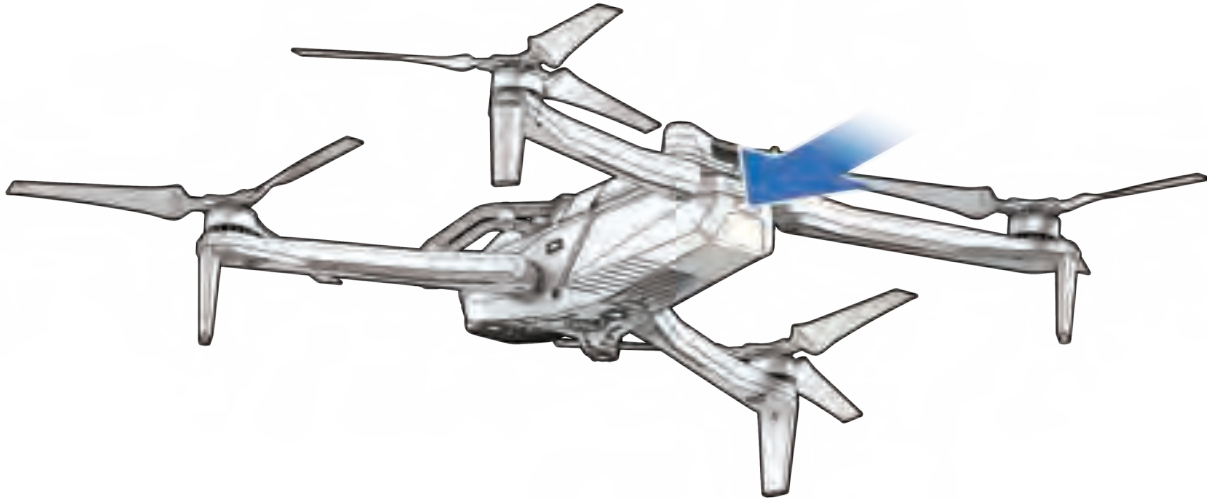
For maximum wireless performance when flying over a direct link, always maintain a direct line of sight between the controller and Skydio X10. Point the controller cover toward the drone, especially when flying at close range high altitude.

Signal strength and maximum control range may be affected when flying in areas with electromagnetic interference.



Connecting Devices

Step 1 - Power on Skydio X10 and the X10 controller



Step 2 - Wait for devices to connect

A drone and controller that were previously paired will automatically connect.



Launching



NOTE: Before your first flight, make sure to set your Return and Lost Connection Behaviors (Global Settings > Return). Scan the QR Code for more information about Return and Lost Connection Behaviors.



Step 1 - Find a clear, safe area to launch

Find a clear, safe area to launch and place your drone on a stable surface. Leave about 10 ft (3 m) of clearance in all directions.

Use the Signal Strength icon to select your connection type (2.4/5 GHz Connect SL or Connect 5G), then select **Fly Now**.



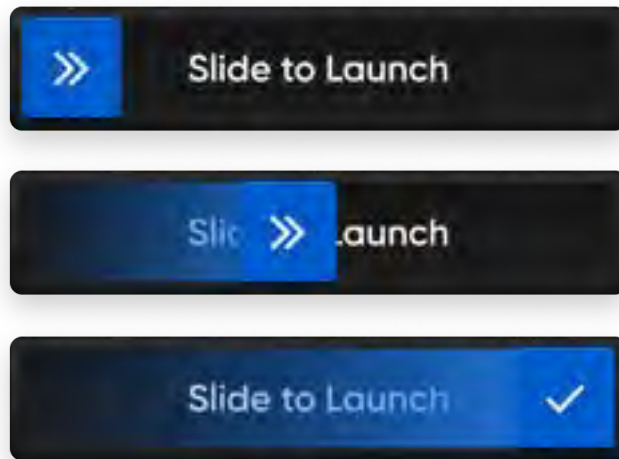
WARNING: Skydio X10 is capable of launching with as little as ~6 ft (2 m) of vertical clearance. However, this capability is intended for constrained environments and should not replace the standard 10 ft (3 m) of recommended clearance in all directions. Operating in tighter spaces may increase the risk of collision or serious bodily injury or damage during launch.

Step 2 - Launch

Your drone will rotate 360° during launch to calibrate its Inertial Measurement Unit (IMU) and navigation system, climb to 10 ft (3 m), and hover.

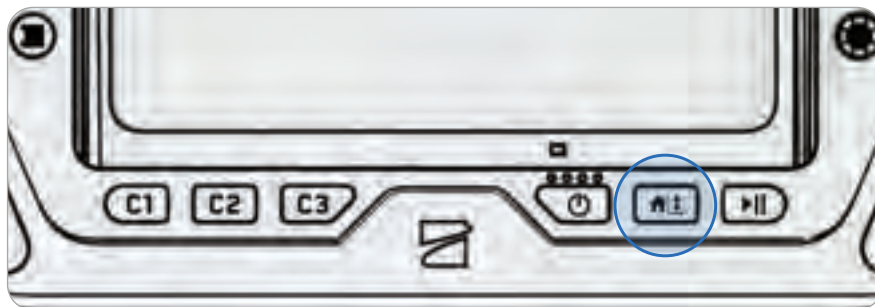
Option 1 - Drag the on-screen slider

The drone will initiate launching when you lift your finger away from the screen.



Option 2 - Press and hold the Launch/Land button on the controller

The drone will initiate launching when you see the on-screen check mark.



NOTE: If obstacle avoidance is enabled prior to launch, it will be active throughout the entire launch sequence.

Step 3 - Acquire GPS Lock

The GPS indicator in the telemetry bar will spin until the drone has acquired GPS Lock and established heading.

When the **Weak GPS Alt Limit** toggle is enabled, your drone will not be able to ascend above 164 ft (50 m) when GPS is weak.

Fly laterally to acquire a GPS lock.



Hand Launching

Launching and landing Skydio X10 from your hand is a quick and convenient way to start or end your flight, particularly if you are not in a clear, level area. For your safety, always use caution when hand launching. This maneuver is dangerous and should not be attempted in unstable environments, such as during high winds or while standing on a moving vehicle or boat.



WARNING: Obstacle avoidance is disabled when the drone is below 10 ft (3 m) during landing. Exercise extreme care to avoid injury or damage. Do not touch spinning propellers.

Step 1 - Ensure you have clearance above and in front of you

Step 2 - Hold the drone away from you (sensor package facing away from your body)

- Lightly grip the battery
- Keep the drone level, still, and at arm's length from your body
- Your fingers should be below the Skydio X10 chassis and away from the propellers at all times
- Ensure your hand steady

Step 3 - Launch

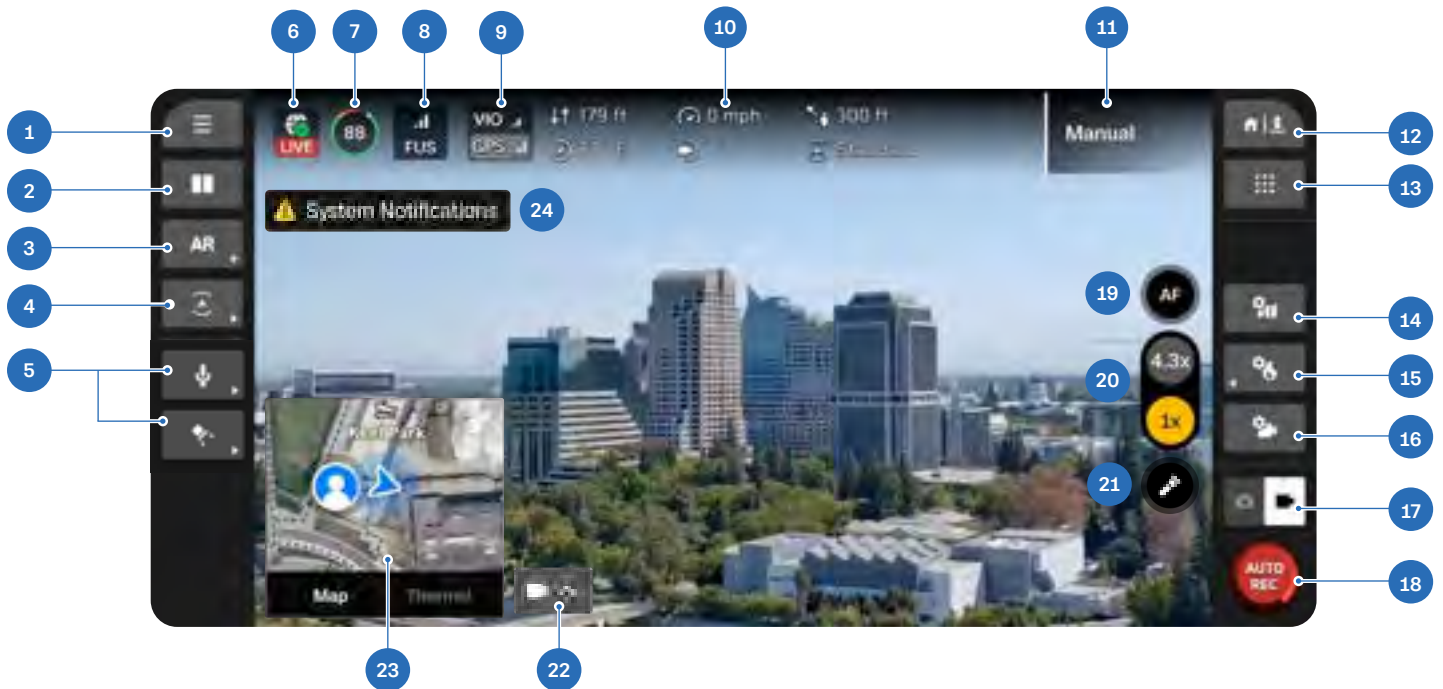
Press and hold the Launch Button on the X10 Controller. Slowly relax your grip as Skydio X10 launches.

- Do not push or throw the drone up in the air
- Keep your hand still - the drone will slide off your palm and take flight on its own



TIP: Quick Launch allows you to use the battery power button to hand launch your Skydio X10 so that you do not have to balance the controller in one hand and your drone in the other. Press the battery button four times to initiate the launch.

Flight Screen



- | | |
|--|--|
| 1. Global Settings | 13. Flight Skills |
| 2. Display Layout | 14. Map Settings |
| 3. AR Quick Actions | 15. Thermal Settings |
| 4. Obstacle Avoidance Quick Actions | 16. Camera Settings |
| 5. Attachments Quick Actions* | 17. Camera Mode |
| 6. Controller Network Indicator | 18. Shutter |
| 7. Drone Battery | 19. Focus Control |
| 8. Signal Strength (Connect Fusion, SL, 5G) | 20. Zoom |
| 9. Navigation Health Indicator | 21. Flashlight On/Off (VT300-L/V100-L Only) |
| 10. Telemetry (customizable) | 22. View Selector |
| 11. Active Flight Skill | 23. Picture-in-Picture (PiP) |
| 12. Return/Land | 24. Notifications |

**Only appears when you are currently using an attachment.*

Controller Network Indicator

This icon at the top of the screen indicates whether or not the Skydio X10 Controller has network connection (WiFi, cellular, or ethernet).

Green check with a LIVE status indicates a healthy network connection and an active stream with ReadyLink



Green check indicates a healthy network connection



Red X indicates no network connection



Battery Indicator

The Battery Indicator **dynamically updates** during flight based on your altitude and distance from the return location. Monitor the indicator to understand how much battery is:

- Available for flight (green)
- Required to return (yellow)
- Required to land (red)

The battery on Skydio X10 has the ability to warm itself in cold environments. For more information, read the Maintenance section.

Green indicates the battery capacity for nominal flight before the time limit required to safely return and land.

- Decreases as battery capacity diminishes
- Adapts based on your altitude and distance from the return location

Yellow indicates how much battery is required to safely return.

- Adapts based on your altitude and distance from the return location

Red indicates how much battery is required to land.

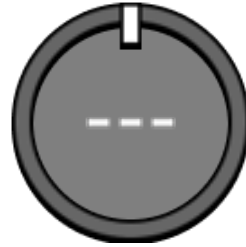
- Adapts based on your altitude and distance from the return location



The **lightning bolt** indicates the battery is connected and charging.



Three **dashes** indicate the battery is disconnected.



Navigation Health Indicator



The Navigation Health Indicator (previously called VIO/GPS Indicator) provides insight into the:

- Navigation source of the drone
- Health of the drone positioning systems
- Reliability of the navigation source
- Current positioning system that is actively being used (highlighted)

This indicator helps you quickly assess whether the drone is relying on Visual Navigation (VIO) or GPS and take action if navigation quality degrades.

- **VIO** - Visual Inertial Odometry. This is the visual navigation system of the drone.
- **GPS** - Global Positioning System

If GPS is lost, Skydio X10 maintains continuous navigation using VIO, which operates alongside GPS and can fully take over all navigation functions.

The drone can maintain stable VIO at altitudes up to 984 ft (300 m).



CAUTION: Monitor your GPS and VIO health in the telemetry bar. If both VIO and GPS become unreliable, your drone will enter Attitude Mode.

The source with a light gray background is the primary navigation source.

White status bars show whether or not VIO/GPS is stable or degraded.

- **One white bar** indicates a degraded state that may be close to failure, meaning you must fly with caution and ensure the other navigation source is in a healthy state
- **Zero bars** means the navigation source has failed and is relying completely on the backup source

Yellow compass indicates GPS is awaiting heading.

- If GPS heading hasn't converged yet, you will receive a notification prompting you to move the drone laterally to establish heading

Yellow highlight means your active navigation source is weak and your backup source is unavailable.

- If the highlighted source fails, the drone will enter **Attitude Mode** and you will have to manually fly



Flight Skills

Skydio offers a range of manual and autonomous controls called **Flight Skills**. Select your desired skill and Skydio X10 will intelligently fly itself to assist with the task at hand.

By default, you will start in the Manual Flight Skill, which provides a traditional flying experience.

Base Skydio Flight Skills

Manual	Traditional flying experience. Obstacle avoidance settings will persist when flying manually, allowing Skydio X10 to route itself around obstacles, modifying any commands that could potentially cause a collision. Fly using Control Mode 1, Mode 2 (default), or Mode 3.
Orbit Point	Rotate around a user-selected point-of-interest in either a clockwise or counter-clockwise direction, keeping the point in the center of the frame. Set a GPS position on a map.
Track in Place	Initiate tracking a person or vehicle and Skydio X10 will hover in a fixed position, as if affixed to a virtual tripod. The drone will automatically yaw and adjust the sensor package pitch to maintain tracking. Tracking in both Color and Thermal is available.
Waypoints	Create and execute multi-waypoint GPS missions, preflight or postflight.



INFO: [Visit our website](#) for more information on additional Flight Skills available for purchase.

Changing Flight Skills

Step 1 - Select the Flight Skills icon

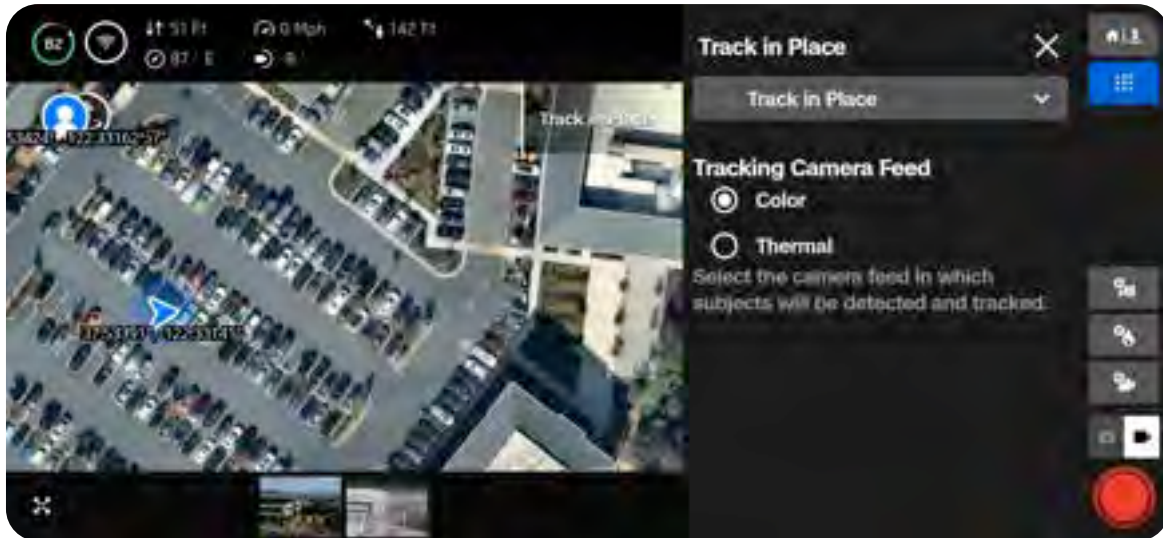


Step 2 - Select your Flight Skill



Step 3 - Adjust settings (optional)

Each skill may have its own adjustable settings which only affect the selected skill. If the selected skill has adjustable settings, they will appear below the skill name within the Flight Skills menu.



NOTE: To begin flying with the selected Flight Skill, you must first exit this menu.

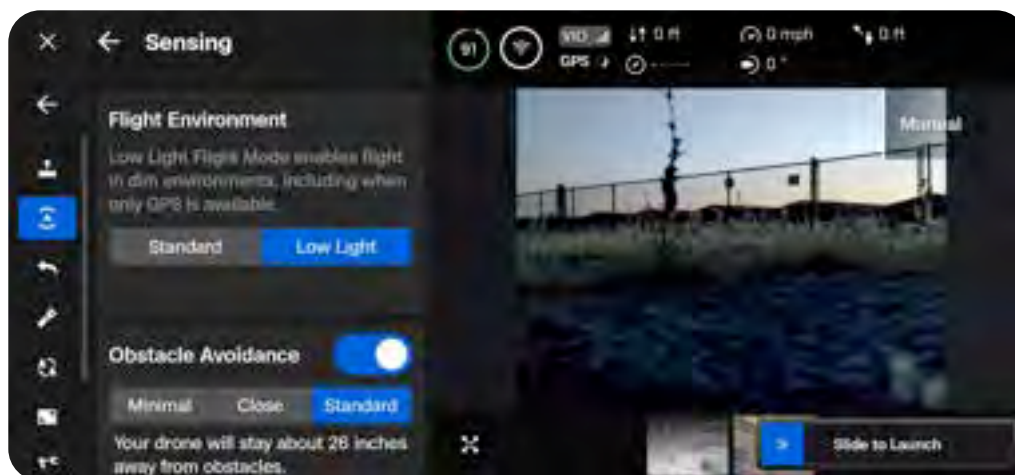
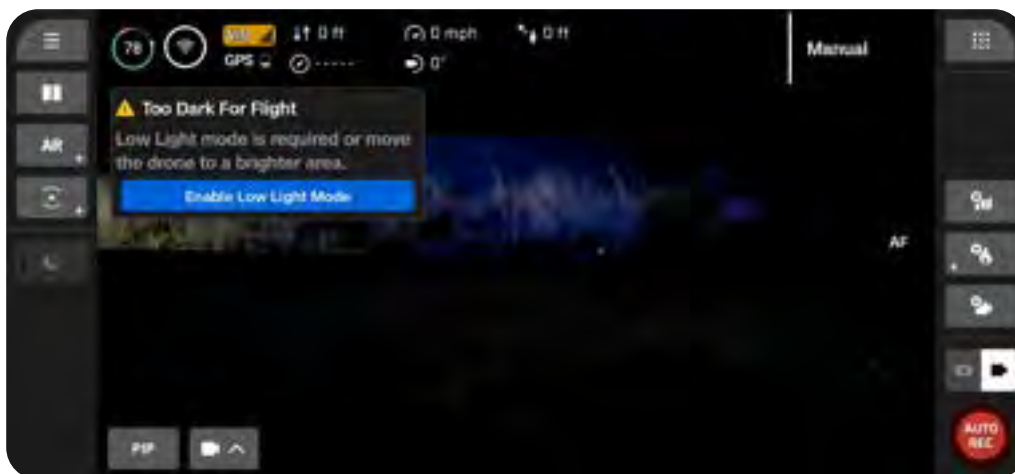
Flying at Night

Flying at Night without the NightSense Attachment

When flying at night or in low light conditions without NightSense, Skydio X10 will primarily use GPS to navigate and obstacle avoidance will be disabled.

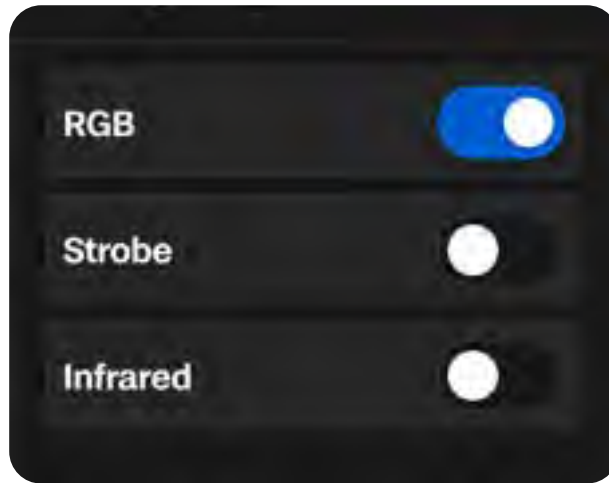
Step 1 - Enable Low Light mode

You will see an on-screen notification to enable Low Light mode if there is insufficient light. You may also select the Global Settings icon, select Sensing, then select Low Light.



Step 2 - Configure your lighting settings

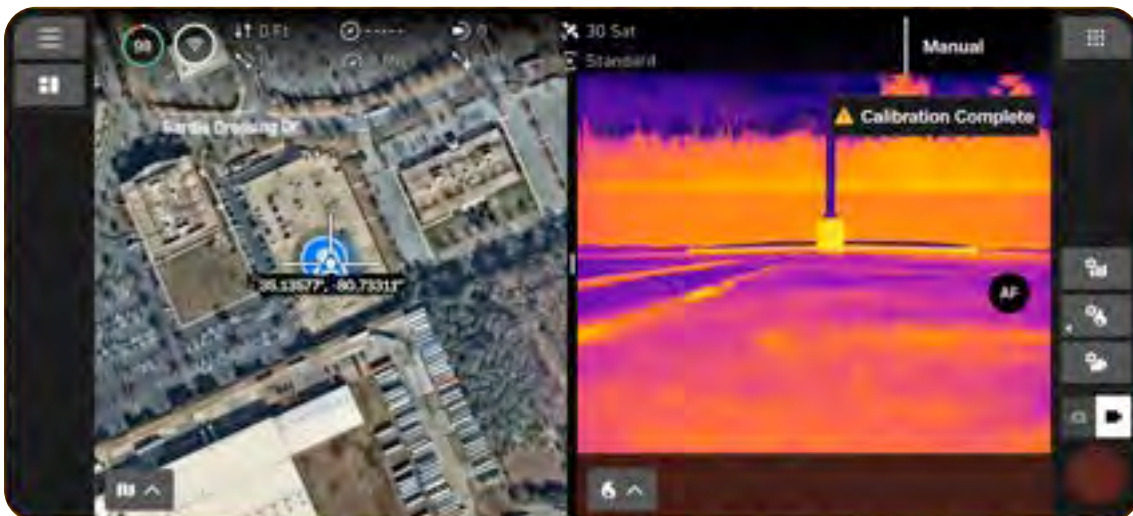
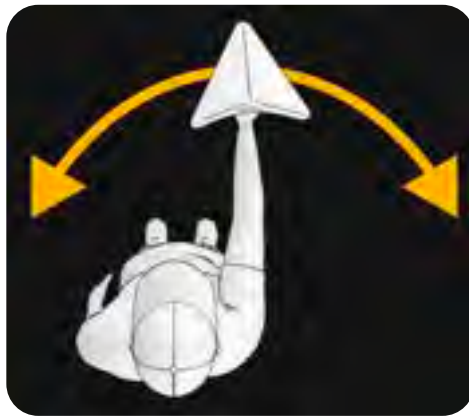
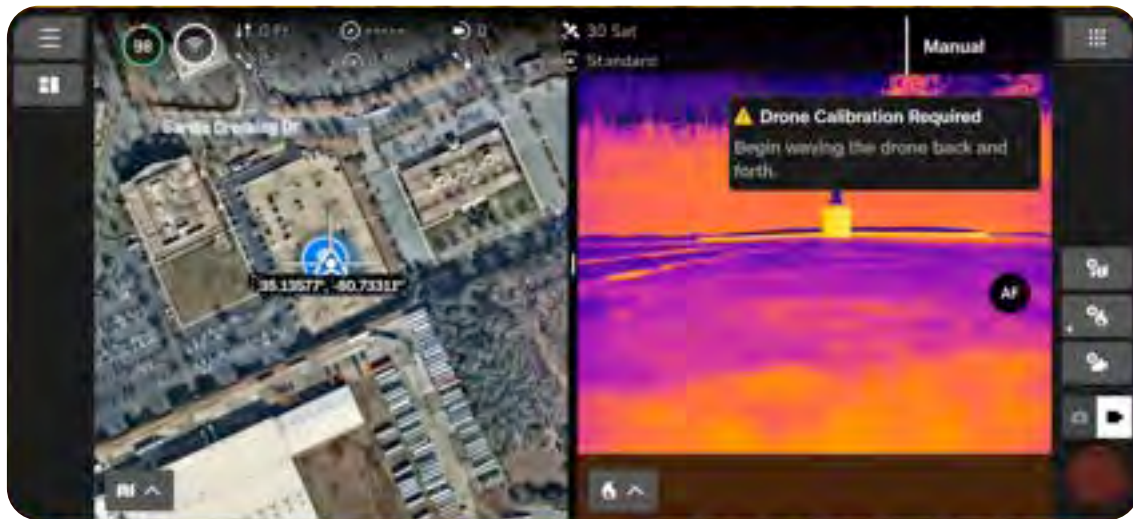
Improve visibility by enabling infrared or visible strobe lights. Select Lighting then toggle on RGB (default navigation lights), Infrared, or Strobe lights. Infrared and Strobe lights cannot be on at the same time.



WARNING: Do not use artificial lighting to brighten the launch or landing zone during night flights. Enhancing illumination in a localized area, such as using vehicle headlights, can cause the drone to misinterpret ambient lighting conditions. This may result in a sudden transition to Attitude Mode after launching, increasing the risk of drift and loss of control. For safe night operations, follow best practices for low-light flight and ensure the drone is prepared for consistent lighting conditions throughout the mission.

Step 3 - Calibrate your drone

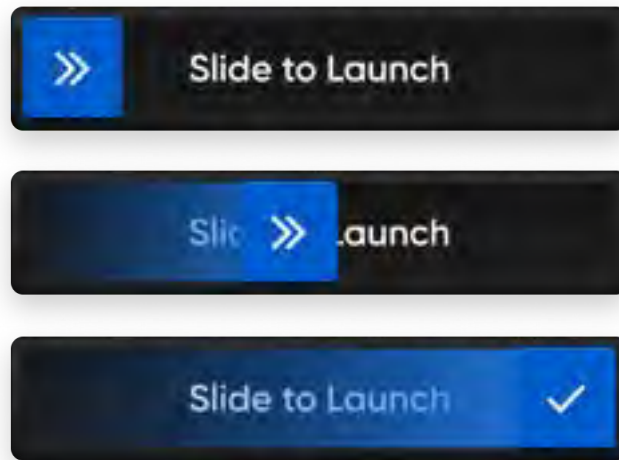
Exit the Global Settings menu, then select Fly Now. Hold your drone from the bottom with the camera facing away from your body and wave from side to side to calibrate. You will see an on-screen message when calibration is complete.



Flight

Step 4 - Launch Skydio X10

Your drone will rotate 360° during launch to calibrate its Inertial Measurement Unit (IMU) climb to 10 ft (3 m), and hover. Obstacle avoidance will be disabled.



WARNING: Obstacle avoidance is disabled in Low Light mode without NightSense. Fly with extreme caution!

Flying at Night with NightSense

The Skydio NightSense attachments and add-on software allow you to leverage Skydio X10 visual navigation and obstacle avoidance capabilities even when flying at night.

The set of two NightSense attachments cast a light above and below your drone. This light illuminates the area, unlocking Skydio visual navigation and obstacle avoidance when flying at night and in low-light conditions.



KEY WARNINGS:

- *After prolonged use of the NightSense attachments, they may be hot to the touch and could burn your hand. After landing, wait for your attachments to cool down before handling.*
- *Do not stare directly into your NightSense modules at close range. NightSense attachments, both Visible and Infrared, may cause eye damage if held closer than an arm's reach for 30 seconds or more.*
- *Due to potential burn risk and eye damage, Skydio does not recommend Hand Landing your drone while using NightSense.*

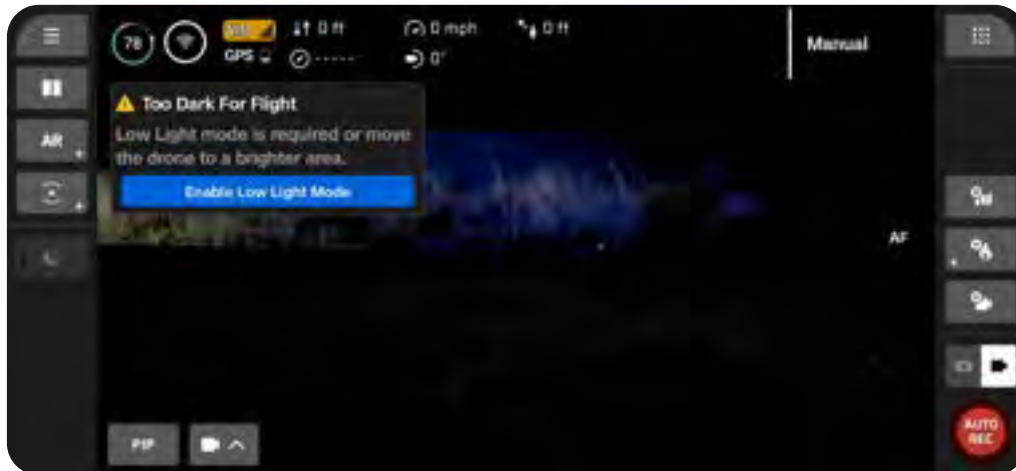
Step 1 - Install your attachments

Visit [Getting Started with NightSense for Skydio X10](#) for step-by-step installation instructions.



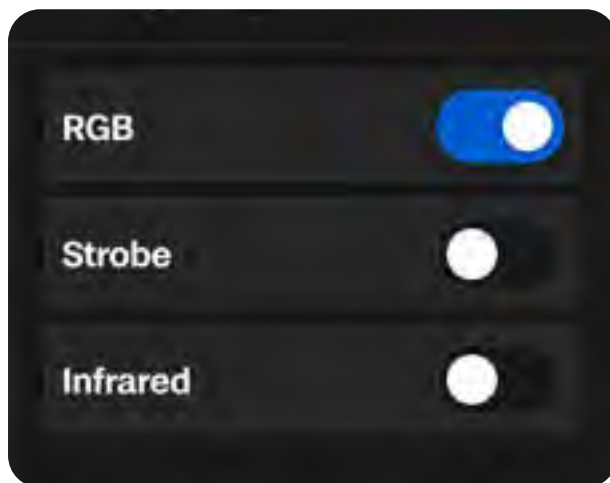
Step 2 - Enable Low Light mode

You will see an on-screen notification to enable Low Light mode if there is insufficient light. You may also select the Global Settings icon, select Sensing, then select Low Light.



Step 2 - Configure your lighting settings

Improve visibility by enabling infrared or visible strobe lights. Select Lighting then toggle on RGB (default navigation lights), Infrared, or Strobe lights. Infrared and Strobe lights cannot be on at the same time.



WARNING: Do not use artificial lighting to brighten the launch or landing zone during night flights. Enhancing illumination in a localized area, such as using vehicle headlights, can cause the drone to misinterpret ambient lighting conditions. This may result in a sudden transition to Attitude Mode after launching, increasing the risk of drift and loss of control. For safe night operations, follow best practices for low-light flight and ensure the drone is prepared for consistent lighting conditions throughout the mission.

Step 3 - Enable NightSense

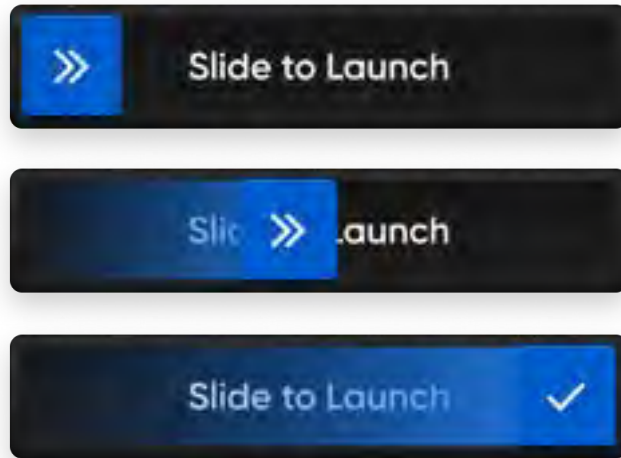
By default, NightSense will automatically turn on when entering Low Light mode. Use the left sidebar quick action to subsequently turn NightSense on or off. The visible or infrared lights from the NightSense attachments will illuminate the area around the drone.



Flight

Step 4 - Launch Skydio X10

Your drone will rotate 360° during launch to calibrate its Inertial Measurement Unit (IMU) climb to 10 ft (3 m), and hover.



NOTE: Skydio X10 is limited to a max speed of 18 mph (8 m/s) with NightSense on.



INFO: For more information about flying with NightSense, including configuring settings visit: [Getting Started with NightSense for Skydio X10](#).

Inflight Safety Considerations

Obstacle Avoidance

- **NightSense ON:** Obstacle avoidance is active when NightSense is on. Enable the AR Depth View (via the **AR Quick Action**) to visually display where Skydio X10 detects obstacles in the environment, especially when using Infrared NightSense attachments, to assist with situational awareness.
- **NightSense OFF:** When in Low Light mode, obstacle avoidance is disabled. Take extra caution when piloting the drone to avoid obstacles and stay clear of people.

GPS Signal

- **NightSense ON:** When NightSense is on, visual navigation (VIO) is the primary navigation method, however if flying at high altitudes the drone will rely on GPS. Monitor your VIO and GPS health status inflight; if both VIO and GPS quality degrade the drone will enter Attitude Mode.
- **NightSense OFF:** Maintaining a strong GPS signal is paramount when operating X10 at night. If Skydio X10 loses GPS while in Low Light mode, and there is not enough ambient light for VIO, it will enter Attitude Mode.

Visibility

Improve visibility by enabling infrared or visible strobe lights. Skydio X10 strobe lights meet the FAA requirement of being visible at a distance of 3 statute miles.

Return Behavior

Review the return behavior height setting in the Global Settings menu.

NightSense OFF: Skydio X10 does not avoid obstacles when in Low Light mode and NightSense off, so you may want to set the drone's return height such that it will be above any potential obstacles.

Landing

When landing, use the controller joystick to descend down to 15 feet (4.6 meters), when you are ready to land, press and hold the LAND button on the screen or the controller. Do not hand launch or hand land at night.



WARNING: Due to potential burn risk and eye damage, Skydio does not recommend Hand Landing your drone while using NightSense.

Flying in Precipitation

Skydio X10 is IP55 rated and able to fly in light to moderate precipitation when **obstacle avoidance** is disabled. Skydio will primarily use GPS to navigate, so it is critical to have a strong GPS signal when flying in precipitation.



CAUTION: *Flight in icing and/or lightning conditions is not supported and may result in the loss of your drone.*

Quick Reference

Flying in precipitation during the day

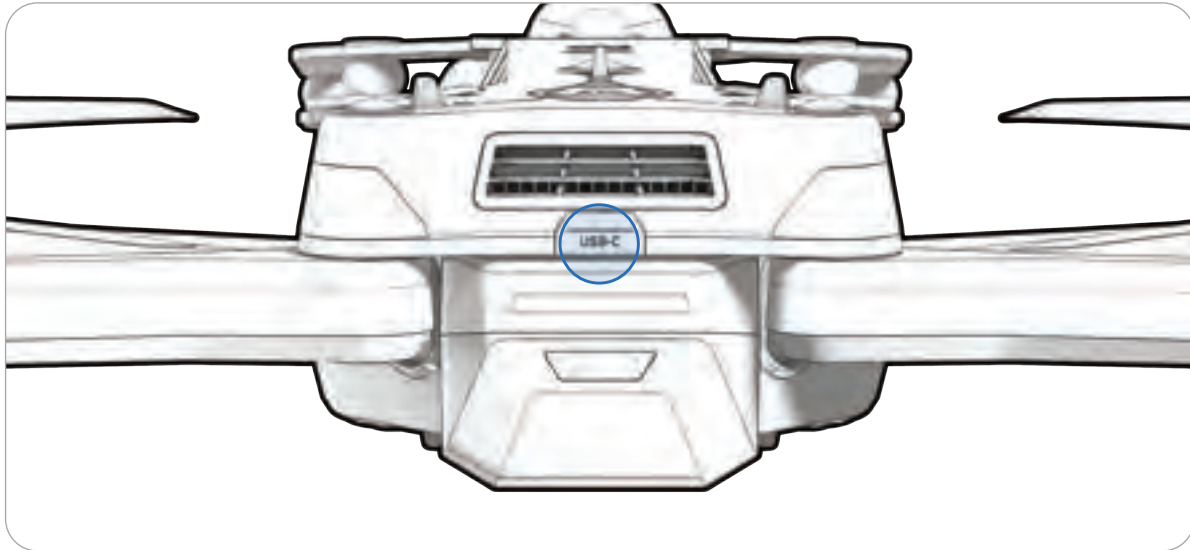
- Select Standard as your Flight Environment
- Disable obstacle avoidance via the quick action or the settings menu
- Only fly with a strong GPS signal

Flying in precipitation at night

- Select Low Light as your Flight Environment
- Disable obstacle avoidance, disable NightSense (if you have NightSense attachments installed)
- Only fly with a strong GPS signal

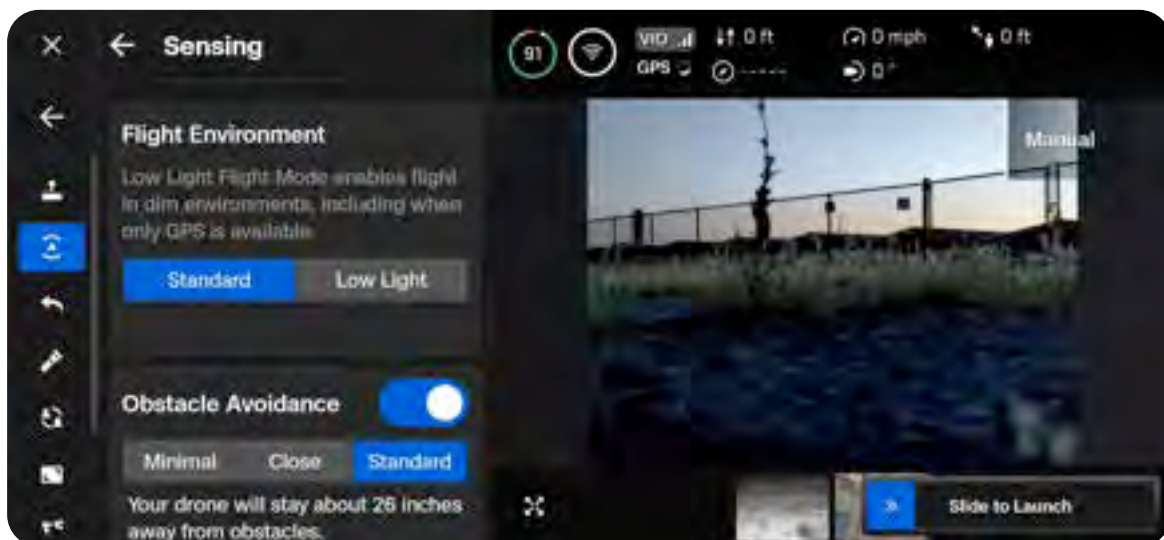
Preflight

Step 1 - Ensure all rubber seals on the drone are securely closed



Step 2 - Select your Flight Environment

Select the Global Settings icon, select Sensing, then select Standard (flying in precipitation during the day) or Low Light (flying in precipitation at night).



Step 3 - Calibrate your drone (if in Low Light mode)

Exit the Global Settings menu, then select Fly Now. Turn off NightSense (if on). Hold your drone from the bottom with the camera facing away from your body and wave from side to side in a straight line to calibrate. You will see an on-screen message when calibration is complete.



Flight

Step 4 - Disable obstacle avoidance

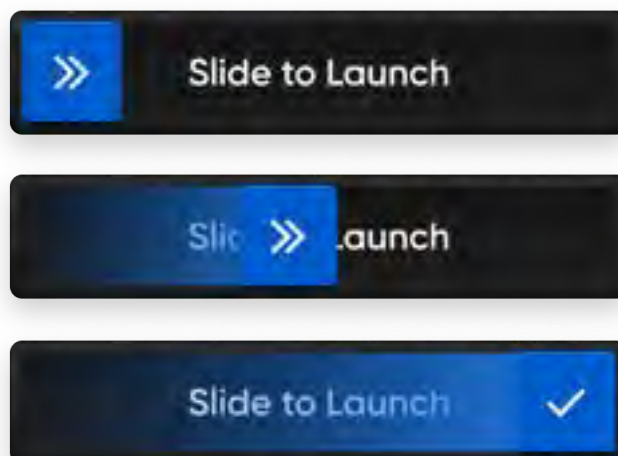
Disable obstacle avoidance from the quick action menu on the Flight Screen. Turn off NightSense if on.



NOTE: If it begins to precipitate during a flight, disable obstacle avoidance inflight.

Step 5 - Launch Skydio X10

Your drone will rotate 360° during launch to calibrate its Inertial Measurement Unit (IMU) climb to 10 ft (3 m), and hover. Obstacle avoidance will be disabled.



Step 5 - Lock controller touchscreen (optional)

To prevent touchscreen interference from precipitation, you have the ability to lock all touchscreen inputs. While holding the back button, press the D-pad down to lock or unlock the screen.



NOTE: You will not be able to make on-screen selections while touchscreen inputs are locked. Controller buttons are still functional and can be customized to perform specific actions (Controls > Input Mapping).

Inflight Safety Considerations

Obstacle Avoidance

When flying in precipitation, obstacle avoidance must be disabled. Take extra caution when piloting the drone to avoid obstacles and stay clear of people.

GPS Signal

Maintaining a strong GPS signal is paramount when flying in precipitation, since Skydio X10 is navigating primarily using GPS. If Skydio X10 loses GPS, it will enter Attitude Mode.

Visibility

Improve visibility by enabling infrared or visible strobe lights. Skydio X10 visible strobe lights meet the FAA requirement of being visible at a distance of 3 statute miles.

Wireless Range

Moisture in the air may significantly reduce wireless range.

Return Behavior

Review the return behavior height setting in the Global Settings menu. Obstacle avoidance must be disabled when flying in precipitation, so you may want to set the drone's return height such that it will be above any potential obstacles.

Landing

When landing, use the controller joystick to descend down to 15 feet (4.6 meters), when you are ready to land, press and hold the LAND button on the screen or the controller. Do not hand launch or hand land in precipitation.

Postflight

After flight operations in precipitation, follow all postflight steps before stowing your drone to ensure it is properly maintained and does not sustain any water damage.

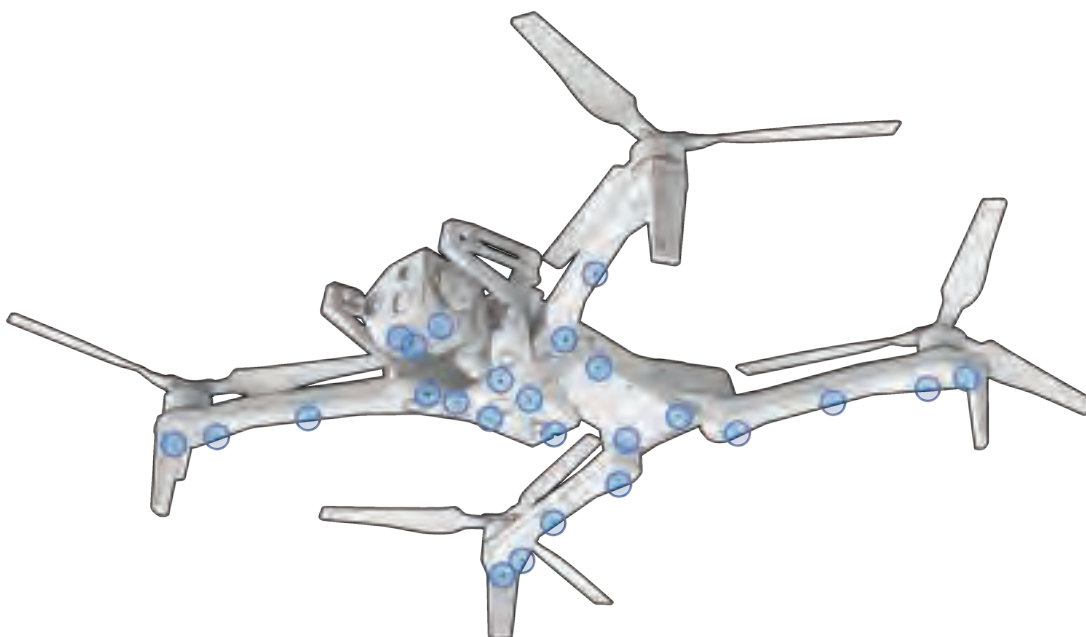


CAUTION: Do not stow Skydio X10 while wet.

Step 1 - Power off Skydio X10

Step 2 - Allow water to drain

Install the sensor package lock to hold the sensor package in place. Rotate the drone to allow water to drip out of all egress areas.



NOTE: Skydio is IP55 rated and able to fly in light to moderate precipitation. It is expected for water to enter different areas of the drone and draining postflight is normal. Sensitive components are protected.

Step 3 - Remove the battery

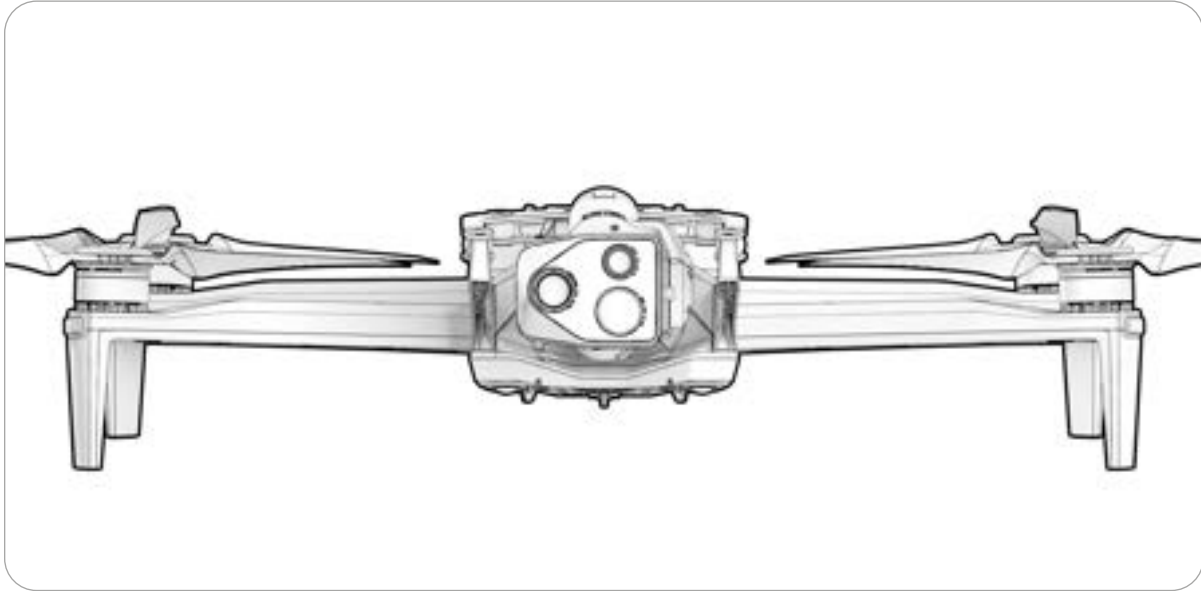


Step 4 - Wipe camera lenses clean

Use a microfiber cleaning cloth to wipe the lenses clean and prevent dried water spots from forming.

Step 5 - Air dry for a minimum of 12 hours

Leave the drone to air dry in a ventilated, temp-controlled environment with the arms deployed in an upright position. Do not open any seals, including the USB-C charge port, until after the drone is dry. Do not remove the sensor package or any attachments while the drone is wet.

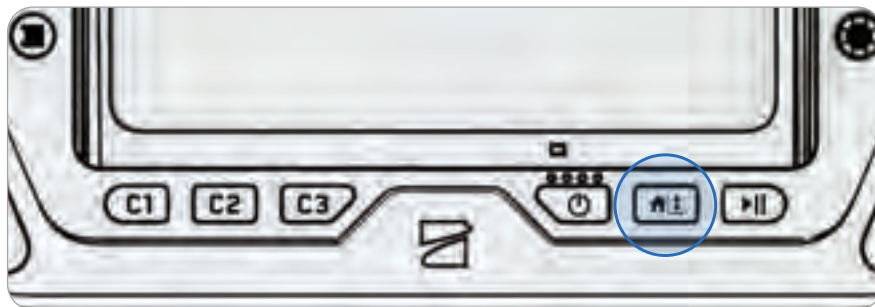


Returning and Landing



Scan for more information about Return and Lost Connection Behaviors.

Step 1 - Select the Return/Land button in the top right of your screen or on the controller



Step 2 - Choose your return location or land in place



Home

Returns to a Home Point previously set on the map (GPS required)



Launch

Returns to the Launch Point



Pilot

Returns to the location of the Skydio X10 Controller



You have three options when landing in place:

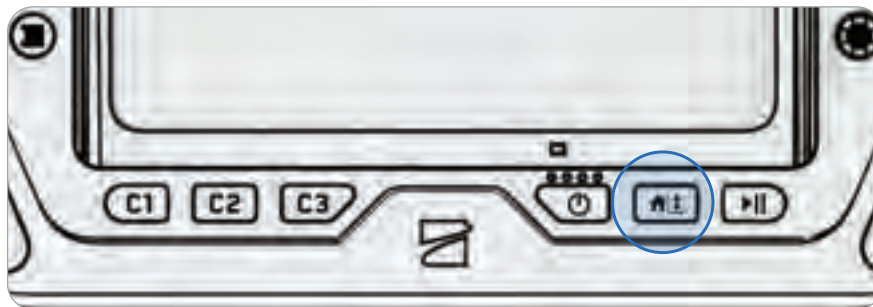
Option 1 - Select and drag the on-screen slider

Landing begins when you lift your finger away from the screen.



Option 2 - Press and hold the Return/Land button on the controller

Landing begins when you see the on-screen check mark.



Option 3 - Press and hold the Return/Land button on-screen

Landing begins when you see the on-screen check mark.



WARNING: Obstacle avoidance is disabled when the drone is below 10 ft (3 m) during landing. Exercise extreme care to avoid injury or damage. Do not touch spinning propellers.

Hand Landing

Landing Skydio X10 in your hand is a quick and convenient way to start or end your flight, particularly if you are not in a clear, level area. For your safety, always use caution when hand launching or landing.



WARNING: Do not hand launch or hand land during windy days, when flying at night, or extreme environmental conditions as serious injury and/or damage may occur.

Step 1 - Position Skydio X10 above a clear area so that you can move underneath it

Ensure the drone is facing away from you and extend your arm away from your body.

- The sensor package should face away from your body
- The back of the chassis and the battery are closest to you

Step 2 - Initiate landing

Skydio X10 will descend vertically with full obstacle avoidance until it is 10 ft (3 m) above the ground.

Once your drone is below 10 ft (3 m), the lights on the drone turn yellow to indicate obstacle avoidance is disabled for the remainder of the landing.



WARNING: Do not attempt to grab or catch Skydio X10 without initiating a landing, the motors will continue to spin at full speed and may cause severe injury. Do not attempt to hand land until the lights turn yellow. Attempting to hand land while obstacle avoidance is active will cause it to attempt to avoid your hand and may result in Skydio impacting yourself or another nearby object.

Step 3 - Lightly grab the drone by the battery from underneath as it touches down on your open palm

Once the battery has made contact with your palm, keep your hand steady until the propellers completely stop spinning.

Do not attempt a hand landing if:

- There are high winds present.
- Skydio X10 is not stable in flight for any reason.
- Skydio X10 is performing an emergency landing after an accident or crash.
- You are in an area where you do not have stable footing.
- You are on a moving vehicle or boat.
- Skydio X10 is in Low Light mode, with or without NightSense.

Hot Sensor Package



WARNING: After prolonged use of the flashlight, your sensor package may be hot to the touch and could present a serious burn risk. After landing, wait for your sensor package to cool down before handling.

If you have the VT300-L or V100-L sensor package and use the flashlight for an extended period of time, the sensor package may be hot to the touch after flight and could present a serious burn risk. If the flashlight is used for the duration of a flight in ambient temperatures of 100°F (38°C), the sensor package can reach temperatures up to 142°C (61°C).

Monitor your controller for any safety notifications and be careful when handling the sensor package after flight.





Postflight

Learn how to access media and properly store your drone.

This section covers

Offloading Media

Logging Out of the X10 Controller

Uploading Skydio Support Logs

Stowing Skydio X10

Offloading Media

Select the **Media** menu within **Global Settings** to view photos, videos and scans from your recent flights.

- Select an image or video to view
- Press and hold on a thumbnail to select multiple or delete

If you captured photos using **Interval**, all photos captured will appear as a single stack. Selecting the stack will allow you to scroll through individual images one by one.

Only standard color and thermal JPGs will display in the Media menu. To access your DNG or RJPG files, you must transfer the files from your drone.



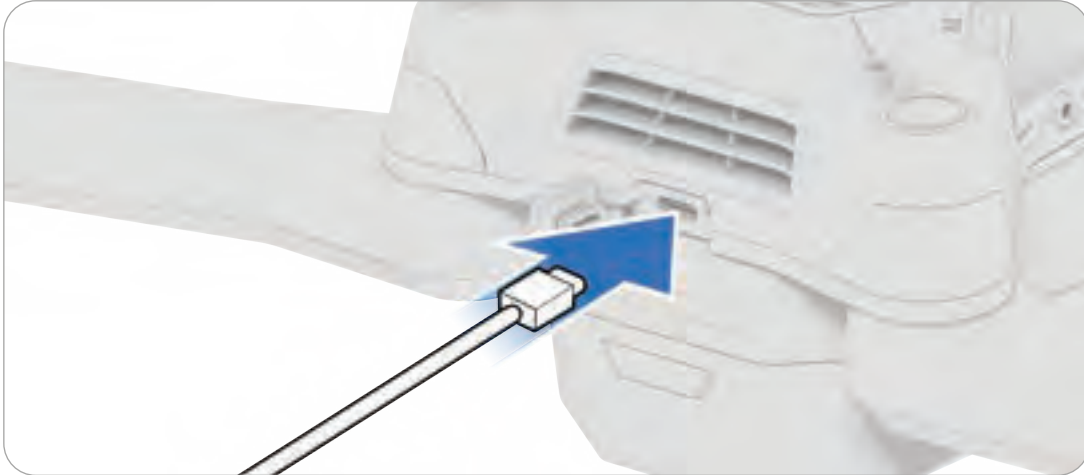
NOTE: *Media is not accessible inflight.*

Transferring Media

Step 1 - Power on Skydio X10

Step 2 - Connect X10 to your computer

Insert the provided USB-C cable into the USB-C charging port on the back of your drone.



Step 3 - Import your media

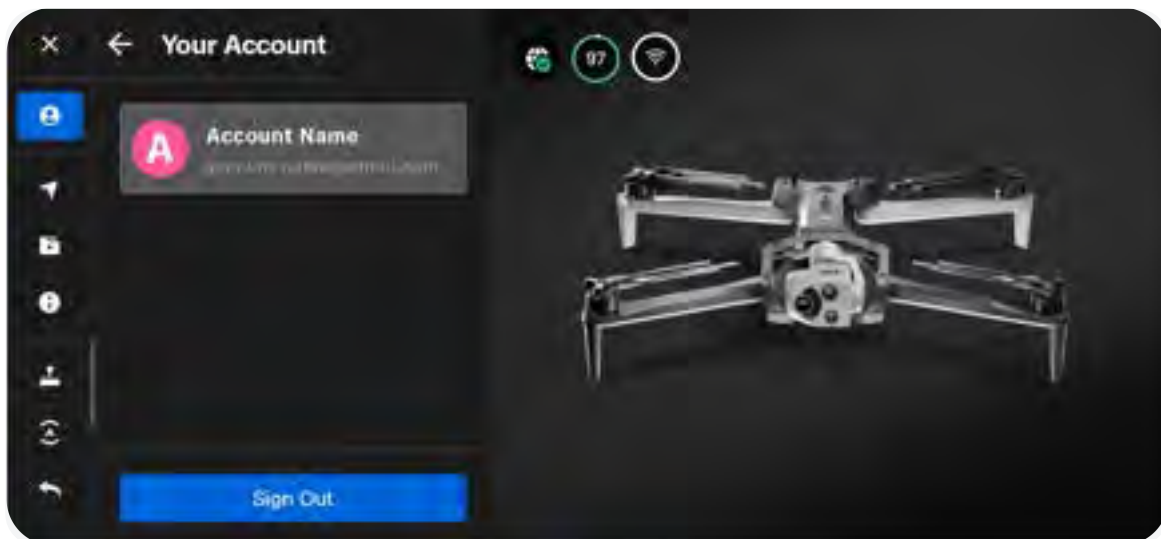
If you are using an Apple product, use the Photos app or the Image Capture app to transfer your files.

Logging Out of the Controller

Easily log out and switch between users of the same organization without factory resetting the controller. Your Skydio X10 and X10 Controller will stay paired between sign-ins.

Settings, media, and support logs are shared and persist between user accounts and logouts.

Navigate to **Global Settings** and select the profile icon in the top left to view the current account that is signed in, and select **Sign Out** to switch users.



NOTE: Always review your settings when logging into a shared controller.

Sending Skydio Support Logs

Uploading your flight logs allows our support team to troubleshoot any issues or questions you may have.

Do not reformat or factory reset your Skydio drone prior to contacting our support team
Skydio will never review your videos or data without your permission.



WARNING: *Powering off the drone immediately after a landing or a crash will result in missing data in the flight logs. Always allow your drone to complete postflight operations before powering off. Removing, swapping, or altering file names on the media card will impact the availability of the flight logs.*



CAUTION: *To maintain operational security and prevent conflicts with controller functionality, never sign in to personal or third-party accounts (such as Google Drive, Gmail, OneDrive, or other cloud services) on the X10 Controller. If you need to import maps or export logs, use the supported workflows provided by Skydio.*

Postflight

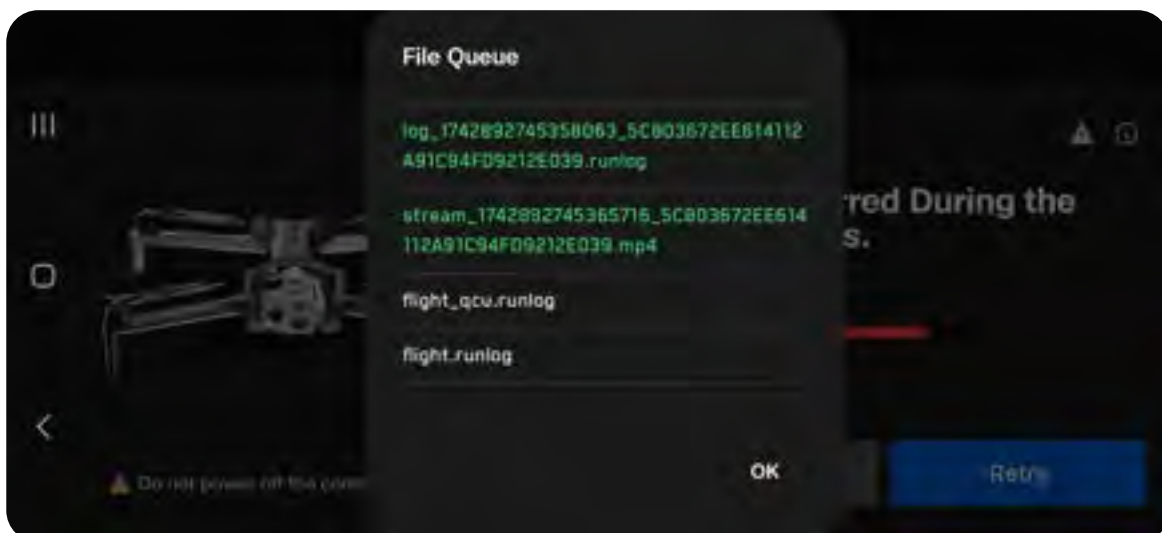
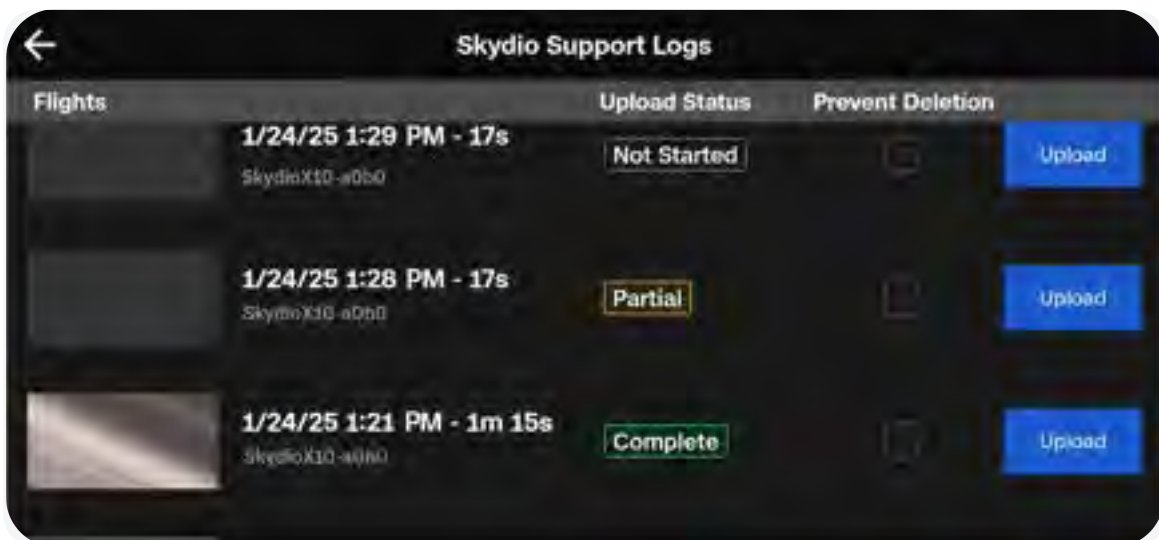
When uploading to Skydio Support, monitor the upload progress of an individual flight directly from the Skydio Support Logs menu:

Individual Flight Log Status

- **Not Started** – Log has not been uploaded
- **Partial** – Some data was uploaded, but the log is incomplete; when you retry, only the missing data will upload
- **Complete** – The log has been fully uploaded

Individual File Status

- **Green** – Upload successful
- **White** – Upload failed; select the error icon (triangle) in the top right to view the files



Uploading Support Logs



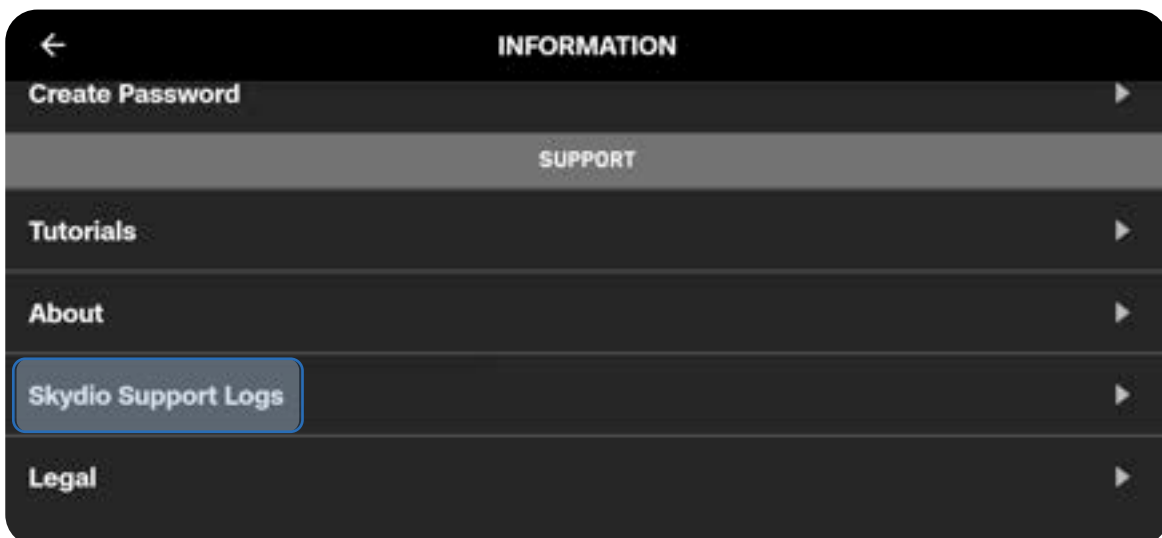
NOTE: The transport of large support logs are not supported over 5G cellular.

Step 1 - Power on Skydio X10 and the X10 Controller

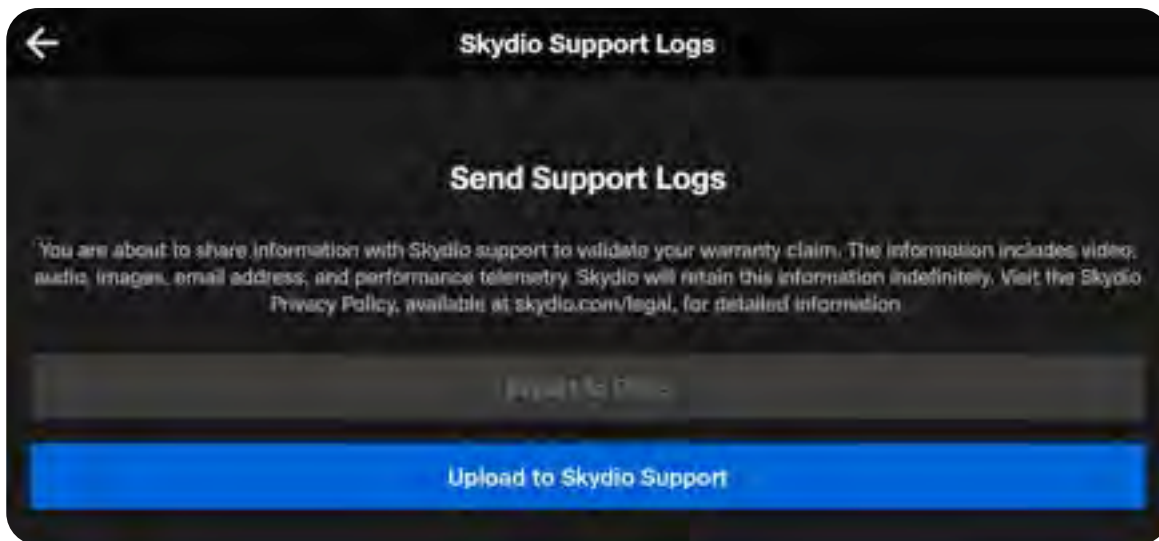
Ensure your X10 Controller is fully charged.

Step 2 - Navigate to Global Settings > Information

Step 3 - Select Skydio Support Logs



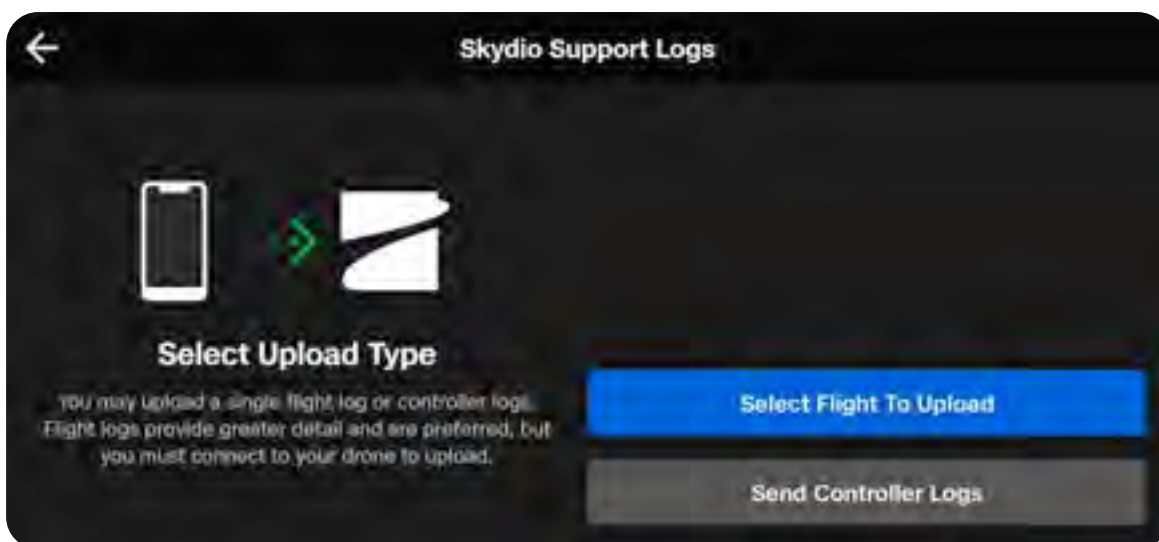
Step 4 - Select Upload to Skydio Support



Step 5 - Choose between a flight or controller logs

Select Flight to Upload includes all logs from a specified flight. This option will show you the history of all flights, organized by date and time. Select the individual flight you wish to upload.

Send Controller Logs uploads all flights saved on the controller from all flight history. This option allows you to sync logs whether or not you are connected to the drone.

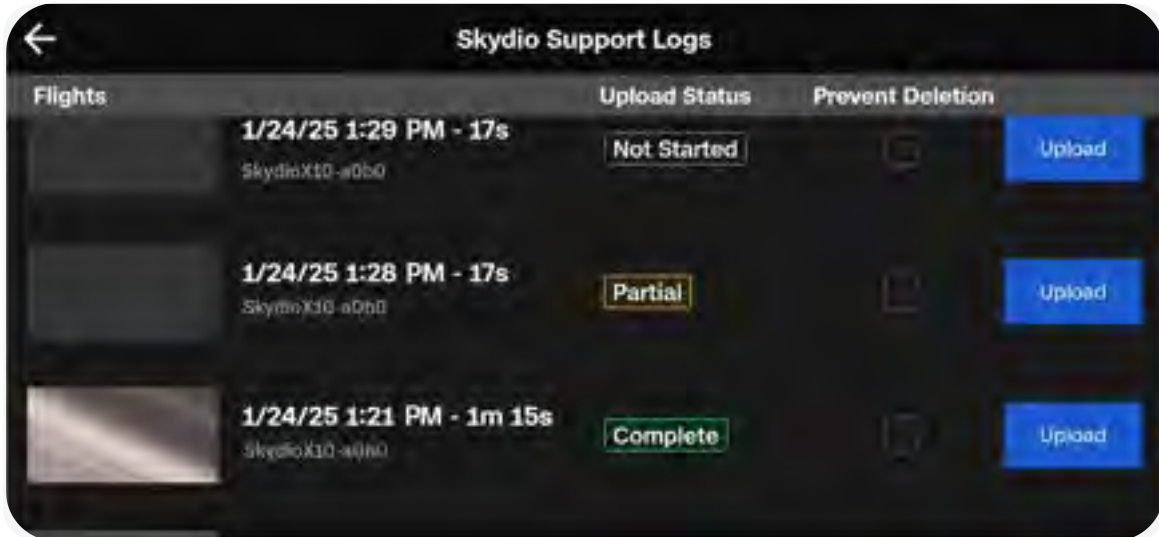


Postflight

Step 6 - Upload Flight Logs

If you selected **Select Flight to Upload** in the previous step, select which flight you wish to send to support.

Wait as your upload completes. You will see a summary after the upload finishes.



Uploading Debug Logs

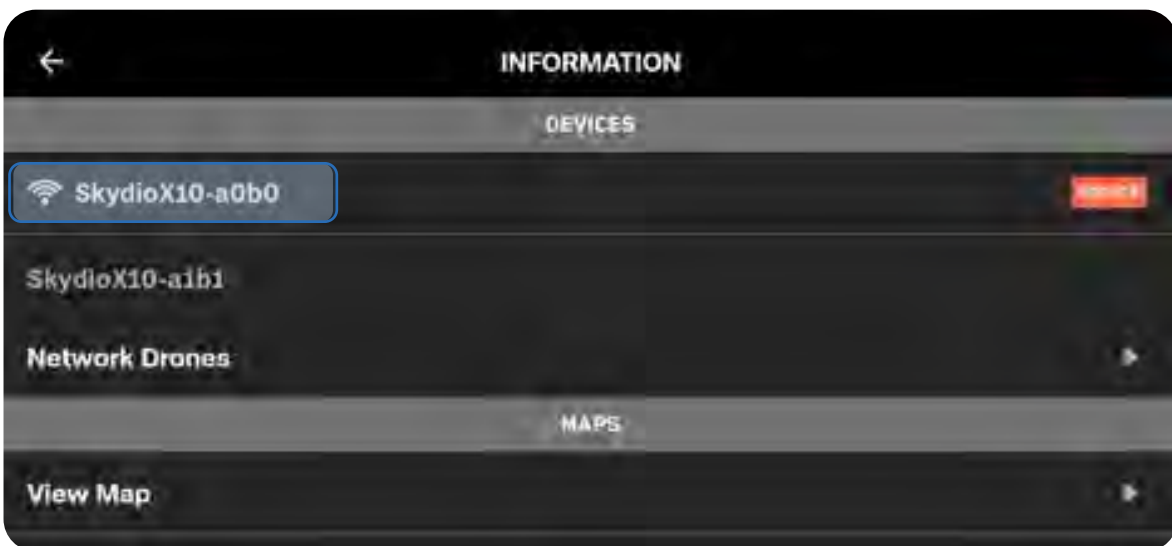
Depending on the issue, Skydio Support may ask you to upload debug logs.

Step 1 - Power on Skydio X10 and the X10 Controller

Ensure your X10 Controller is fully charged.

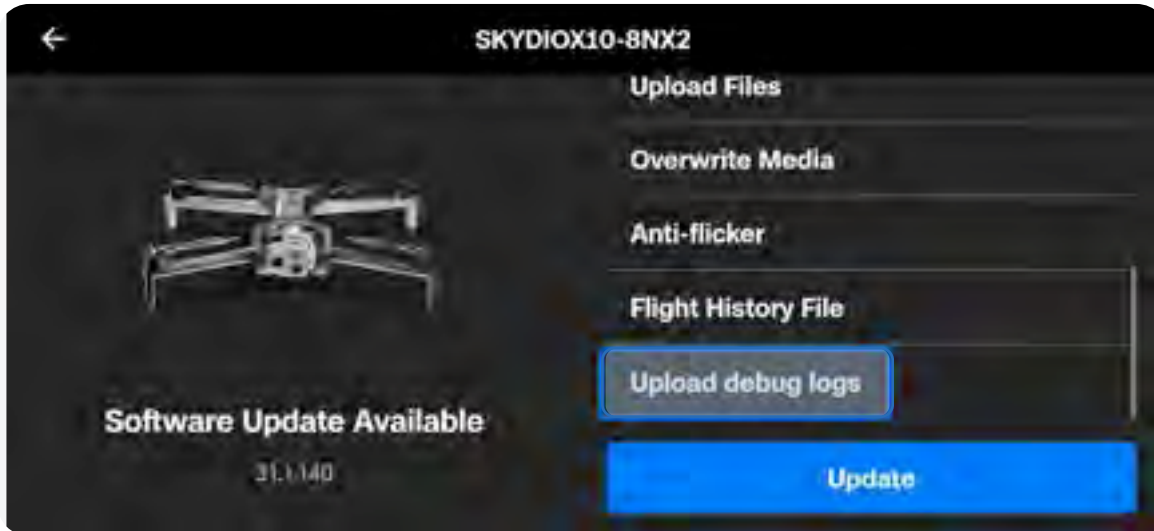
Step 2 - Navigate to Global Settings > Information

Step 3 - Select the name of your X10 under Devices



Step 4 - Select Upload debug logs

It may take a few minutes for debug logs to upload to Skydio.



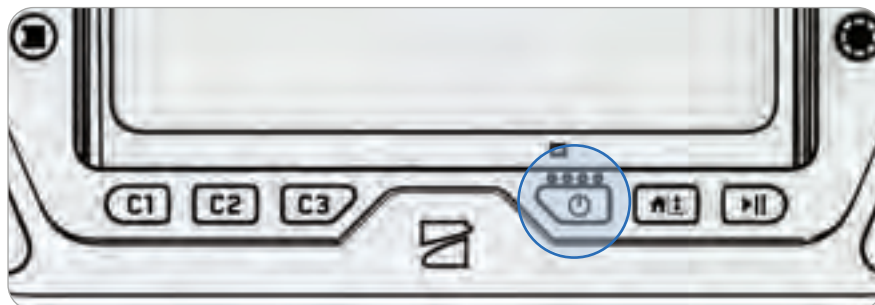
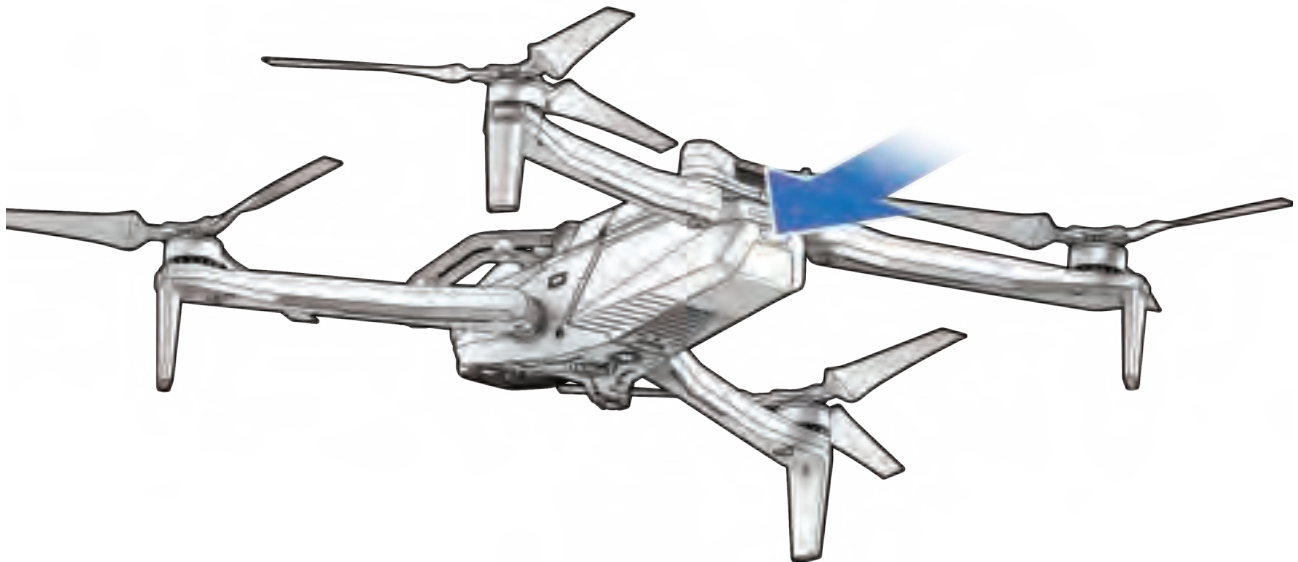
Stowing Skydio X10

Step 1 - Wait for postflight tasks to complete

If the battery is low while performing a longer postflight task, such as Onboard Modeling with Map Capture, ensure the drone is plugged into a power source.

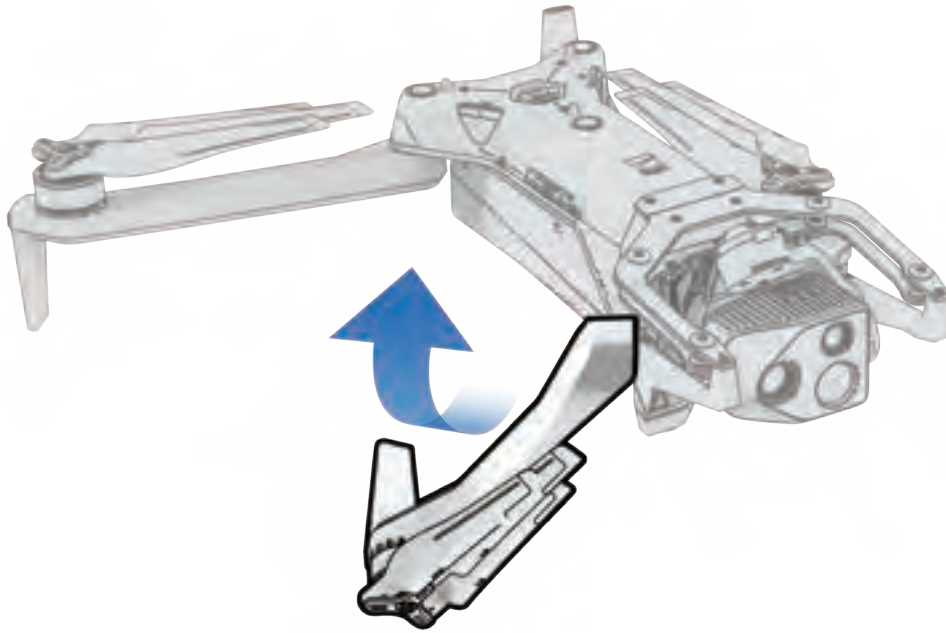
- Powering off or removing the battery during postflight tasks will result in loss of data

Step 2 - Power off Skydio X10 and the X10 controller



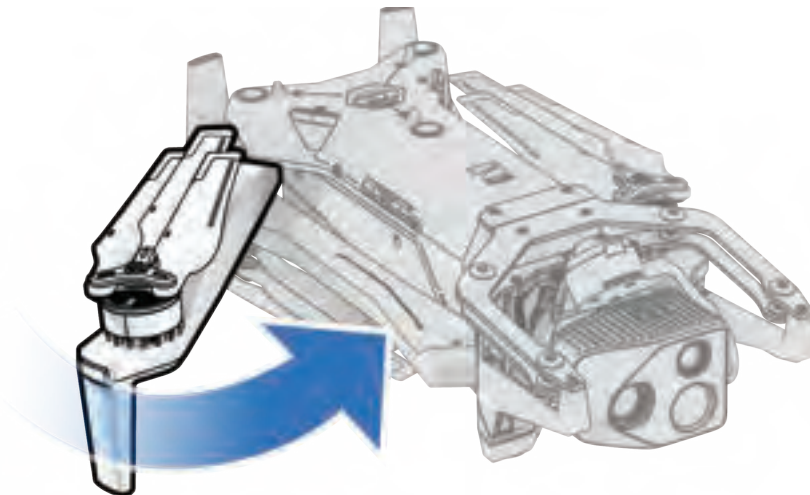
Step 2 - Fold in the front arms

Hold the drone with the sensor package facing away from you. Gently pull the arm toward the back of the drone and rotate until it is tucked into place.



Step 3 - Fold in the rear arms

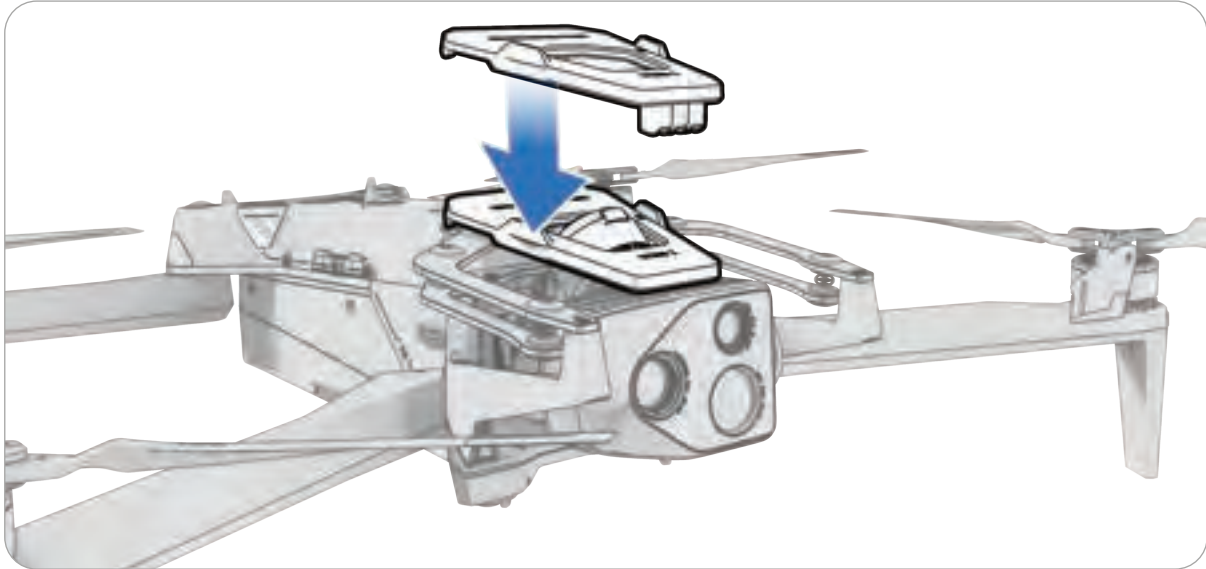
Push laterally toward the chassis. Gently continue until you meet resistance.



Postflight

Step 4 - Attach the sensor package lock

Hold the sensor package and gently reattach the lock to the top of your drone.



NOTE: Refer to the *Flying in Precipitation* section above for steps on how to properly store your drone after flight in precipitation.



Contingency Behaviors

Learn about the behaviors during events such as lost connection or low battery. Always monitor Skydio Flight Deck for in-app alerts.

This section covers

Low Battery

Lost Connection

Lost GPS

Reduced Performance State

Controller Overheating

Emergency Landing and Attitude Mode

Flight Termination

Contingency Behaviors



WARNING: While flying, always monitor Skydio Flight Deck for in-app alerts relating to battery levels, signal quality and other inflight notifications.



Scan for more information about Contingency Behaviors.



Before flying, ensure you have configured return settings, such as an automatic return on low battery, in the Return menu (**Global Settings > Return**).

Low Battery

The **Battery Indicator dynamically updates** during flight based on your altitude and distance from the return location. Monitor the indicator while flying to understand how much battery is:

- Available for flight
- Required for return
- Required to land

Skydio X10 will assess the altitude and distance from the Launch or Home Point and alert you when it is time to return and land. It is **strongly recommended you initiate a return or land at this time**.

Contingency Behaviors



WARNING: You may choose to cancel an automated landing or delay your return when the Return Battery capacity has been reached at your own risk – you are solely responsible for the potential loss of the drone and/or any serious bodily harm and property damage that may result.

Skydio X10

You will receive a series of notifications alerting you about the current status of the drone battery:

Return Battery Capacity Reached

The drone only has enough battery to return and land. This is based on conditions such as weather or attachment usage. **If you delay your return, you may not have sufficient battery power to reach the return point.**

- An audible beep will play over the X10 Controller
- If the drone is within 32 ft (10 m) of the return location, it will begin returning without displaying this notification
- If you selected **Ask to Return (default)** in your **Return** settings, you will be prompted to choose **Return** or **Keep Flying**
- If you selected **Auto Return on Low Battery** in your **Return** settings, the drone will begin returning automatically when this notification appears; if you select **OK**, it dismisses the notification but does **not** cancel the return
- If no Home Point is set, the drone will return to the Launch Point



Ask to Return (default)



Auto Return on Low Battery

Land Soon

The drone will continue flying, but this notification alerts you one minute before the drone will begin to land in place. Autonomous actions, such as Waypoints, 3D Scan, or Subject Tracking, are disabled. **You may choose to continue flying, however, it is strongly recommended that you fly to a safe location and land soon.**



Drone Battery Low

The battery level is too low to continue flying and the drone will begin looking for a clear and flat landing surface. The drone will land in place and may make minor adjustments to land in the flattest spot. You have the ability to make small, nudging adjustments. **It will not return to any designated return location.**

- An audible beep will play over the X10 Controller
- Mute this using the sound icon within the notification, or turn down all notifications using the Sound menu (Global Settings > Sound)



Contingency Behaviors

Land Immediately

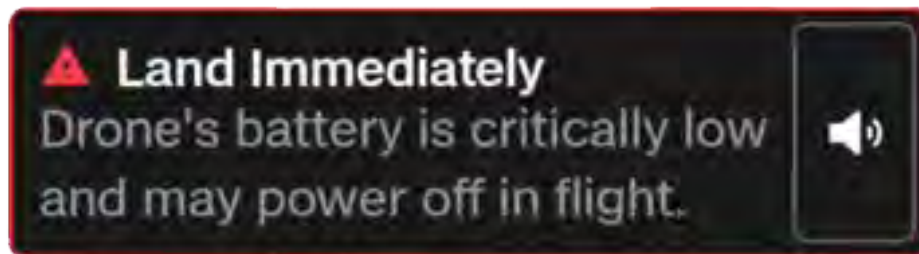
The battery level is critically low and the drone may power off mid-flight without warning. This warning appears if you chose to cancel the earlier **Drone Battery Low** automatic landing and continue flying.



WARNING: Continuing to fly greatly increases the risk of a crash, loss of the drone, or serious bodily harm or injury.

If the drone is descending toward an unsafe area (e.g., water, traffic), you may cancel the landing, move to a safer location, and then resume the landing process.

- A series of audible beeps will play over the X10 Controller
- Mute this using the sound icon within the notification, or turn down all notifications using the Sound menu (Global Settings > Sound)



Skydio X10 Controller

To help ensure a safe and uninterrupted flight, always launch with sufficient controller battery. If your battery drops below 20%, you will be notified on the Gate screen.

You will receive a series of notifications alerting you about the current status of the controller battery:

Controller Battery Low

When the X10 Controller battery reaches 10%, a **Controller Battery Low** notification will appear.

Controller Battery Critical

When the X10 Controller battery reaches 5%, a **Controller Battery Critical** notification will appear.

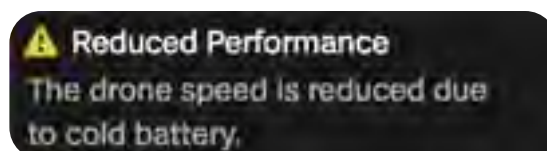
- If the controller powers off, the drone will follow the **Lost Connection** behaviors you have configured
- An audible beep will play over the X10 Controller
- Mute this using the sound icon within the notification, or turn down all notifications using the Sound menu (Global Settings > Sound)

Reduced Performance State due to Battery Capability Limitation

At times of reduced battery performance, the drone will reduce its top speed and acceleration in order to enhance safety:

- Max ascent speed: 4.5 mph (2 m/s)
- Max ground speed: 27 mph (12 m/s)

Scenarios where the drone will have reduced battery performance include extreme cold, extremely low battery level, overheating, and cell imbalance.



Lost Connection



WARNING: Before flying, ensure you have set your Lost Connection Return Behaviors. This is a critical step that ensures your drone returns safely and lands in an accessible location.

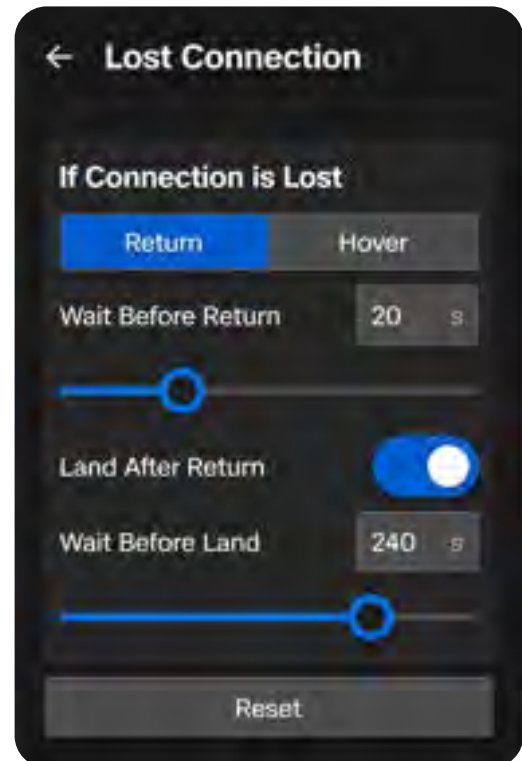
If connection is lost, Skydio X10 will default to the **Lost Connection** settings. Select between **Return** (default) and **Hover** upon lost connection.

Return (default)

Wait Before Return - set the amount of time you want Skydio X10 to wait before it initiates a return flight, allowing time to reconnect

Land After Return - when enabled, your drone will return, hover for a specified amount of time, then land.

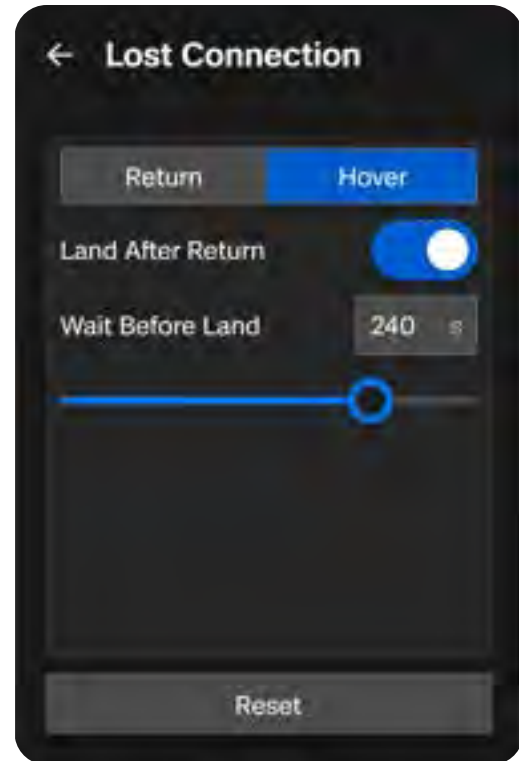
Wait Before Land - the amount of time between 0 to 300 seconds (default is 240 seconds) that you want your drone to wait above the landing location before landing. This setting is only enabled when Land After Return is toggled on.



Hover

Land After Hover - when enabled, Skydio X10 will hover for a specified amount of time, then use visual navigation to find a safe area to land.

Wait Before Land - the amount of time between 0 to 300 seconds (default is 240 seconds) that you want your drone to wait before landing. This setting is only enabled when Land After Hover is toggled on.



Skydio X10 will continue hovering as it tries to regain connection. If it fails to reconnect and reaches low battery:

- If you have an automatic return set, your drone will return to either the Launch Point or Home Point (if set)
- If you do not have an automatic return set, your drone will use visual navigation to find a safe area to land
- If you do not have strong VIO (e.g. you are flying in Low Light mode without NightSense), your drone will be unable to use visual navigation and will descend vertically and land

Lost GPS

If Skydio X10 loses GPS signal, the drone will continue flying using the vision system. Actions that require GPS will be disabled.

If visual navigation (VIO) is also unavailable, the drone will enter Attitude Mode, a mode of flying that relies on the drone barometer to maintain altitude. By default, if there are no joystick inputs for 5 seconds, the drone will begin emergency landing.

Controller Overheating

If the Skydio X10 Controller reaches critical temperatures and overheats in flight, it will shut down and lose connection to the drone, triggering the X10 Lost Connection behavior.

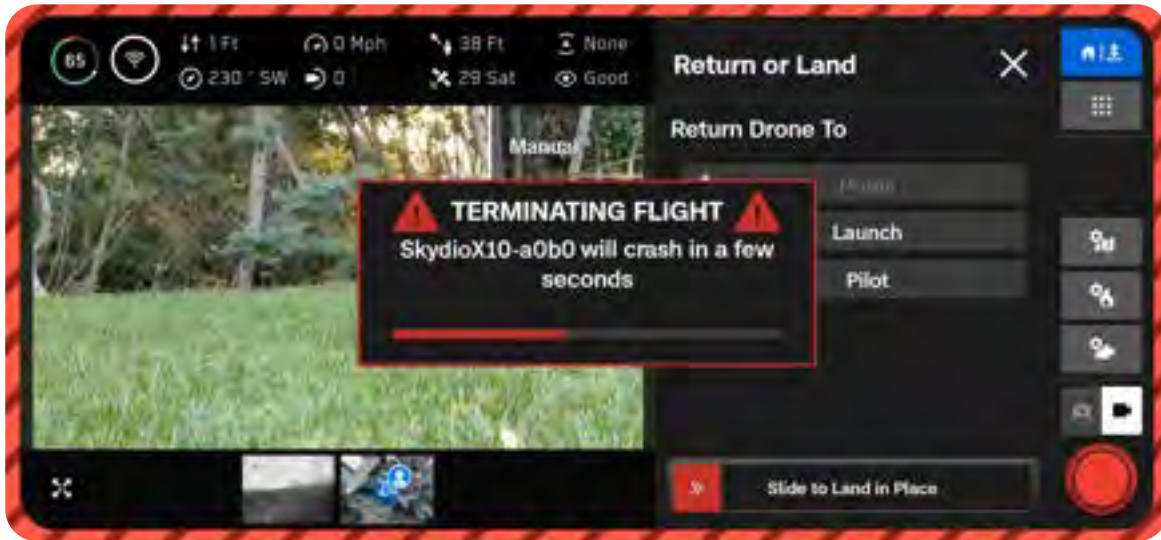


To mitigate overheating and reduce the risk of lost connection during flight, the controller will issue two alerts and provide guidance to the operator.

Alert	Operator Action
Controller is Heating Up	Take mitigation actions: - Move the controller screen out of direct sunlight - Move to a shaded or cooler area if possible - Continue flight - this alert is cancellable
Controller Too Hot to Operate	Take immediate action to land: - Launch will be prevented - If the drone is in flight the signal between the controller and the X10 will be lost and your drone will default to the operator-defined lost connection settings - This alert is non-cancellable

Emergency Landing and Attitude Mode

If both GPS and the visual navigation system (VIO) become unreliable, Skydio X10 will enter Attitude Mode.



NOTE: Monitor your GPS and VIO health in the telemetry bar. If VIO and GPS both fail, your drone will enter Attitude Mode.

In this mode, the drone will use internal barometer readings to maintain altitude when the throttle joystick is centered.

The drone will drift, in which case you will need to adjust roll and pitch movements to maintain the drone's position. The drone will not automatically hold position or automatically brake when the joysticks are centered.



CAUTION: Obstacle Avoidance is not available in Attitude Mode.

Contingency Behaviors

If the drone regains GPS and/or VIO while in Attitude Mode, it will switch out of Attitude Mode and use whichever navigation system is strongest.

- To have the best chance of recovering VIO, descend below approximately 60 feet above the ground and fly within sight of visual features.
- When GPS becomes available again, a prompt will be displayed. Fly the drone horizontally (either left and right, or back and forth) to regain GPS heading and resume GPS navigation.

Navigate to **Global Settings > Flight Controls > Landing** to configure landing behavior while in attitude mode.

Auto Land in Attitude

Toggled ON (default) - After 5 seconds of inactivity in Attitude Mode (i.e. the joysticks are centered and not touched), Skydio X10 will automatically initiate an emergency landing.

Toggled OFF - The drone will remain in attitude mode under joystick command until GPS/VIO navigation systems are healthy or you land.



After 5 seconds of inactivity in Attitude Mode (i.e. the joysticks are centered and not touched in a neutral position and not engaged), Skydio X10 will automatically initiate an emergency landing and descend autonomously. An alert notification will display that Skydio X10 is initiating an emergency landing.

If you input any joystick command while the drone is emergency landing, it will stop descending landing and you can continue to fly in Attitude Mode.

Low Battery in Attitude Mode

The drone will not return or land automatically at low battery while flying in Attitude Mode. It is your responsibility to monitor battery level and manually fly the drone to a safe landing location and land the drone when the battery is low. When the battery is low and the throttle stick is centered, the drone will descend to remind you that it is time to land.

Lost Connection in Attitude Mode

If you lose connection with the drone while flying in attitude mode, the drone will descend and emergency land in place.

Landing in Attitude Mode

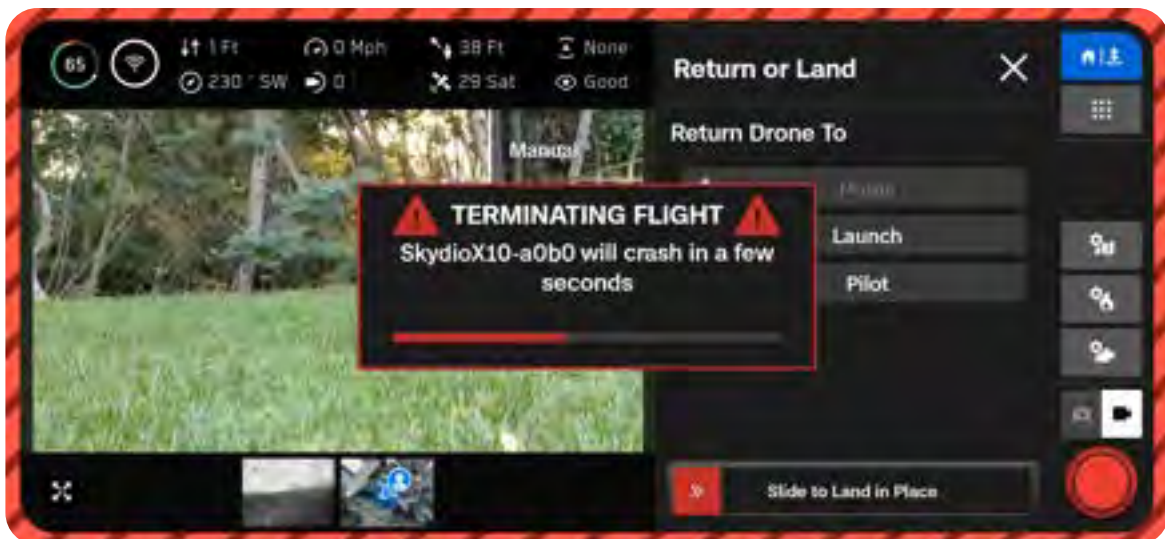
To land the drone in attitude mode, you can press or hold the Land button to autonomously descend and touch down, or you can manually descend and touch down. The drone will automatically disarm and spin down the propellers after a few seconds when it detects that the drone is safely on the ground and the throttle joystick is held in a full down position.

Flight Termination



WARNING: Terminating a flight will cause your drone to crash. Damage resulting from Flight Termination is not covered under warranty and may result in injury or damage. Use only in extreme situations.

In the event of an extreme emergency, you have the option to immediately terminate your flight. **Simultaneously press and hold the C3 button and Launch/Return/Land button for three seconds** while in flight to immediately stop the motors.





Maintenance

Learn how to replace your propellers and best practices for battery and equipment storage.

This section covers

Monitoring Your System in Skydio Cloud

Updating Your System

Replacing Propellers

Cleaning Your System

Storage

Maintenance Schedule

Monitoring Drones and Batteries in Skydio Cloud

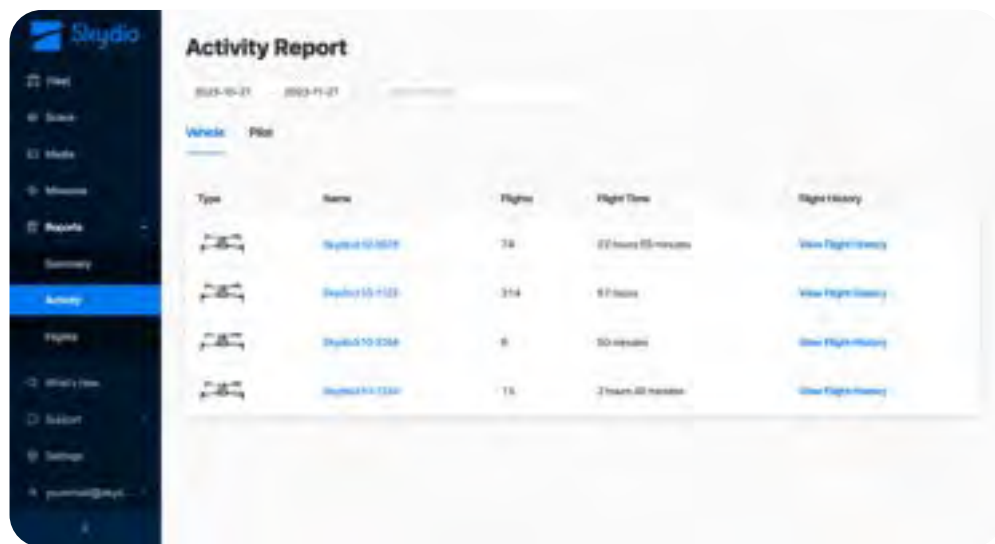
If you've claimed your Skydio X10 drones and batteries in Skydio Cloud, you can track the total number of flights and flight hours for each device.



NOTE: Skydio X10 drones and batteries must be claimed in Skydio Cloud to view this flight data. Visit the Skydio Cloud Setup section of this manual for instructions on how to claim your devices.

Skydio X10

Navigate to Reports > Activity. Here you can view the total number of flights and total flight time for each drone in your fleet.



Type	Name	Flights	Flight Time	Flight History
	Skydio X10-1010	78	22 hours 55 minutes	View Flight History
	Skydio X10-1020	314	87 hours	View Flight History
	Skydio X10-1030	6	50 minutes	View Flight History
	Skydio X10-1040	15	2 hours 30 minutes	View Flight History

Maintenance

Under the Flights tab you can also view a list of all flights, their duration, launch time, and more.

Flights						
2023-10-27		2023-11-21		2023-12-05		Event CDF
Pilot	Vehicle ID	Duration	Launch Time	Battery	More Details	
user1@skydio.com	Skydio X10-1234	8 minutes 4 seconds	11/27/23 10:37 AM	P2089G4043990171	View Flight	Export CSV
user1@skydio.com	Skydio X10-1234	14 minutes 20 seconds	11/27/23 10:20 AM	P2089G4043990201	View Flight	Export CSV
user1@skydio.com	Skydio X10-1234	25 seconds	11/22/23 3:48 PM	P2089G4043990087	View Flight	Export CSV
user1@skydio.com	Skydio X10-1234	1 minutes 7 seconds	11/22/23 3:57 AM	P2089G4043990179	View Flight	Export CSV
user1@skydio.com	Skydio X10-1234	8 minutes 58 seconds	11/22/23 3:47 AM	P2089G4043990179	View Flight	Export CSV
user1@skydio.com	Skydio X10-1234	16 minutes 8 seconds	11/21/23 5:04 PM	P2089G4043990395	View Flight	Export CSV

Skydio X10 Batteries

Navigate to Settings > Devices > Batteries. Here you can view the total number of flights, total flight time, and cycles for each battery in your fleet.

Settings

- General
- Ports
- Serial
- Devices**
- Sites
- Network
- Media Upload
- Limits
- Alerts
- Developer
- MOT Tools
- Miscellaneous

Devices

Vehicles Docks Batteries

Battery ID	Lifetime Flights	Lifetime Flight Time	Battery Config	Min/Max Voltage	Min/Max Temperature
0000000000000000	38	2 hours 52 minutes	0	3.55v - 4.75v	(3 °C) - 40 °C
0000000000000001	34	2 hours	0	3.95v - 4.54v	(12 °C) - 52 °C
0000000000000002	101	2 hours 12 minutes	0.1	3.75v - 4.42v	0 °C - 60 °C
0000000000000003	40	2 hours 58	1.4	3.75v - 4.46v	0 °C - 54 °C

Updating Your System

Skydio will not force an update for your system, however, for optimal performance, we recommend that you keep your Skydio system up-to-date. If flying your Skydio X10 as part of a larger fleet or organization, follow your organization's update guidelines.

Visit **Updating Skydio X10** in the preflight section of this manual for detailed instructions.

Replacing Propellers

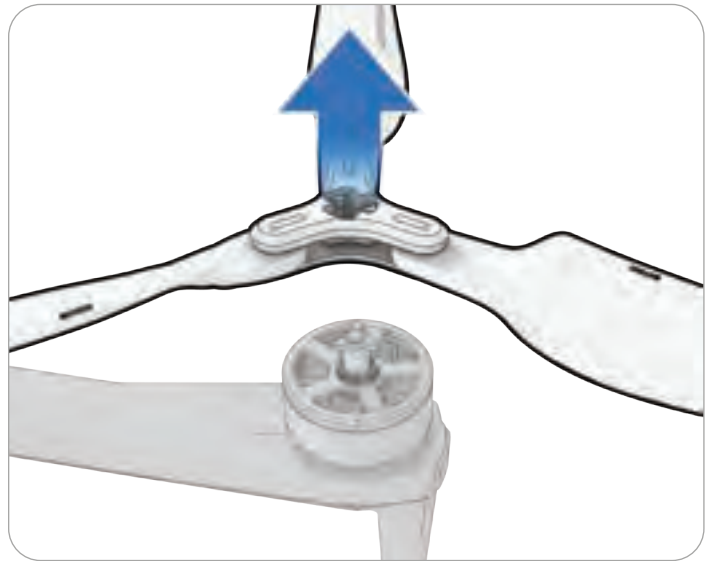
For optimal performance, Skydio recommends replacing your propellers **after 250 hours of flight time** or whenever you notice any damage.



WARNING: Propellers with hairline cracks or large breaks, chops, or bends should be replaced immediately. Do **NOT** fly Skydio X10 with propellers that are not in good condition as serious bodily harm or injury may occur.

Step 1 - Remove old or damaged propeller set

Hold onto the motor with one hand and take the propeller hub in the other. Press down on the propeller hub and twist to release.



NOTE: You will need to twist either clockwise or counterclockwise depending on the motor.

Maintenance

Step 2 - Identify the propeller set that matches the motor

Match the replacement set of propellers to the color on the motor.

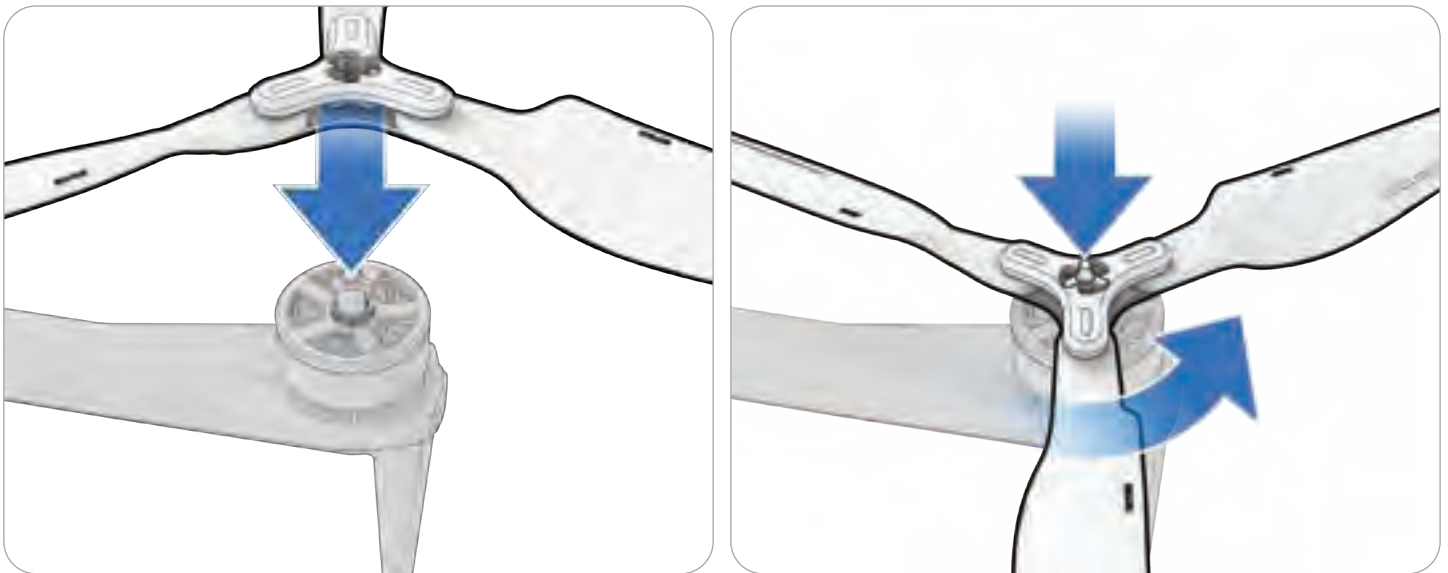
Step 3 - Inspect for any debris

Inspect the motor to ensure there is no dirt or debris.

Step 4 - Install new propeller set

Hold onto the motor with one hand and take the propeller hub in the other. Flip the propeller hub so the opening on the hub aligns with the the motor. Press down and twist to lock in place.

Be gentle and do not force the fitment. If the propeller set does not fit on the motor it might be the incorrect set or side.



NOTE: You will need to twist either clockwise or counterclockwise depending on the motor.



TIP: The total number of flight hours for your drone is tracked in Skydio Cloud. Consider replacing all propellers at the same time for ease of tracking total propeller flight time.

Cleaning Your System

It is recommended to wipe down your drone after flights in environments with significant dust or debris.



WARNING: Do not submerge your drone or batteries or place them under running water. Water volumes from flight in precipitation are much lower than those from a faucet or hose. Water may get into areas where the drone is not designed to withstand and you may compromise the sensors.

- Wipe down your drone with a dry or water-damp microfiber cleaning cloth.
- Only use lens cleaner on cameras.
- Do not submerge your drone or batteries.
- Do not place your drone or batteries under running water.
- Use a compressed air canister to remove any debris in hard to reach areas.
- If needed, mild soap and water may be used to remove heavier dirt or debris. Avoid getting any soap near ingress areas on the drone.
- Keep all the drain holes on the drone and battery clear. If any debris is blocking a drain hole (e.g., mud), use compressed air or gently scrape with a toothpick to remove.

Storage

- Do not store Skydio X10 while wet. After flying in precipitation, allow the drone to air dry in a dry, temperature-controlled environment before stowing. Visit the **Flying in Precipitation** section of this manual for more information.
- Store batteries at room temperature 71°F - 82°F (22°C - 28°C) for optimal performance and longevity.
- Store batteries in a cool, dry area with less than 75% relative humidity. Do not store your batteries in extreme environmental conditions.
- Batteries in an idle state (14 days of idle time with no flights) will start to self-discharge in an effort to retain capacity. This may take several days to complete and it is normal for the battery to be slightly warm during this discharge process.

Maintenance Schedule

To optimize the performance of your Skydio X10 it's important to keep your drone updated, inspect your equipment, store your equipment properly, and occasionally replace your propellers and batteries.

Action	Interval
Update system	When an update is available (per your organization's guidelines).
Clean drone navigation cameras	Before each operational session, and after sessions in dust or precipitation. If flying in areas with high amounts of dust or debris, you may be prompted to clean your camera lenses before each flight.
Replace propellers	Per 250 hours of flight time
Replace battery	Per 300 battery cycles A battery cycle is the depletion of at least 80% of the charge. An undamaged and properly stored battery can be safely used beyond 300 cycles.
Inspect and/or replace sensor package isolators	Every 40 flights or 30 hours of flight time. If you are storing the drone in hot temperatures (122°F/50°C to 131°F/55°C), then we recommend inspecting every 20 flights (or every 15 hours of flight time).



Specifications

This section covers

Skydio X10

Skydio X10 Controller

Sensor Packages

Navigation Camera System

Flight Battery

Skydio Connect

System Security

Skydio X10 Dual Charger and Power Supplies

Specifications

X10 Drone

Applicable Model Numbers	SR47PI, SR47PV, SR47PCI, SR47PCV SR47PC9V
Dimensions (unfolded, with propellers)	31.1" x 25.6" x 5.7"
Dimensions (folded, without battery)	13.8" x 6.5" x 4.7"
Weight (incl. batteries)	Connect SL: 2.11 kg / 4.65 lbs Connect SL + 5G1: 2.14 kg / 4.72 lbs
Max Launch Weight	2.49 kg / 5.49 lbs
Operation Frequency	Connect SL: 2400-2483.5MHz, 5150-5850Mhz
Transmitter Power (EIRP)	Connect SL: 34.3dBmi (2.4GHZ) Connect SL: 33.7dBmi (5GHZ)
Hovering Accuracy (windless or breezy)	VIO: +/- 10cm
Max Angular Velocity	GNSS: +/- 1m
Max Tilt Angle	Yaw: 100 deg/s Roll / Pitch: 225 deg/s 40 degrees
Max Ascent/Descent Speed	Ascent: 6 m/s / 13.4mph Descent: 4 m/s / 9.0mph
Max Non-Vertical Descent Speed	6 m/s / 13.4mph
Max Horizontal Speed (at sea level)	20 m/s / 45mph
Max Horizontal Speed with Obstacle Avoidance	16 m/s / 36mph
Max Service Ceiling Above Sea Level (without other payload)	15,000 ft density altitude
Max Gust Handling	Under 12.8 m/s / 28 mph
Max Hover Time	35 minutes
Max Flight Time ¹	40 minutes

Specifications

Processors	NVIDIA Jetson Orin SoC Qualcomm QRB5165 SoC
Ingress Protection Rating	IP55
GNSS	GPS + Galileo + GLONASS + BeiDou
Operational Temperature Range	-20C to +45C / -4F to 113F
Wireless Range (no interference, line of sight operation)	Connect SL: 12km / 7.5 miles
Wireless Networking (media offload)	Connect SL: WiFi6
Obstacle Avoidance Coverage	True 360°

¹In optimal, controlled conditions; completely depleting a fully charged battery

X10 Controller

Controller Dimensions	10" x 5" x 3"
Dimensions	10.5" x 5" x 3"
Screen	6.6" Dynamic AMOLED touchscreen 120Hz Adaptive Refresh Rate Resolution: 2340 x 1080 pixels Brightness: 1750 nits (outdoor peak) 392ppi
Weight	1135 grams
Max Range	Connect SL: 12km / 7.5 miles
Operating Frequencies	Connect SL: 2400-2483.5MHz, 5150-5850MHz
Transmitter Power (EIRP)	Connect SL: 34.7dBmi (2.4GHz) Connect SL: 35.9dBmi (5GHz)
Ingress Protection Rating	IP54
Operating Time	Approx. 5 hours
Battery	9600mAH
GNSS	GPS + Galileo + GLONASS + BeiDou
Operational Temperature Range	-20C to +45C / -4F to 113F
Wired Outputs	HDMI & USB-C
Wireless Networking	WiFi, Cellular LTE/5G 1
Security	NDAA compliant AES-256 encrypted data link Encrypted internal disk storage Password protected Root of trust Trusted boot Secure update

Specifications

VT300-Z Sensor Package

Angular Vibration Range	+/-0.01°
User Controllable Range	+/-90° pitch
Mechanical Range	+/-140° pitch, +/-90° yaw, +75° to -230° roll

VT300-L and V100-L Sensor Package

Angular Vibration Range	+/-0.01°
User Controllable Range	+/-90° pitch
Mechanical Range	+/-140° pitch, +/-90° yaw, +75° to -230° roll
Flashlight Illumination	22 lux at 3 meters

Specifications

Telephoto Camera (VT300-Z only)

Sensor	1/2" 48MP CMOS
Diagonal Field of View	13°
Focal Length	35 mm (190 mm equivalent)
Aperture	f/2.2
Focus	hybrid PDAF, 5 m to ∞
Exposure Compensation	+/-3
Electronic Shutter Speed	1/30 to 1/8000
ISO Range	100 to 16000
Max Video Resolution	3840 x 2880
Max Photo Size	8000 x 6000

Wide Camera (VT300-L and V100-L only)

Sensor	1" 50.3MP CMOS
Diagonal Field of View	93°
Focal Length	8 mm (20 mm equivalent)
Aperture	f/1.95
Focus	100% focus pixel, 1 m to ∞
Exposure Compensation	+/-3
Electronic Shutter Speed	1/30 to 1/8000
ISO Range	100 to 16000
Max Video Resolution	3840 x 2880
Max Photo Size	8192 x 6144

Narrow Camera (VT300-Z, VT300-L, and V100-L)

Sensor	1/1.7" 64MP CMOS
Diagonal Field of View	50°
Focal Length	10 mm (46 mm equivalent)
Aperture	f/1.8
Focus	hybrid PDAF, 1 m to ∞
Exposure Compensation	+ -3
Electronic Shutter Speed	1/30 to 1/8000
ISO Range	100 to 16000
Max Video Resolution	3840 x 2880
Max Photo Size	9248 x 6944

Thermal Camera (VT300-Z and VT300-L)

Thermal Imager	Flir Boson+ Uncooled VOx Microbolometer
Diagonal Field of View	41°
Focal Length	13.6 mm (60 mm equivalent)
Aperture	f/1.0
Focus	5 m to ∞
Thermal Sensitivity	<30mK NEDT
Infrared Temperature Measurement Accuracy	Larger of +- 5°C or 5%
Image Processing	Adreno 650 GPU accelerated ISP pipeline
Max Video Resolution	640 x 512
Photo Size	640 x 512
Photo Format	JPEG, RJPEG
Pixel Pitch	12 um
Temperature Measurement Method	Spot Meter, Region of Interest
Temperature Measurement Range	-40° to 150° C (-40° to 350° C low gain)
Palette	White hot Black hot Ironbow Rainbow

Specifications

Vision Systems / Navigation Cameras

Configuration	6x cameras in trinocular configuration top and bottom
Sensor	1/2.8" 32MP color CMOS
Light Sensitivity	Visible Light
Aperture	f/1.8
Diagonal Field of View	200°
Obstacle Sensing Range	20 meters
Environment Coverage	True 360°

Flight Battery

Capacity	8419 mAh
Voltage	18.55 V
Battery Type	Rechargeable Lithium Ion Polymer
Energy	156.17 Wh
Net Weight	1.56 lbs +/- 0.003 lbs
Operational Temperature Range	-20C to 60C
Storage Temperature Range	-20C to +45C (storage less than 3 months)
Charging Temperature Range	5C to 45C
Chemical System	Lithium Ion Polymer

Specifications

Skydio Connect

Operating Frequency	Connect SL: 2400-2483.5MHz, 5150-5850MHz
Transmitter Power (EIRP)	Connect SL: 34.7dBmi (2.4GHz) Connect SL: 35.9dBmi (5GHZ)
Antenna Configuration	Connect SL: 2Tx, 4Rx

System Security

Wireless Encryption	Connect SL: AES-256
NDAA Compliance	NDAA Compliant
Root of Trust	HSM protected keys
System Integrity	Secure boot
Secure Update	AES-256 encrypted, signed, & verified
Internal Disk Storage	Encrypted
SD Cards	Unencrypted
Pairing	Secure wired pairing

X10 Dual Charger and Power Supplies

Dimensions	180 x 75 x 48mm
Weight (w/o Battery)	0.73 lbs
Charge Time 230W	1 hour (0-100%)
Charge Time 100W	1 hour 45 minutes (0-100%)
Weatherproofing	No ingress protection
Power input (100W USB-C)	5VDC, 3A / 20VDC, 5A (USB PD)
Power input (230W DC Barrel)	20VDC, 11.5A



Legal

This section covers

Safety

Battery

Skydio One (1) Year Limited Warranty

Skydio Care

California Prop 65 Warnings

FCC Compliance Statement

FAA Compliance Statement

Skydio X10

Before operating Skydio X10, review the *Getting Started* information including the *Operator Manual* available at www.skydio.com/manuals. Retain documentation for future reference.

Safety

Review the *Skydio Safety and Operating Guide* available at www.skydio.com/safety.

Battery

Handle the battery with extreme care and refer to the Operator Manual and to the *Skydio Safety and Operating Guide* for additional information.

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the equipment manufacturer. Carefully dispose of batteries according to manufacturer's instructions and to your local environmental laws and guidelines.

Risque d'explosion si la batterie n'est pas correctement remplacée. Remplacer uniquement par un type identique ou équivalent recommandé par le fabricant de l'équipement. Jeter les batteries conformément aux instructions du fabricant et aux lois et directives environnementales locales.

Charging

Do not use the X10 Dual Charger near wet locations. To avoid the risk of electric shock, use only in dry locations. Do not allow anything to rest on the power cord. Do not locate this device where the cord will be abused by persons working on it. Do not overload wall outlets and extension cords as this can result into fire or electric shock.

N'utiliser pas le X10 Dual Charger à proximité d'endroits humides. Pour éviter tout risque de choc électrique, utiliser uniquement dans des endroits secs. Ne laisser rien reposer sur le cordon d'alimentation. Ne placer pas ce moniteur dans un endroit où le cordon pourrait être maltraité par les personnes travaillant dessus. Ne surcharger pas les prises murales et les rallonges car cela pourrait provoquer un incendie ou un choc électrique.

Skydio One (1) Year Limited Warranty

Skydio warrants the included hardware product against defects in materials and workmanship in hardware under normal use in accordance with published guidelines including but not limited to the *Terms of Use*, *Operator Manual* and the *Skydio Safety and Operating Guide* for one year from the date of delivery (the “Limited Warranty”). The Limited Warranty does not warrant against normal wear and tear or damage caused by accident or abuse and is not applicable to any software provided with the hardware product. The Limited Warranty is subject to the full terms and detailed information about how to obtain service available at

www.skydio.com/legal/limited-warranty. If you submit a valid claim under this Limited Warranty, Skydio will either repair, replace, or refund your hardware product at its sole discretion. You may be required to furnish proof of purchase details when making a claim under this Limited Warranty.

Skydio Care

Skydio offers Skydio Care as a subscription service at an additional cost that provides protection from collisions, water damage, or lost drones which are not covered under the Limited Warranty. Skydio Care can be purchased as a one (1) year plan co-extensive with the Limited Warranty, or as a three (3) year plan, which includes a two (2) year extension to the one (1) year term of the Limited Warranty. Skydio Care is subject to the full terms and detailed information about how to obtain service available at <https://www.skydio.com/legal/skydio-care-terms-of-service>. If you submit a valid claim under Skydio Care, you may be eligible to purchase discounted drone replacements for otherwise uncovered damage or losses. The Skydio Care benefits are in addition to the rights provided under the Limited Warranty.

California Prop 65 Warnings

Skydio X10 uses lithium-ion batteries. Exposure to lithium-ion, containing cobalt lithium nickel oxide, and nickel, is known to the State of California to cause cancer and birth defects, or other reproductive harm. For more information visit:

www.P65Warnings.ca.gov

Skydio X10 Controller contains chemicals including cadmium, which is known to the State of California to cause cancer and birth defects, or other reproductive harm. For more information visit: www.P65Warnings.ca.gov

Skydio X10 Dual Charger contains chemicals including BPA and nickel, which are known to the State of California to cause cancer and birth defects, or other reproductive harm. For more information visit: www.P65Warnings.ca.gov

FCC and ISSED Compliance Statement

These devices comply with Part 15 of the FCC Rules and with ISSED Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) these devices may not cause harmful interference, and (2) these devices must accept any interference received, including interference that may cause undesired operation. The X10 drone should be operated with a minimum distance of 8.5 in (22 cm) between the X10 drone antennas and your body.

Le drone X10 devrait être utilisé à une distance minimale de 8.5 po (22 cm) entre les antennes de drone X10 et votre corps. Ces appareils sont conformes aux normes RSS exemptes de licence d'ISDE Canada. Leur fonctionnement est soumis aux deux conditions suivantes: (1) ces appareils ne doivent pas causer d'interférences nuisibles, et (2) ces appareils doivent accepter toutes interférences reçues, y compris les interférences susceptibles d'entraîner un fonctionnement indésirable.

Changes or modifications not expressly approved by Skydio could void the user's authority to operate these devices.

These devices have been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when these devices are operated in a commercial environment. These devices generate, use, and can radiate radio frequency energy and, if not installed and used in accordance with the Operator Manual and Safety and Operating Guide, may cause harmful interference to radio communications. Operation of these devices in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

This Class A digital apparatus complies with Canadian ICES-003.

Cet appareil numérique de classe A est conforme à la norme Canadienne NMB-003.

FAA Compliance Statement

Unless specifically exempt, this product complies with 14 CFR Part 89 regulations on Remote Identification per ASTM F3411-22a-RID-B and ASTM F3586-22.

Software License

The *Skydio Software End-User License Agreement* available at www.skydio.com/legal/eula governs the use of any Skydio software that is pre-installed, downloaded, installed, or otherwise provided in connection with any included hardware.

Additional Resources

For all the latest information about Skydio and our products visit: www.skydio.com

For Skydio legal information and product terms of use visit: www.skydio.com/legal

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